Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: DESIGN CALCULATION FOR SUBSTATION FOUNDATION -49W1 Raw Water Treatment Plant

Document Number:	**MD-216-49W1-EG-CV-CAL-1003**
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TABLE OF CONTENTS

1. Intent 4
2. Scope 4
3. Material property 4
4. Soil bearing capacity 4
5. Substation foundation 5
   5.1. Structure geometry 5
6. Load calculations 7
   6.1. Design loads 7
   6.2. Load combinations 7
7. Analysis results 23
   7.1. Design of columns 23
   7.2. Global deformation 28
   7.3. Check of soil pressure 30
   7.4. Check of overturning and sliding 31
   7.5. Check of punching shear 32
   7.6. Raft design 33
8. Transformer room 37
   8.1. Structure geometry 37
   8.2. Load cases 37
   8.3. Analysis results 43
      8.3.1. Global deformation 43
      8.3.2. Column design 45
      8.3.3. Plinth beam design 47
      8.3.4. Ground slab design 48
      8.3.5. Pad footing design 50
      8.3.6. Stability check 52
9. Reference 54

Attachment 1 Input drawing 55

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Attachment 2 Comment resolution sheet..... 56

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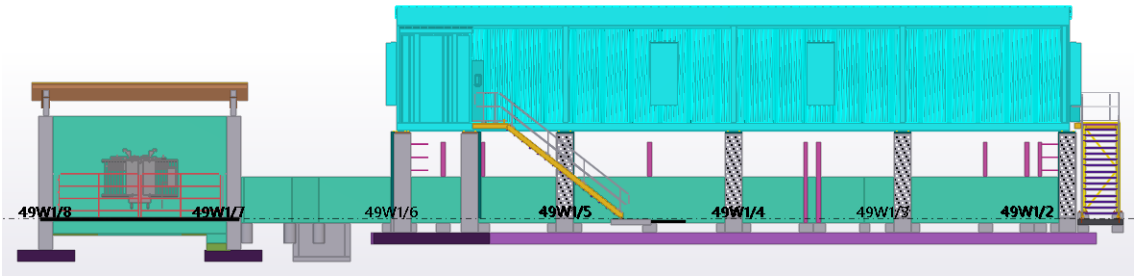
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1. INTENT

This calculation report is to design and analyze substation foundations and transformer room in Raw Water Treatment Plant area. The report describes the methodology and the design parameters used in the design calculation.

2. SCOPE



The design follows and verified with requirements in Saudi Building Code:

- SBC_301 (Structural – Loading and Forces)
- SBC_304 (Structural – Concrete Structures)
- SBC_303 (Structural – Soil and Foundations)

Concrete structures are modeled as surface and member in Dlubal RFEM 5.21, both strength and allowable stress calculations are included.

3. MATERIAL PROPERTY

Concrete compressive strength at 28 days: f_{ck} = 35 N/mm^2

Reinforcing bars ASTM A615- Grade 60, yield strength f_{yk} = 420 N/mm^2

4. SOIL BEARING CAPACITY

Vertical modulus of subgrade reaction $K_s = 120q_a$ = 33,750 kN/m^3 is calculated and applied in RFEM as spring constant in z direction. Horizontal modulus of subgrade reaction K_s = 155,000 kN/m^2 is applied as line support in x and y directions. Recommended allowable bearing capacity q_a = 375 kN/m^2 is given in geotechnical report. Therefore, the contact stress results shall not exceed the limit.

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5. SUBSTATION FOUNDATION

5.1. Structure geometry

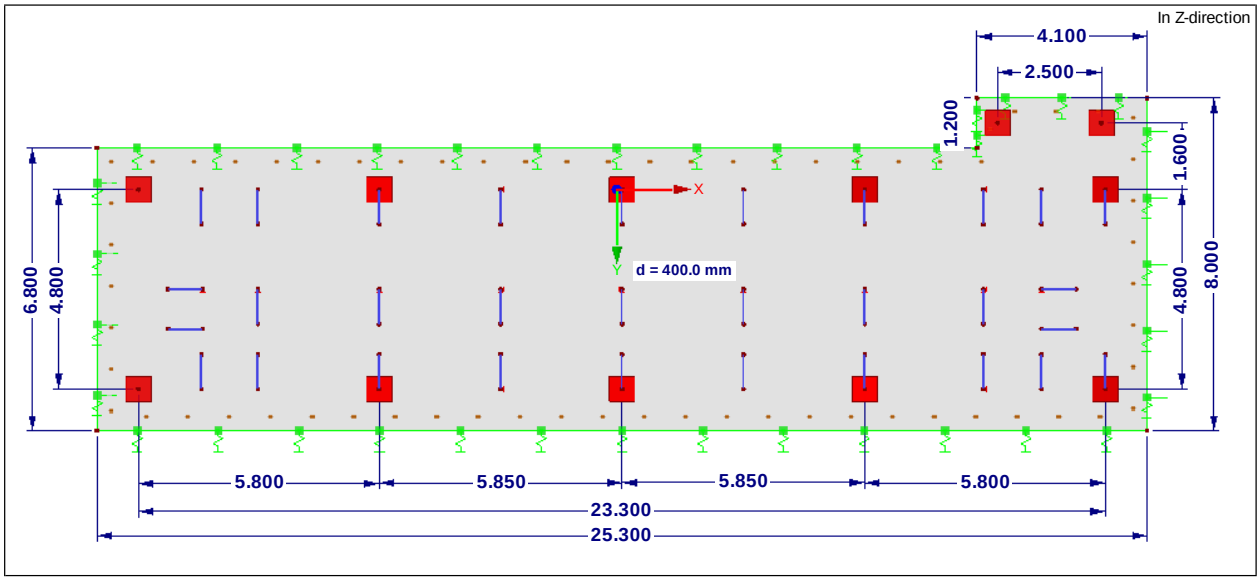
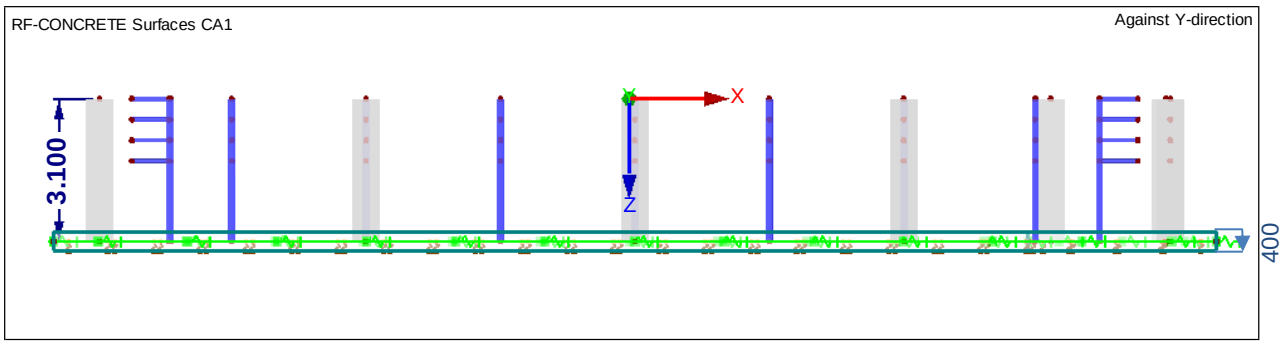


Figure 1 Calculation model in RFEM



C01

Figure 2 Section A-A

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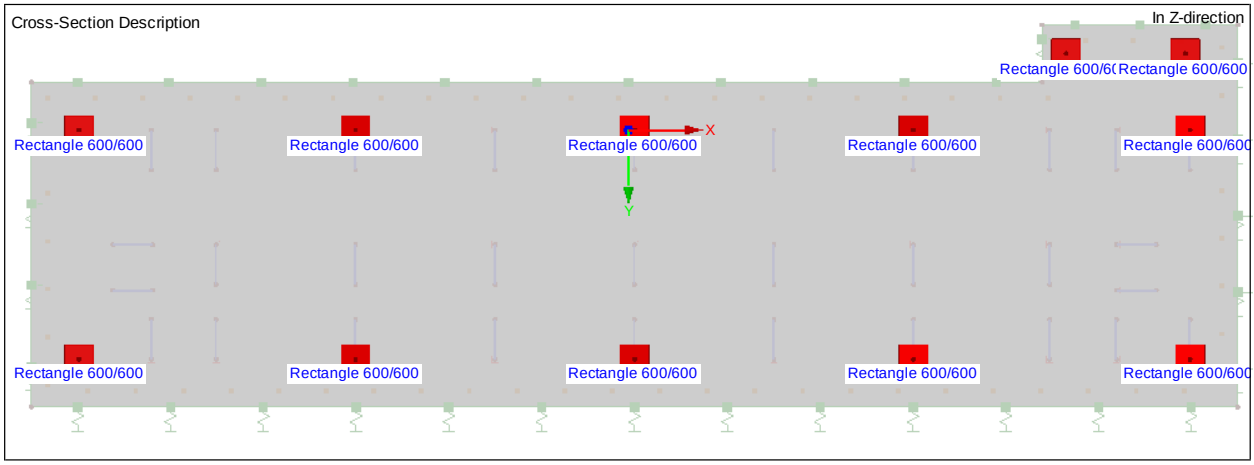


Figure 3 Concrete column dimension

600mmx600mm for all rectangular concrete columns.

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6. LOAD CALCULATIONS

Load data is based on drawing number MD-216-49W1-EG-EL-GAD-V079-1004-A02, see Attachment 1.

6.1. Design loads

1	Dead load (D_1)
2	Live load (L)
3	Equipment load (D_2)
4	Seismic load (E)
5	Wind load (W)

6.2. Load combinations

For strength calculation:

Load Combination	Expression
101	1,40 (D_1 +D_2)
102	1,40 (D_1 +D_2) + 1,70 L
103	1,20 (D_1 +D_2) + 1,60 W + 1,00 L
104	1,20 (D_1 +D_2) + 1,0 E
105	0,90 (D_1 +D_2) + 1,60 W
106	0,90 (D_1 +D_2) + 1,0 E

For allowable stress calculation:

Load Combination	Expression
201	D_1 +D_2
202	D_1 +D_2 + L
203	D_1 +D_2 + W
204	D_1 +D_2 + 0,70 E
205	D_1 +D_2 + 0,75 W + 0,75 L
206	D_1 +D_2 + 0,75 (0,70 E) + 0,75 L
207	0,60 (D_1 +D_2) + W
208	0,60 (D_1 +D_2) + 0,70 E

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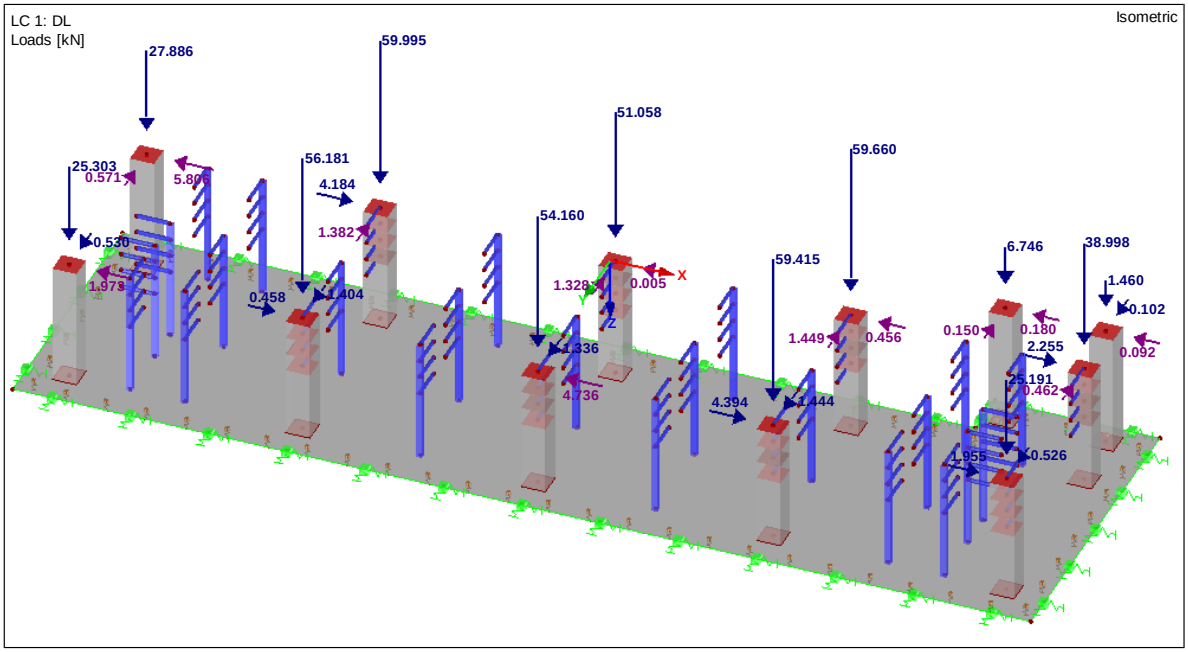


Figure 4 Dead load

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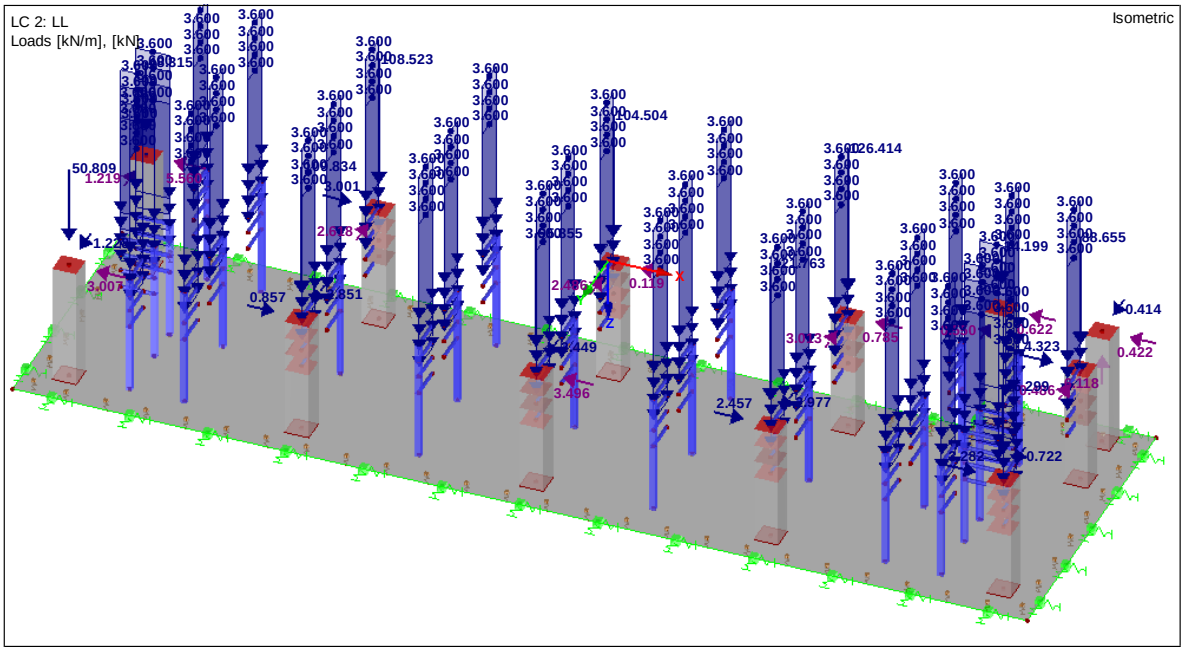


Figure 5 Live load (including cable tray weight)

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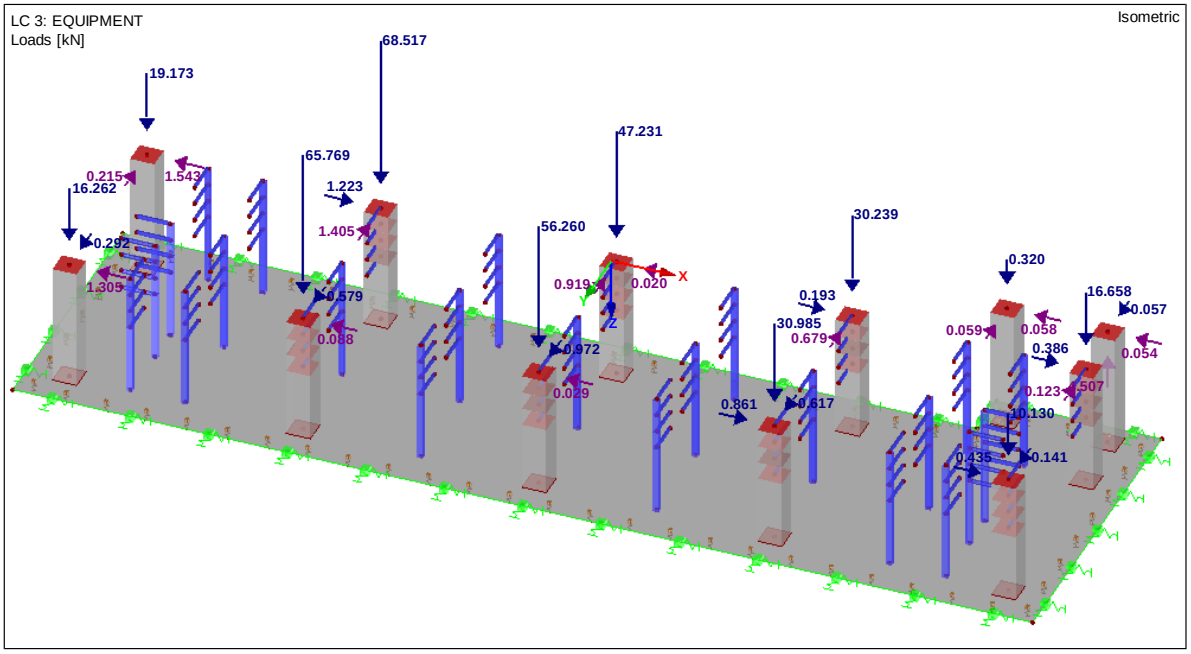


Figure 6 Equipment load

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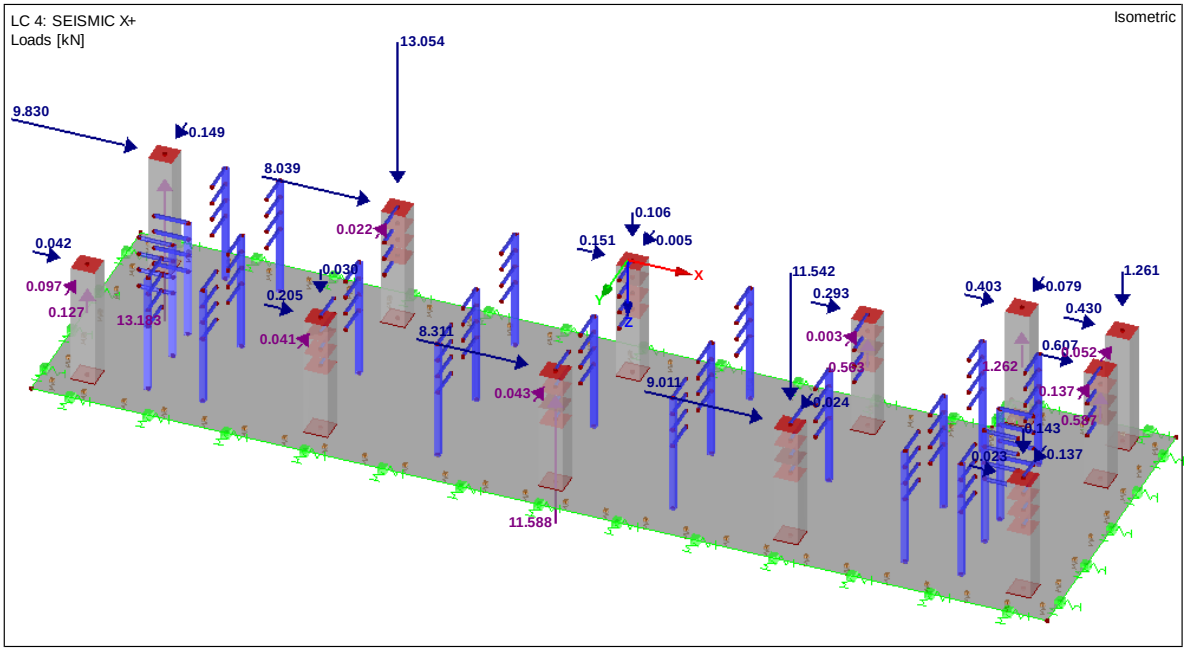


Figure 7 Seismic load X+

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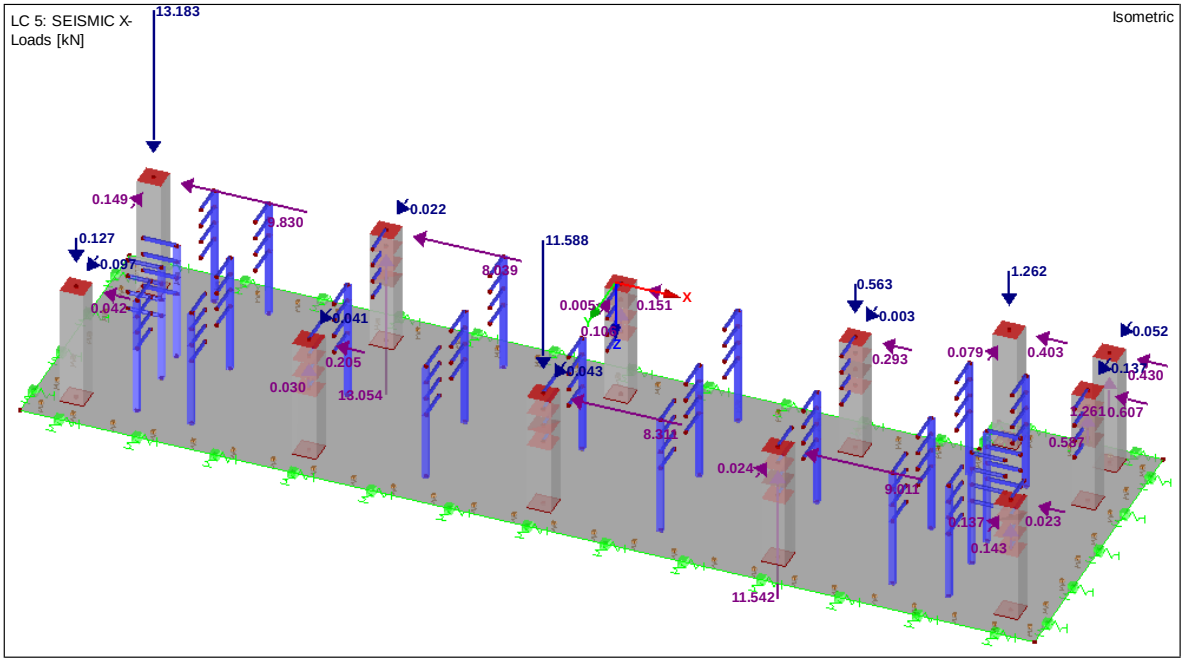


Figure 8 Seismic load X-

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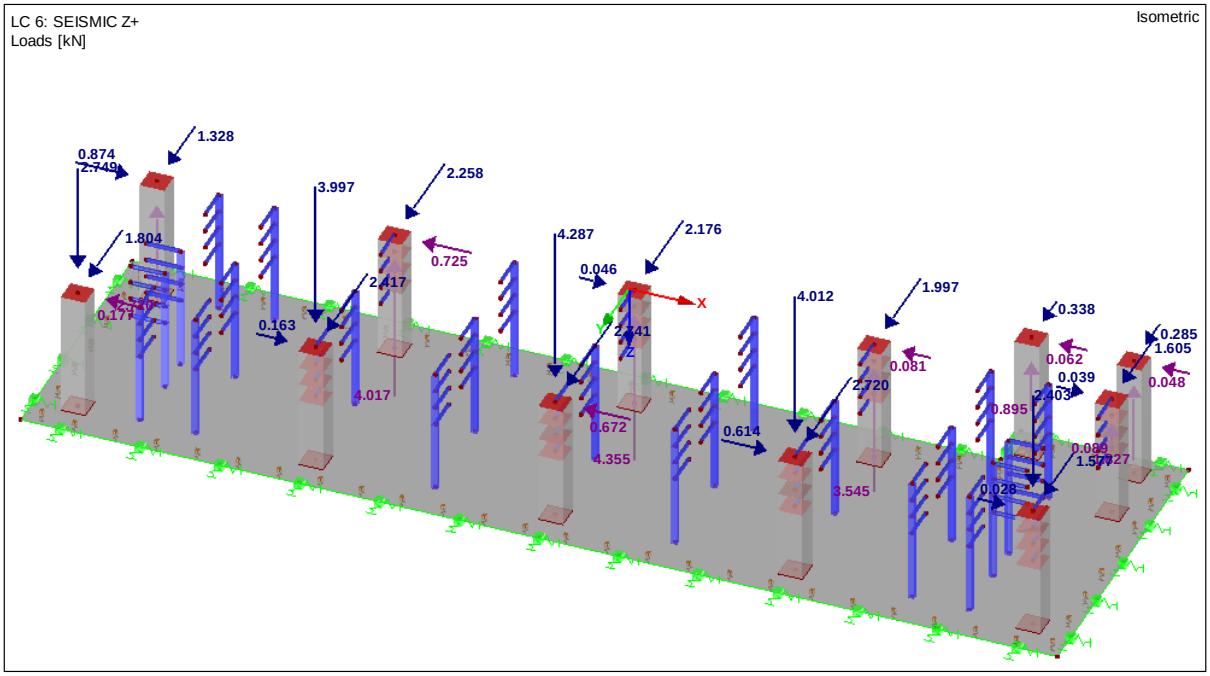


Figure 9 Seismic load Z+

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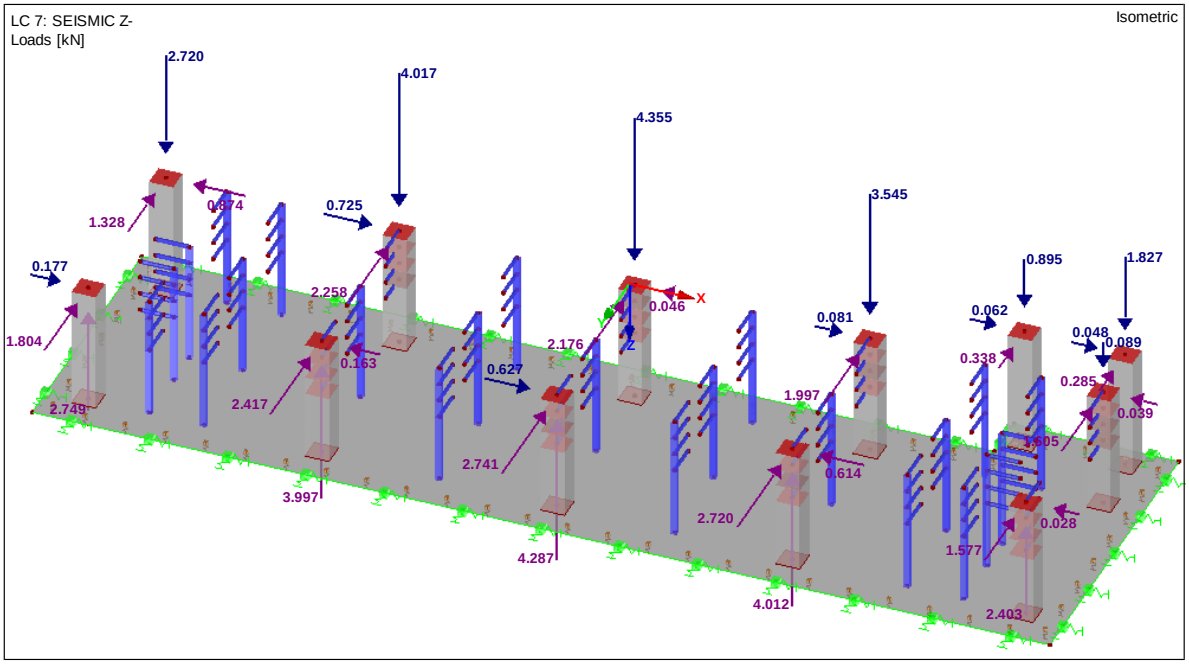


Figure 10 Seismic load Z-

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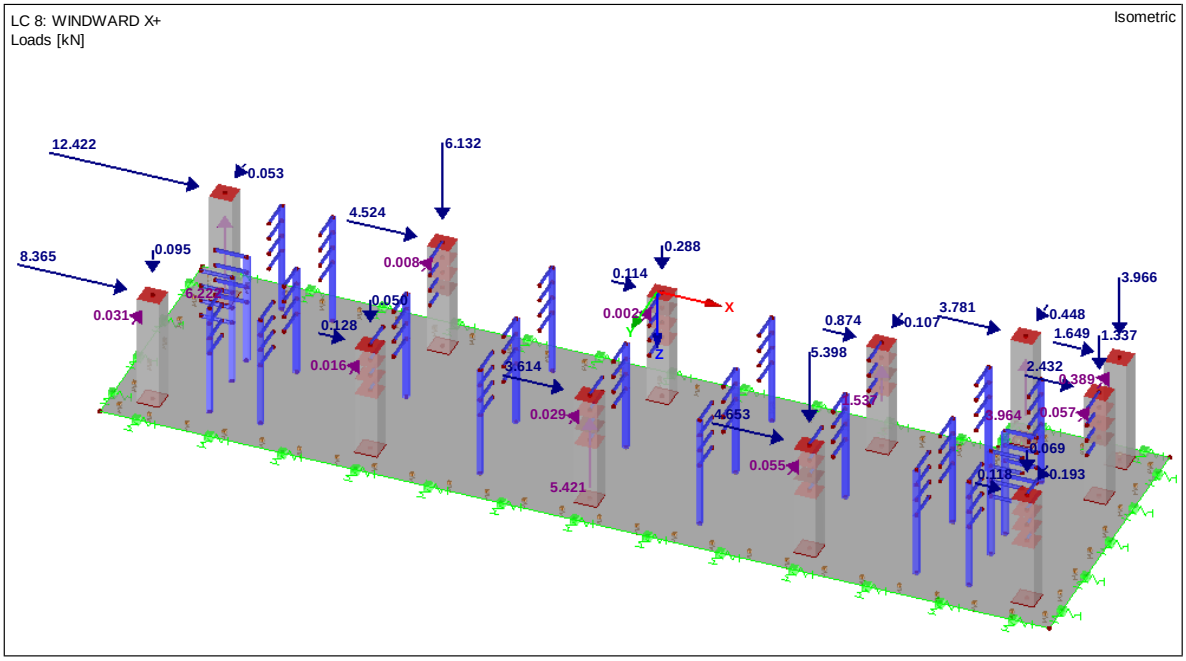


Figure 11 Windward load X+

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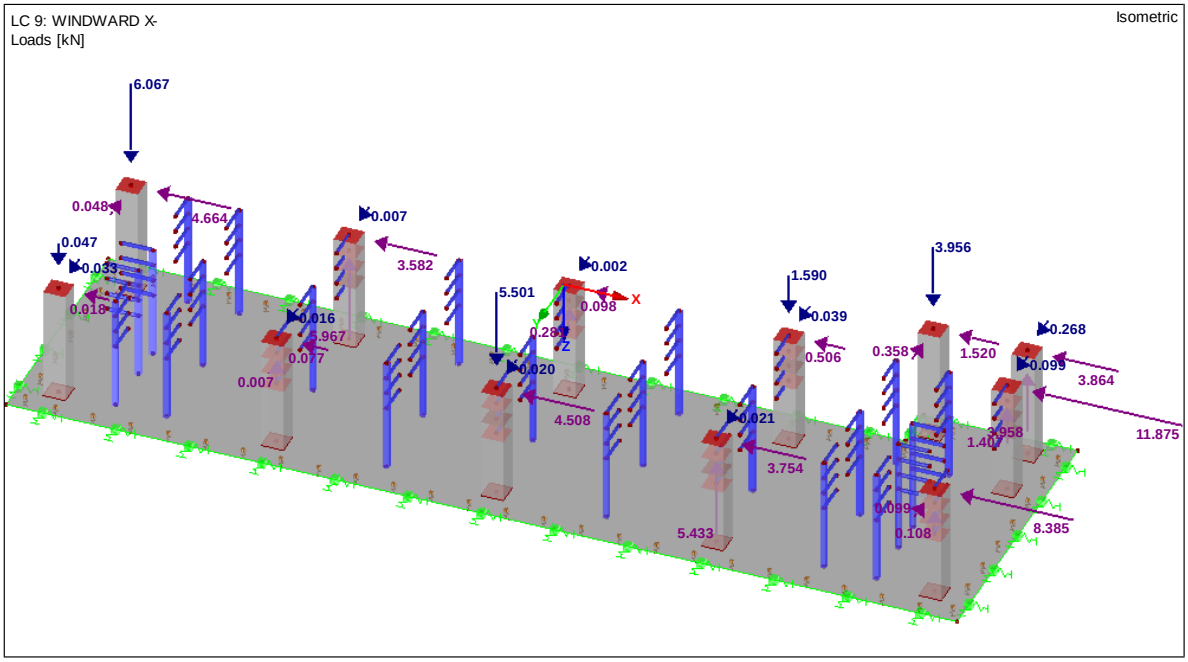


Figure 12 Windward load X-

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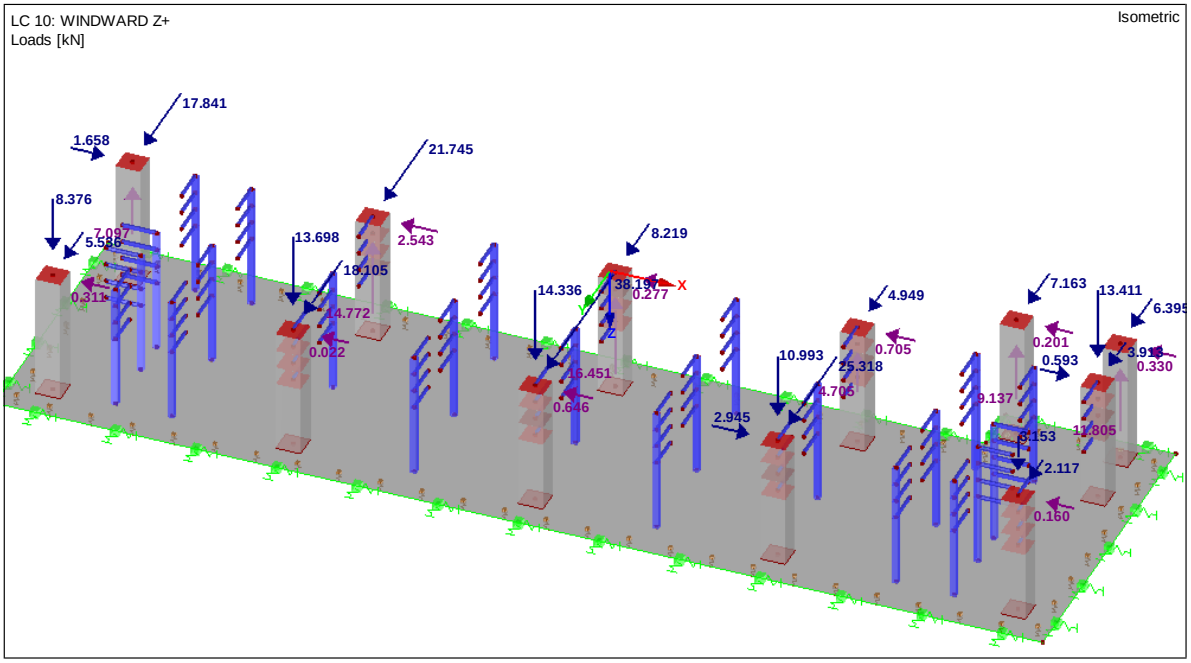


Figure 13 Windward load Z+

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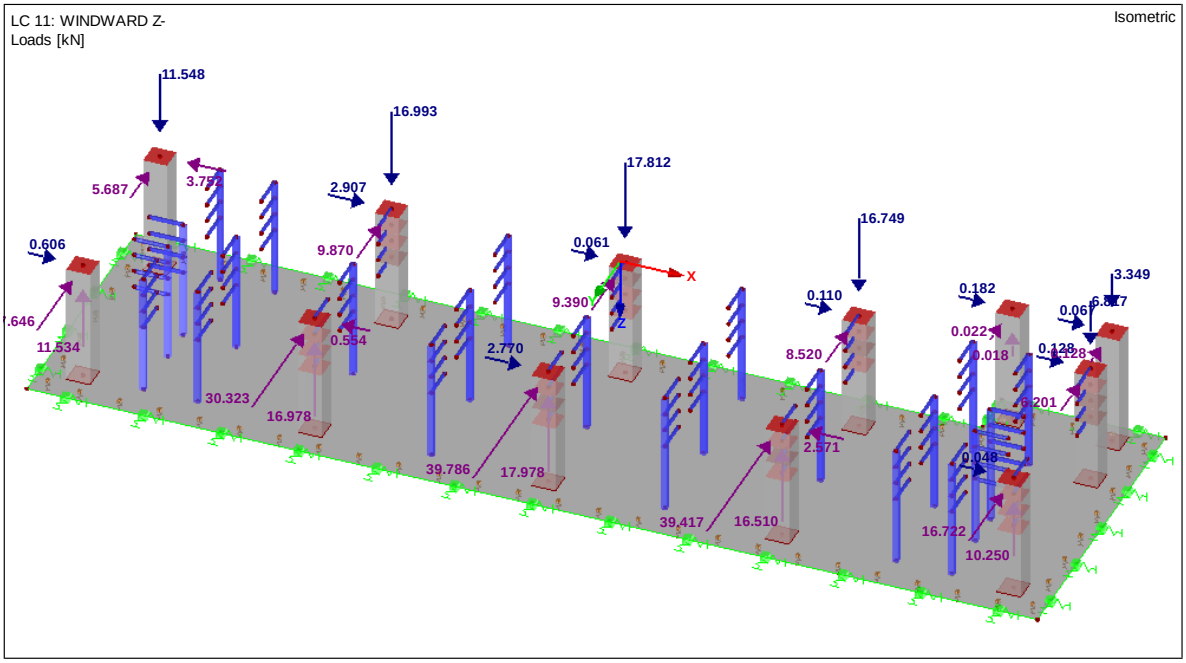


Figure 14 Windward load Z-

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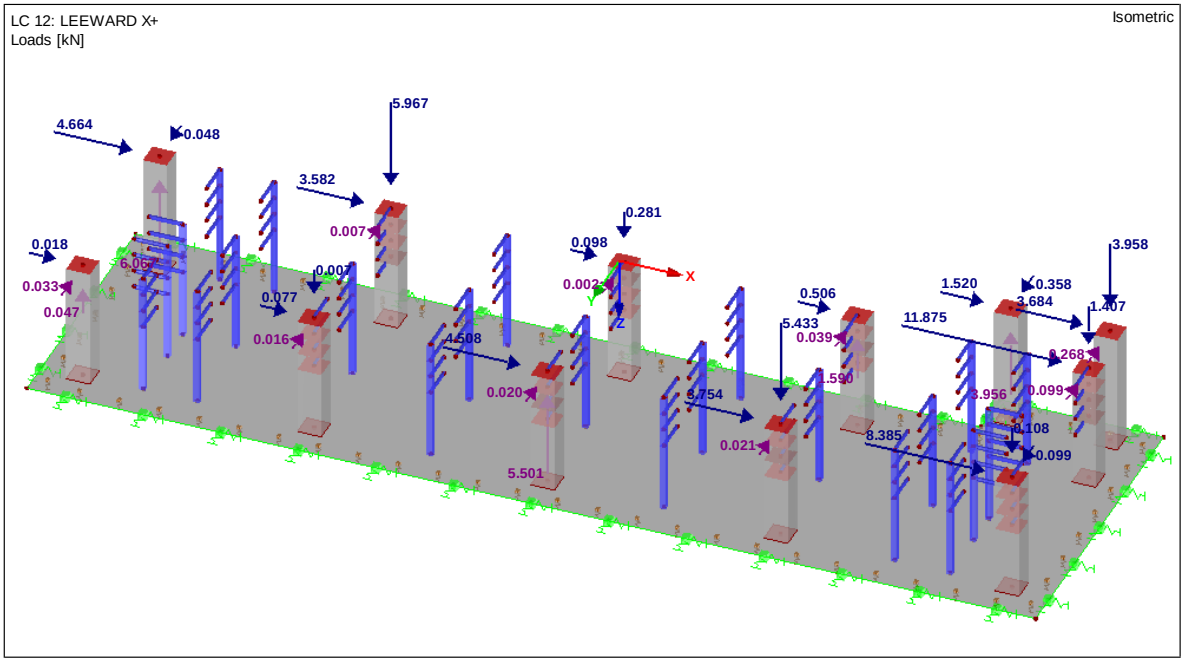


Figure 15 Leeward load X+

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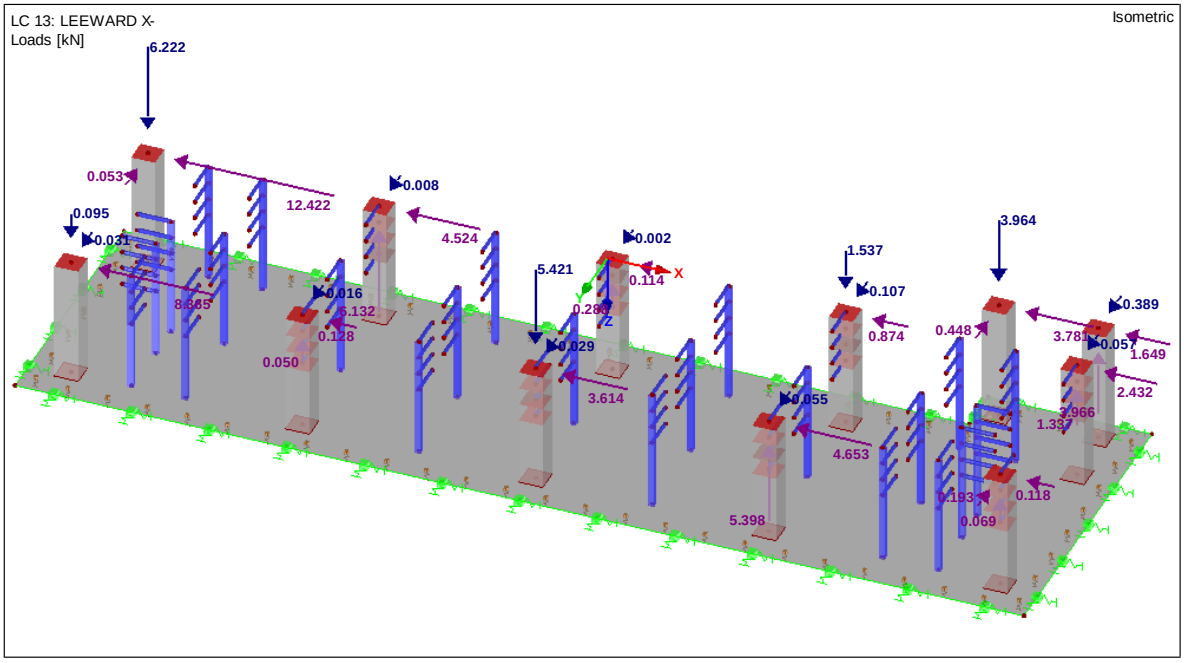


Figure 16 Leeward load X-

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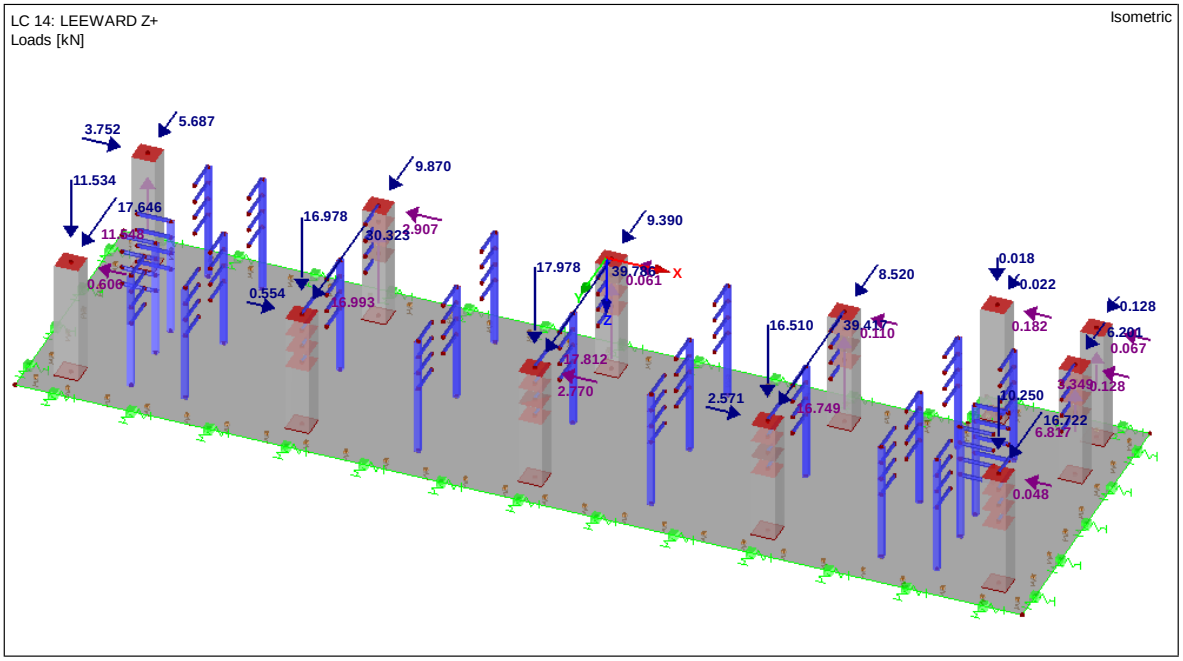


Figure 17 Leeward load Z+

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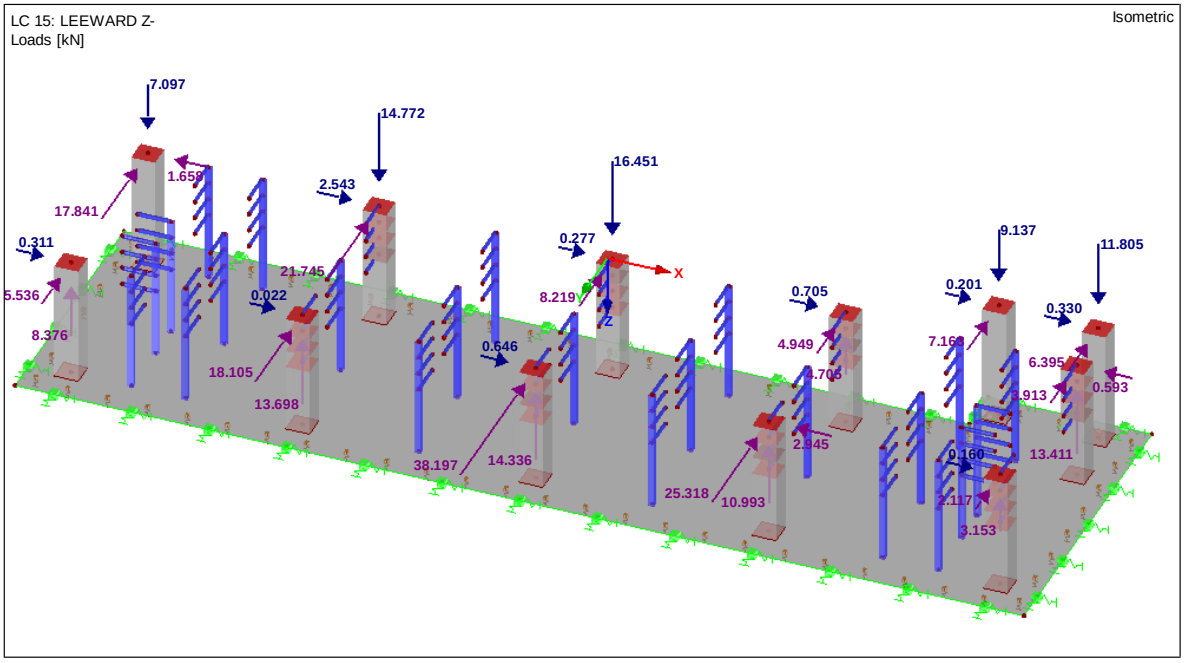


Figure 18 Leeward load Z-

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7. ANALYSIS RESULTS

7.1. Design of columns

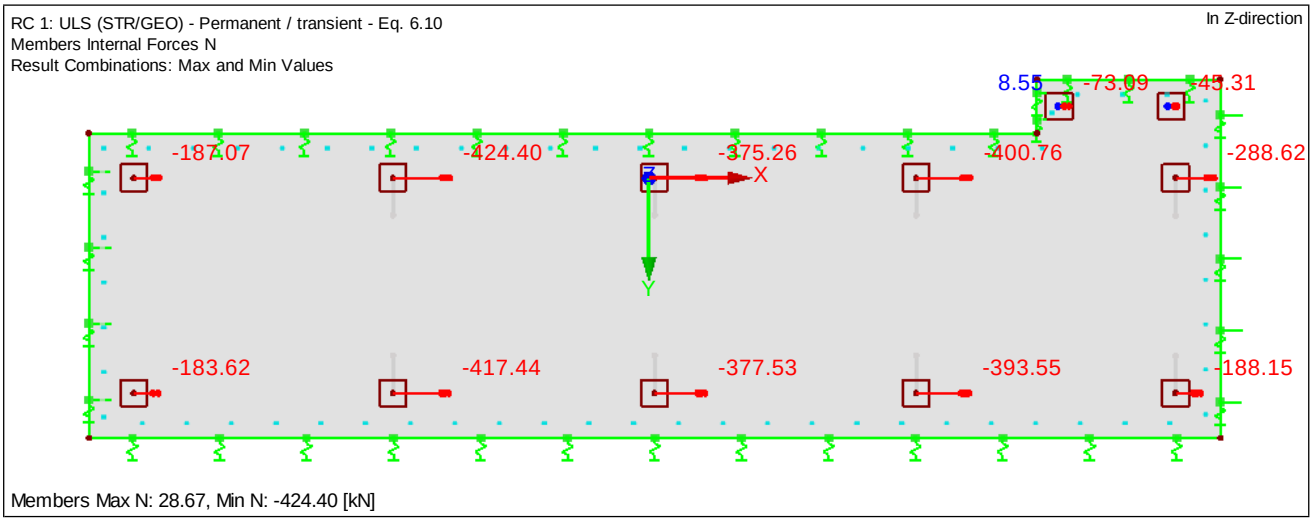


Figure 19 Axial force N_{Ed}

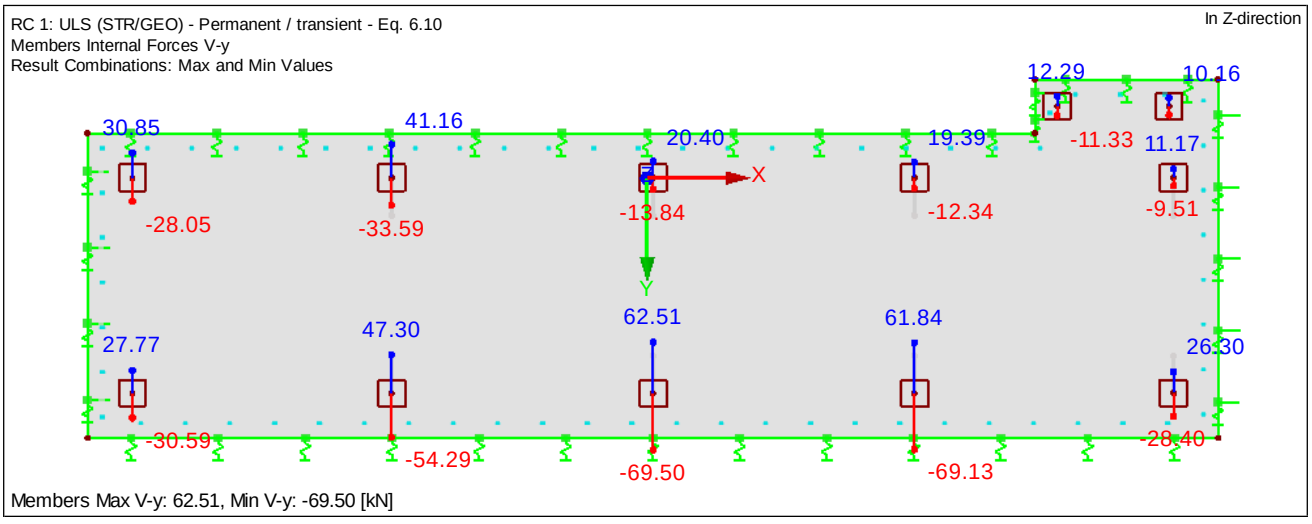


Figure 20 Shear force Vy

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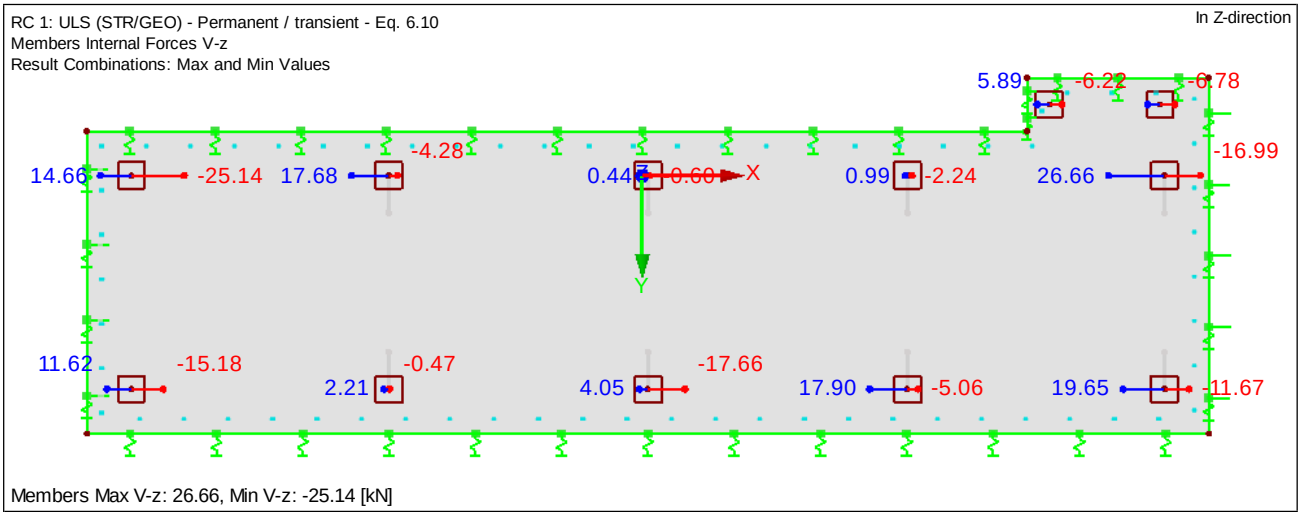


Figure 21 Shear force Vz

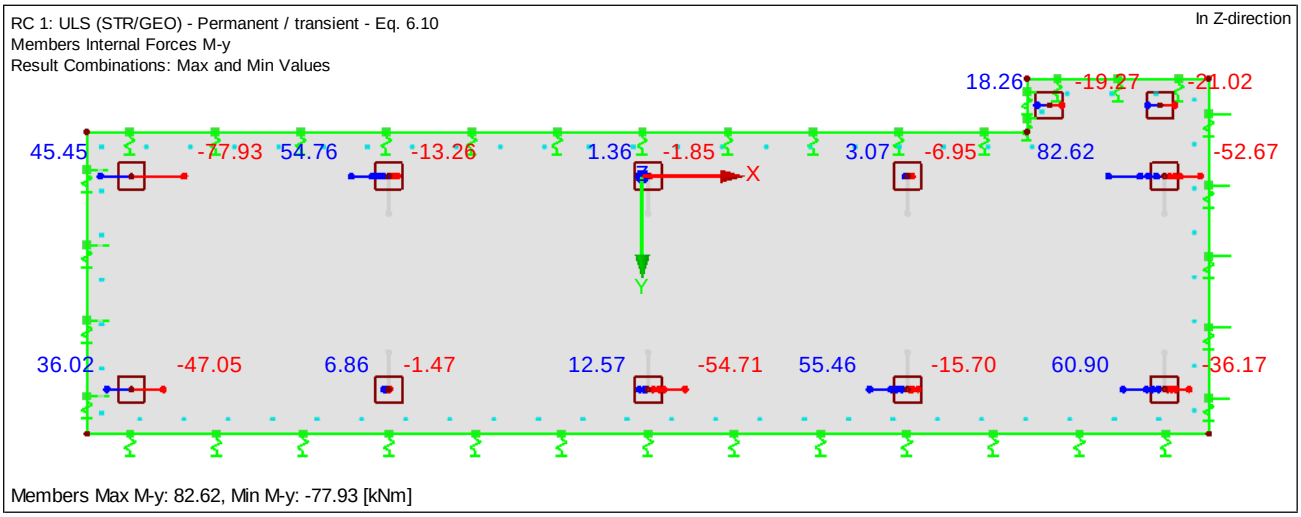


Figure 22 Bending moment My

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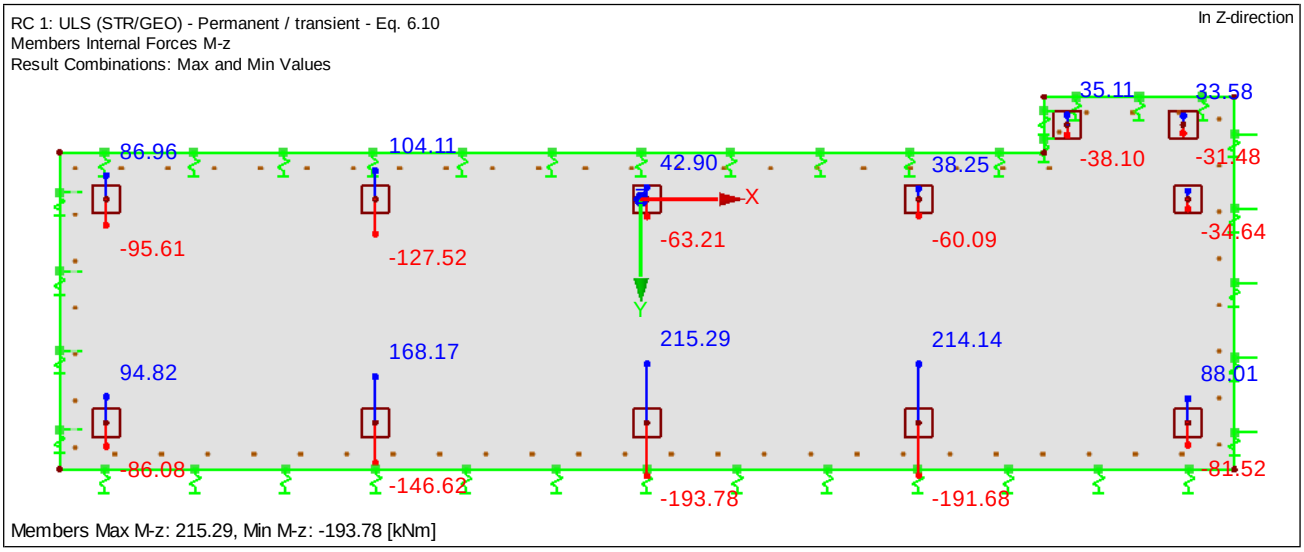


Figure 23 Bending moment Mz

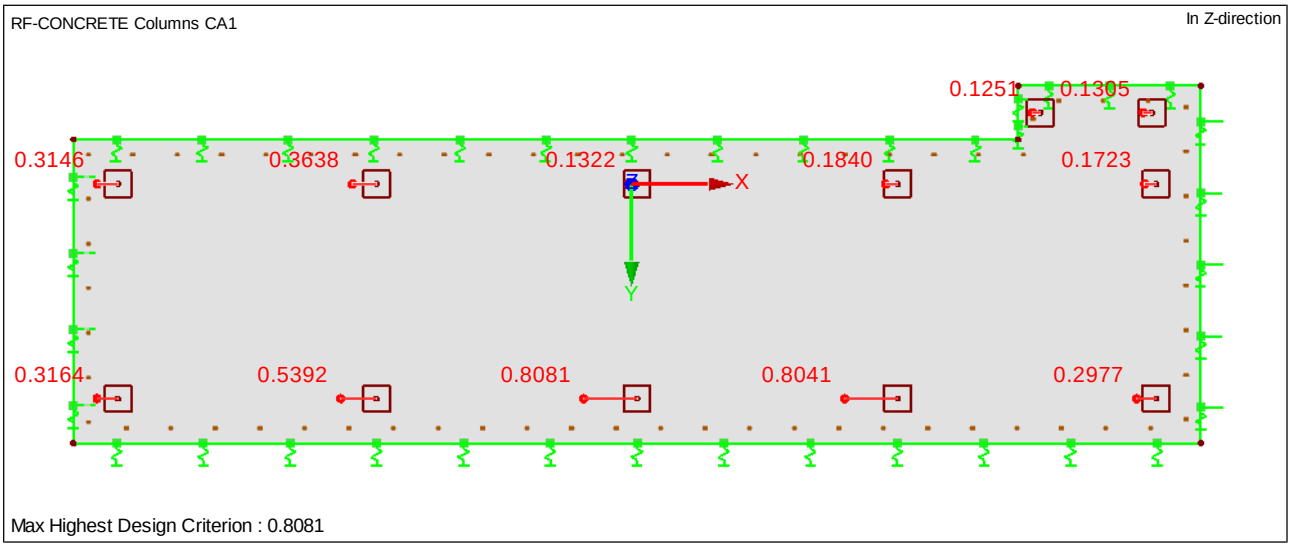


Figure 24 Highest utility ratio

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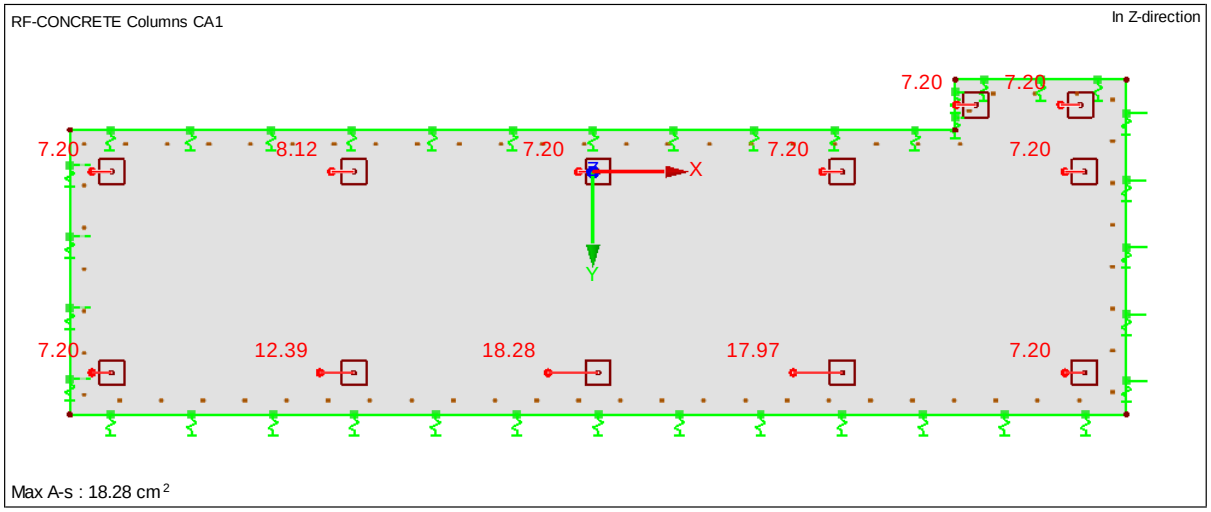


Figure 25 Required longitudinal reinforcement area As

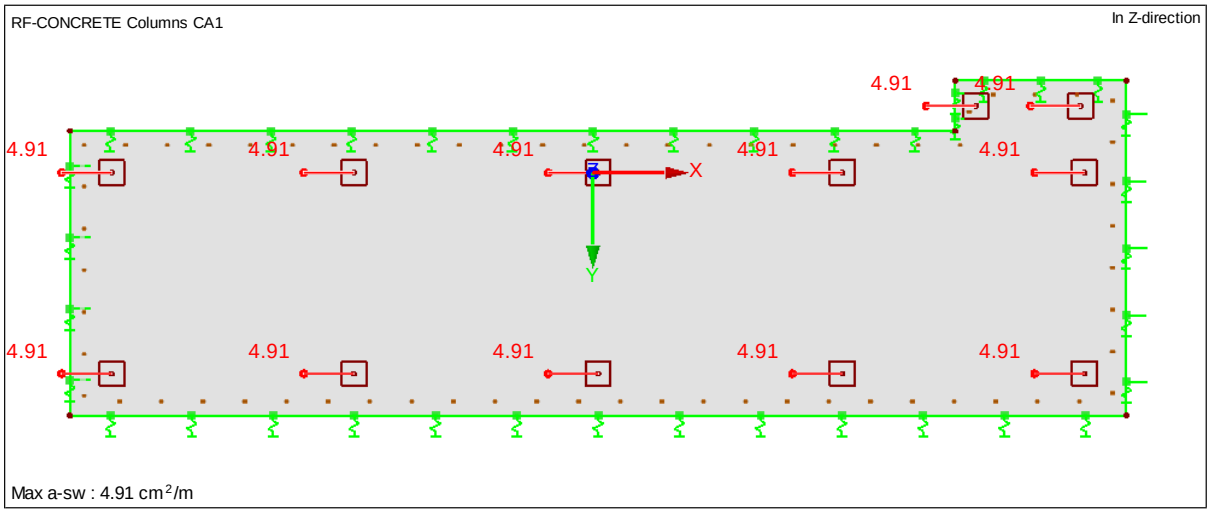


Figure 26 Required and provided shear reinforcement area a_{sw}

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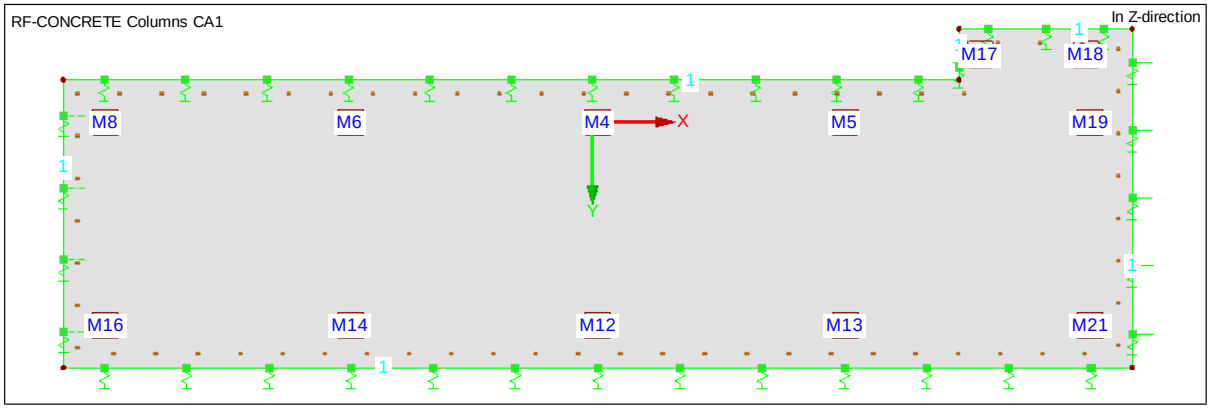


Figure 27 Column number

Table 1 Column reinforcement summary

column nr.	cross-section [mm]	height [mm]	main bars	$A_{s,provided}$	shear reinforcement
M17	600x600	3100	12T16	24.13	T12-300mm
M18	600x600	3100	12T16	24.13	T12-300mm
M8	600x600	3100	12T16	24.13	T12-300mm
M6	600x600	3100	12T16	24.13	T12-300mm
M4	600x600	3100	12T16	24.13	T12-300mm
M5	600x600	3100	12T16	24.13	T12-300mm
M19	600x600	3100	12T16	24.13	T12-300mm
M16	600x600	3100	12T16	24.13	T12-300mm
M14	600x600	3100	12T16	24.13	T12-300mm
M12	600x600	3100	16T16	32.16	T12-300mm
M13	600x600	3100	16T16	32.16	T12-300mm
M21	600x600	3100	12T16	24.13	T12-300mm

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7.2. Global deformation

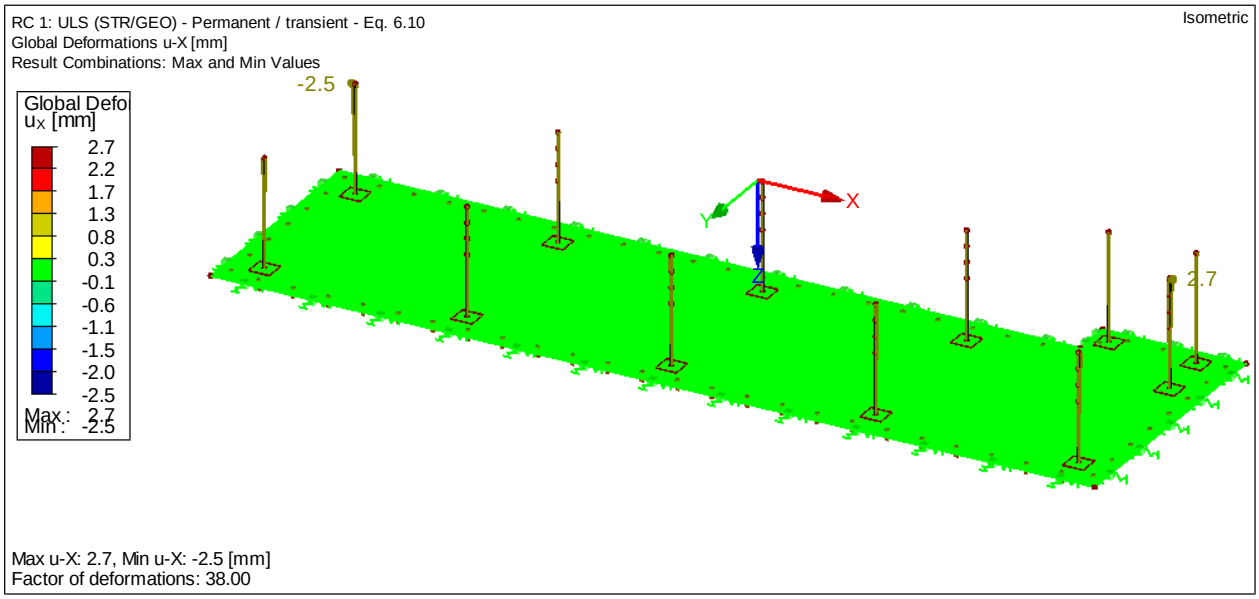


Figure 28 deformation ux

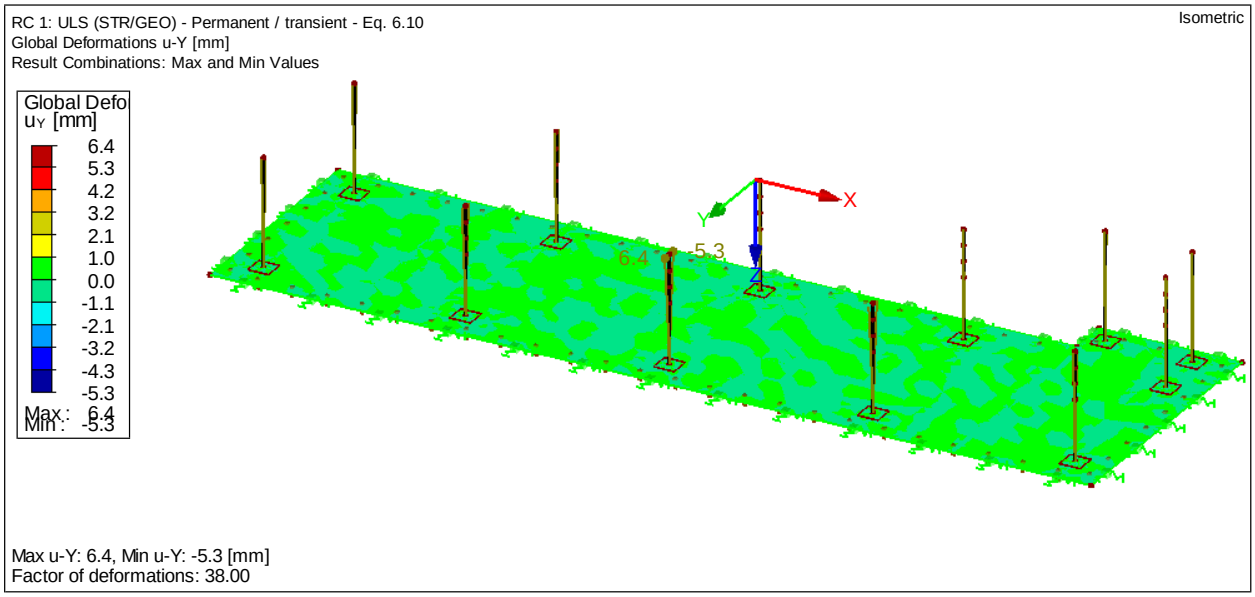


Figure 29 deformation uy

C01

u< u_{max}=H/300=10.3mm, ok

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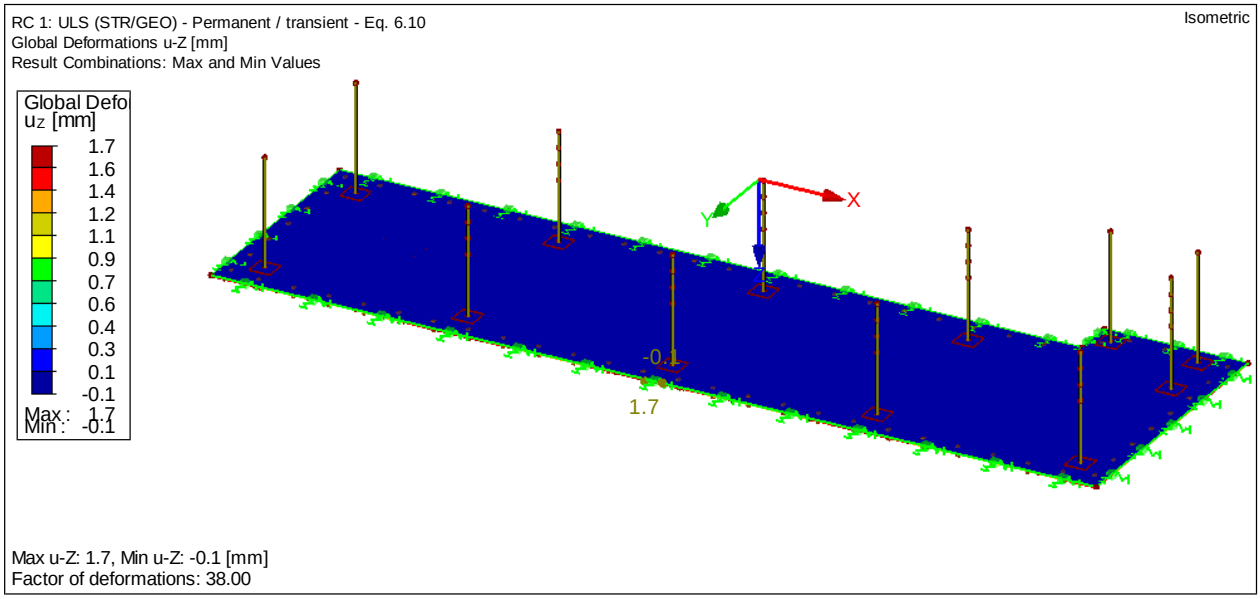


Figure 30 settlement uz

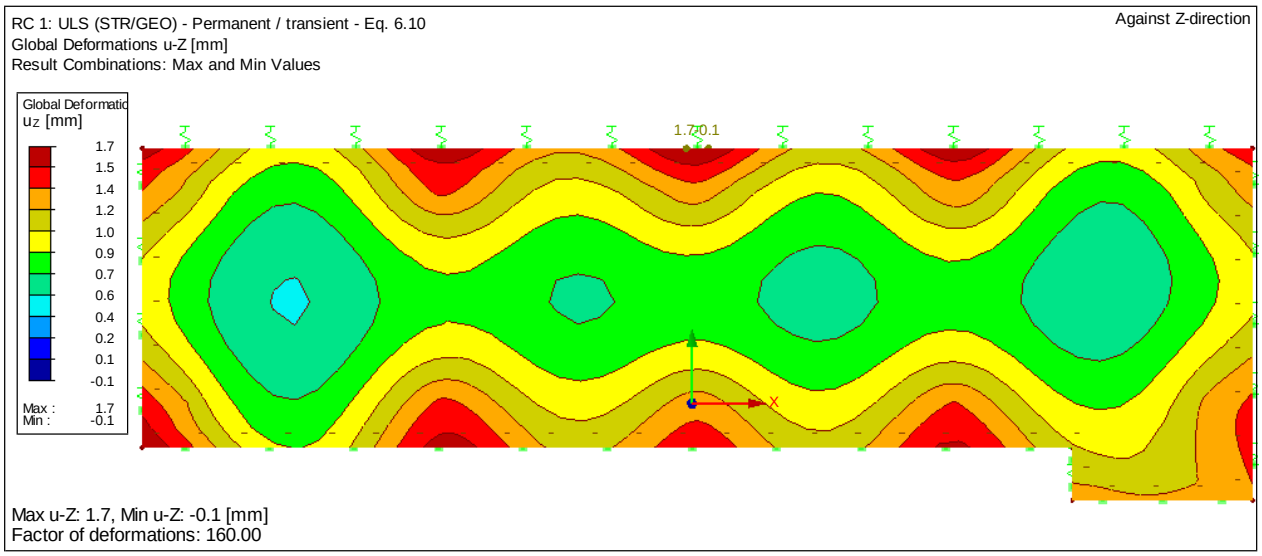


Figure 31 settlement uz from bottom view

C01

u< u_{max}=25mm (according to MD-101-SMEM-EG-ST-CRT-0001), ok

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7.3. Check of soil pressure

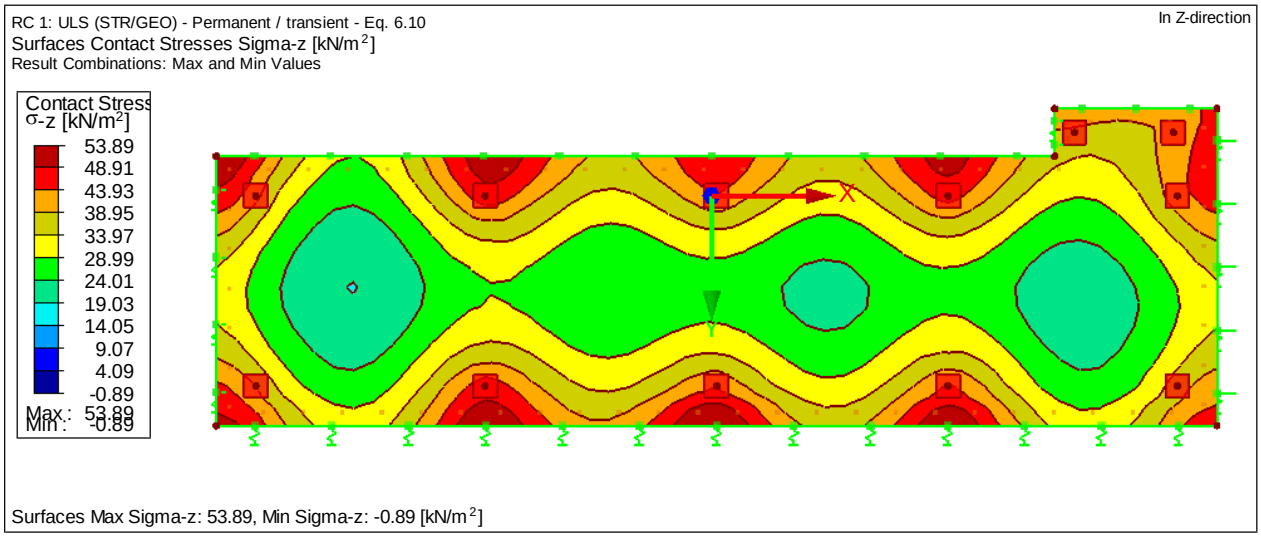


Figure 32 Soil contact pressure at ULS

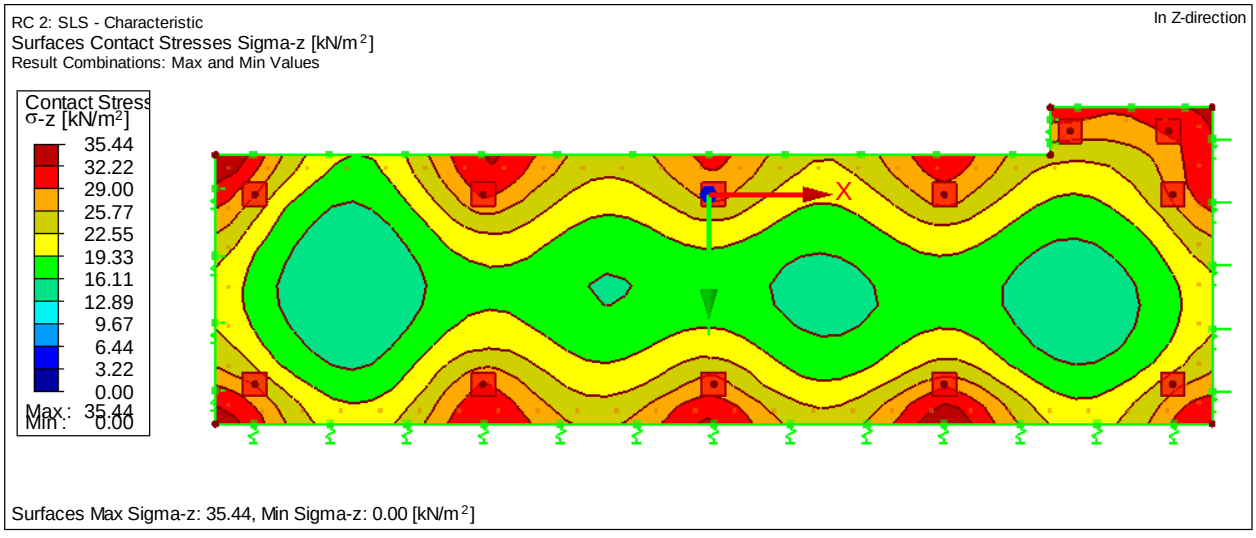


Figure 33 Soil contact pressure at SLS

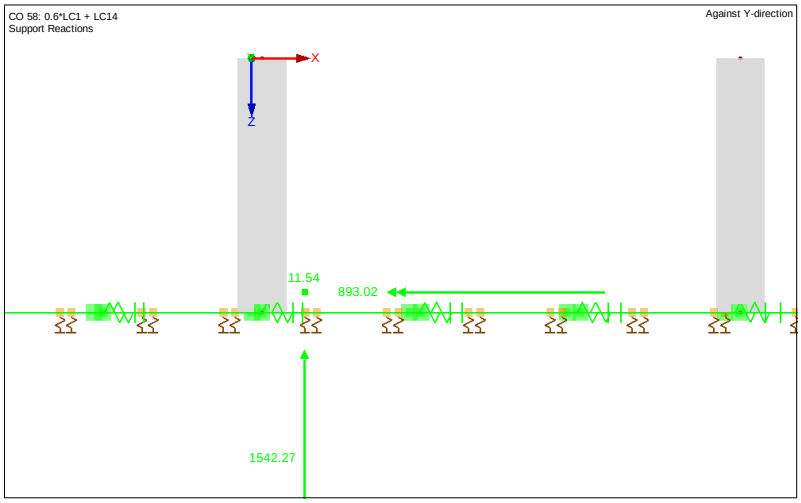
$\sigma_s < \sigma_{s,max}$ = 375kPa, ok

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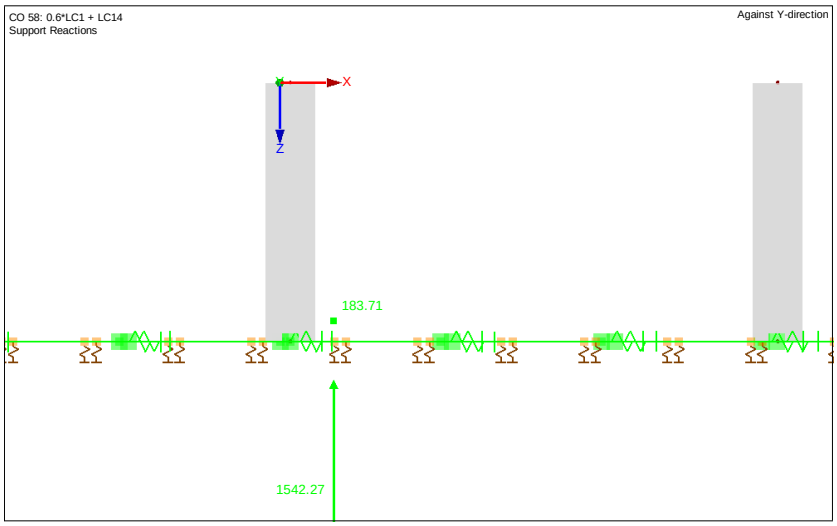
7.4. Check of overturning and sliding



for critical load combination CO58: 0.60D+ 1.00W,

eccentricity e= Mx/Ned= 893.02kNm/1542.27kN = 0.58m < B/2 = 6.8m/2 = 3.4m,

F.O.S = (B/2) / e = 5.86 > 1.5, overturn OK!



resultant horizontal force on foundation

Hd = $[(Fx)^2 + (Fy)^2]^{1/2}$ = 183.71kN

resistance to sliding Rd = μ (Ned) = 0.40* 1542.27kN = 616.91kN

F.O.S = Rd/ Hd= 3.36 > 1.50, sliding ok!

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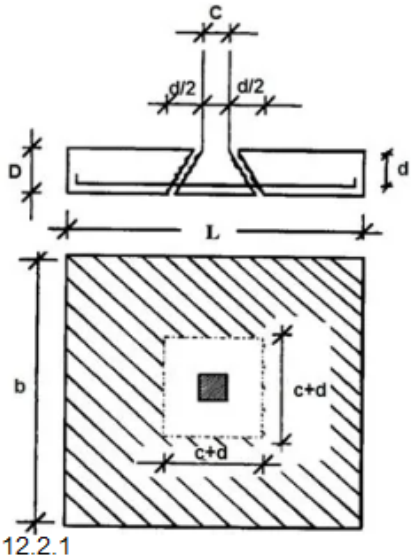
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7.5. Check of punching shear

PUNCHING SHEAR CHECK AT DISTANCE d/2 FROM COLUMN FACE

material property in [MPa]

$f_{ck} := 35 \quad f_{yk} := 420$



effective depth in [m]

$d := 0.400 - 0.075 - \frac{0.016}{2} = 0.317$

column dimension in [m]

$c_x := 0.6 \quad c_y := 0.6$

perimeter for critical section in [m]

$b_0 := 2 \cdot 2\left(c_x + 2 \cdot \frac{d}{2}\right) = 3.668$

ratio of long side to short side of the column

$\beta_c := 1$

for corner columns

$\alpha_s := 20$

concrete Shear strength in [kN], SBC 304 Ch. 11.12.2.1

$$V_c := \min\left[\left(1 + \frac{2}{\beta_c}\right) \cdot \frac{\sqrt{f_{ck}} \cdot b_0 \cdot d}{6} \cdot 10^3, \left(\frac{\alpha_s \cdot d}{b_0} + 2\right) \cdot \frac{\sqrt{f_{ck}} \cdot b_0 \cdot d}{12} \cdot 10^3, \frac{\sqrt{f_{ck}} \cdot b_0 \cdot d}{3} \cdot 10^3\right] = 2.137 \times 10^3$$

reduction factor for shear

$\phi := 0.75$

Design value of concrete shear strength in [kN]

$\phi \cdot V_c = 1.603 \times 10^3$

max applied shear force for column M6 in [kN]

$V_{Ed} := 403.46$

factored shear force at section in [kN]

$V_u := 1.35 \cdot V_{Ed} = 544.671$

punching shear check

$\frac{V_u}{\phi \cdot V_c} = 0.34$ ok!

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7.6. Raft design

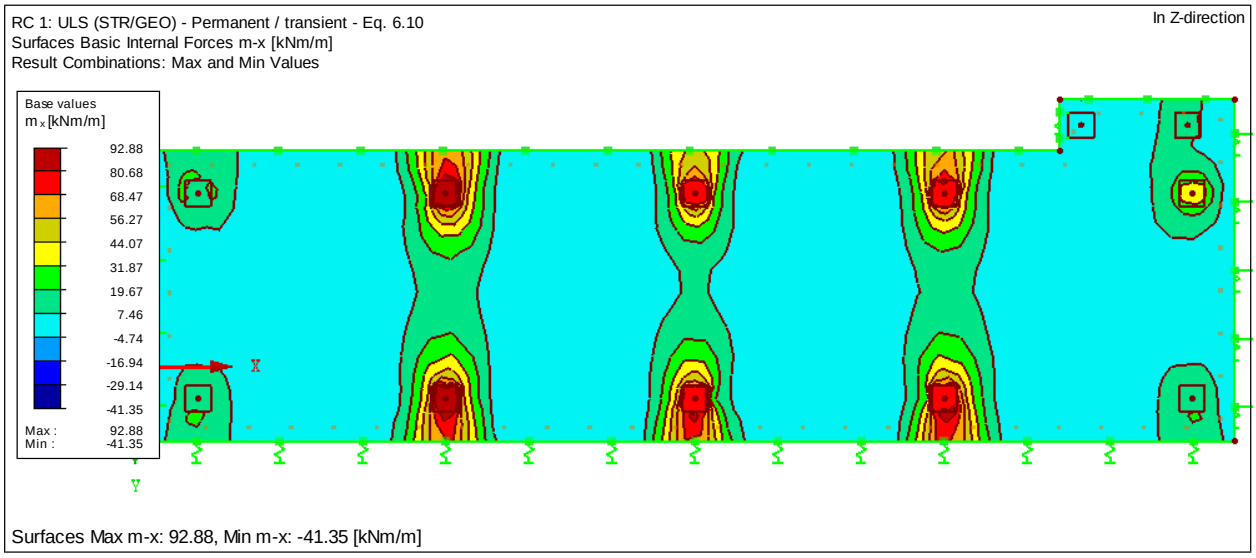


Figure 34 Bending moment Mx (with average regions)



Figure 35 Bending moment My (with average regions)

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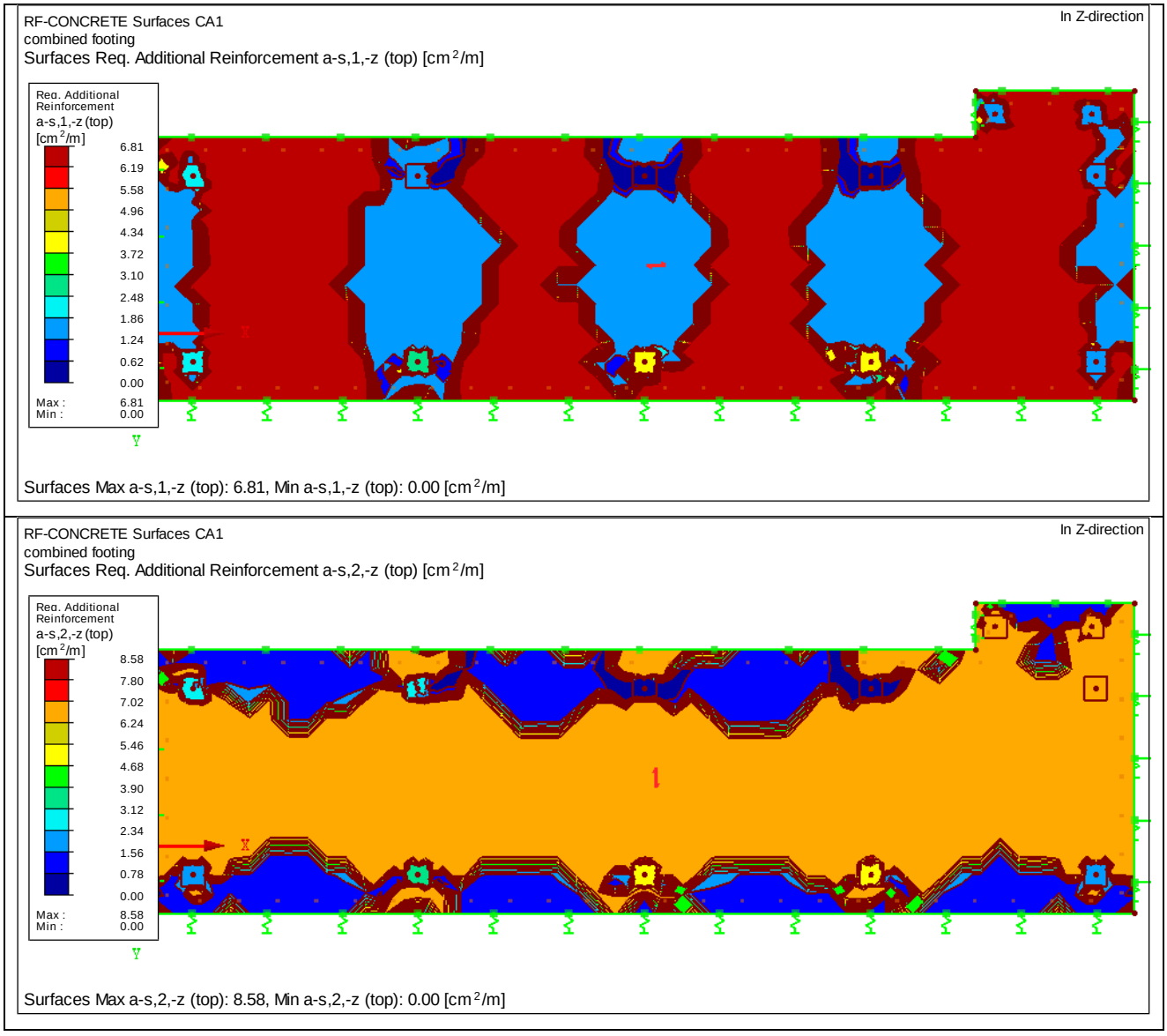


Figure 36 Required reinforcement at top

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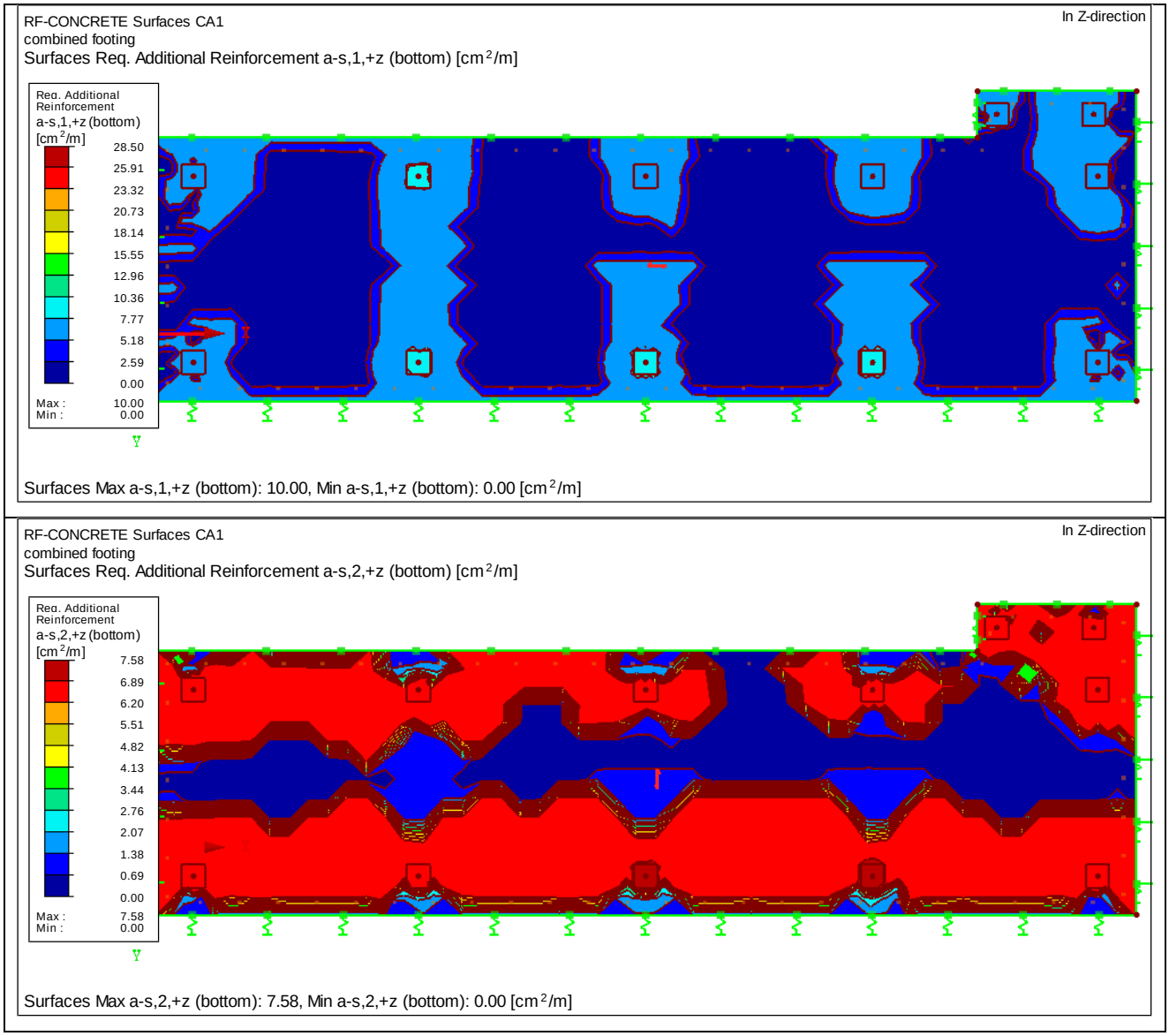


Figure 37 Required reinforcement at bottom

Provide T16@150mm, $A_{s,p}$ = 13,40cm^2/m

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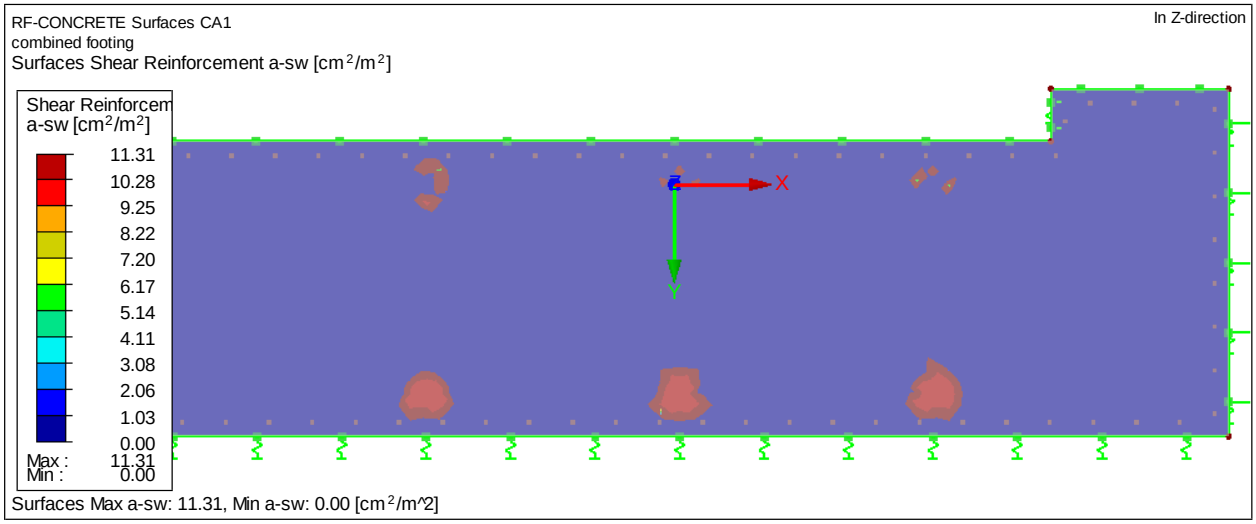


Figure 38 Required shear reinforcement a_{sw}

Starter bars 12T16, $a_{s,p}$= 24,12cm^2 > a_{sw}, ok

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8. TRANSFORMER ROOM

8.1. Structure geometry

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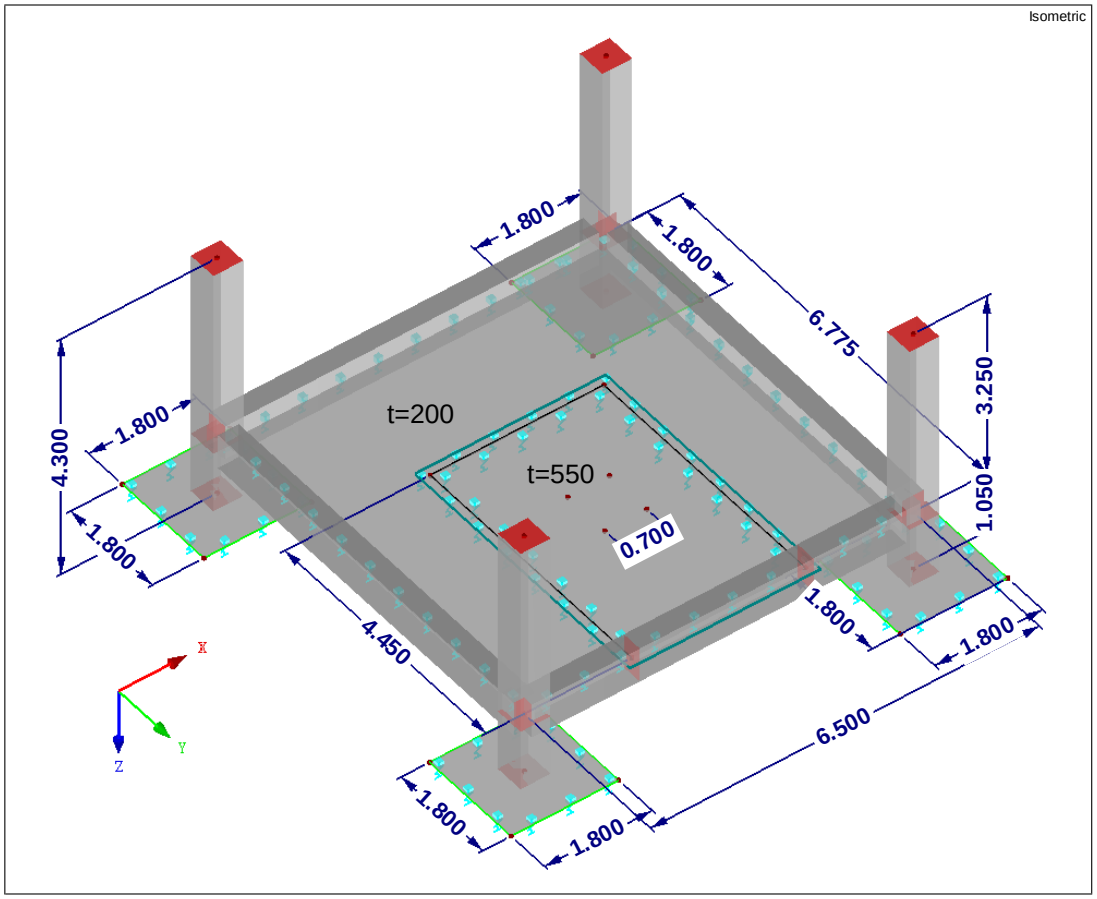


Figure 39 Structure geometry

Column 500mmx500mm, footing 1.8mx1.8mx350mm, plinth beam 350x600mm, ground slab under equipment t = 550mm, other areas t = 200mm.

8.2. Load cases

Load cases include self-weight of the structure, dead load and wind load.

Dead load includes brick wall weight 20 kN/m^3 * 3.5m * 350 mm= 24.5 kN/m on concrete plinth beam, cladding and steel beams total weight 250kN/4 = 63kN on each concrete column. Equipment load 37.50kN/4 = 9.375kN on each roller.

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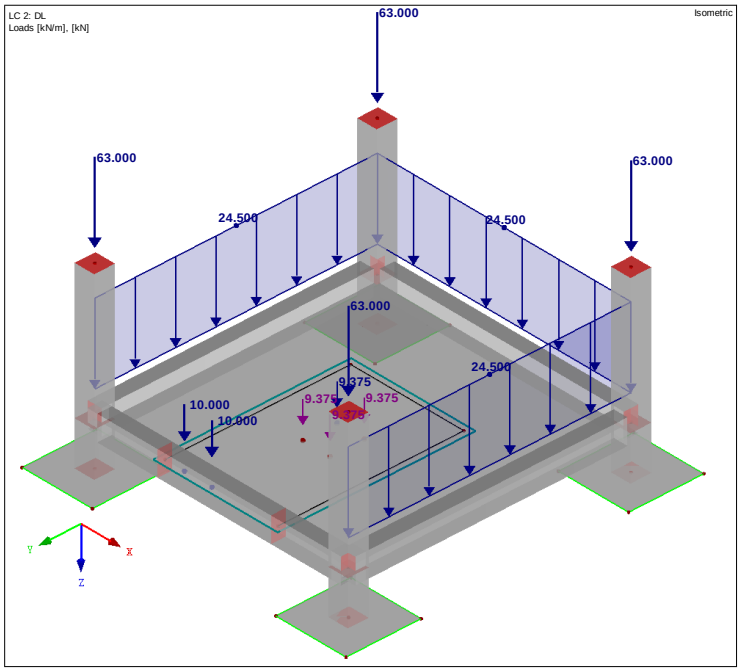


Figure 40 Dead load

Wind load is calculated according to SBC-301, Section 7. Basic wind speed 43m/s and Site class C are given in project design criteria.

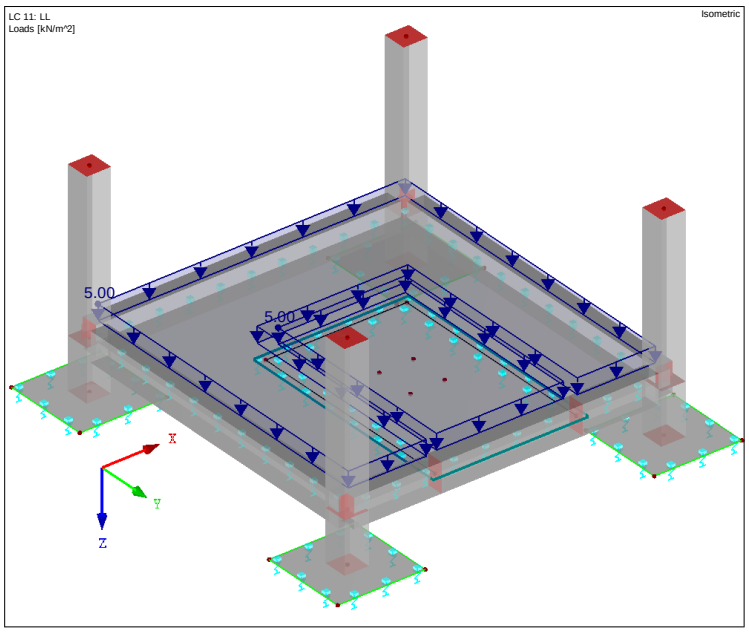


Figure 41 Live load 5kPa

C01

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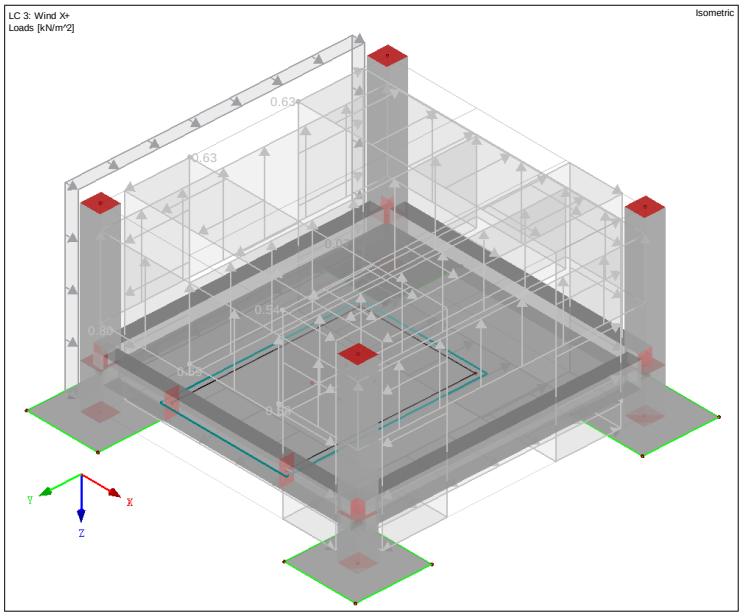


Figure 42 Wind in x+, +Cpi

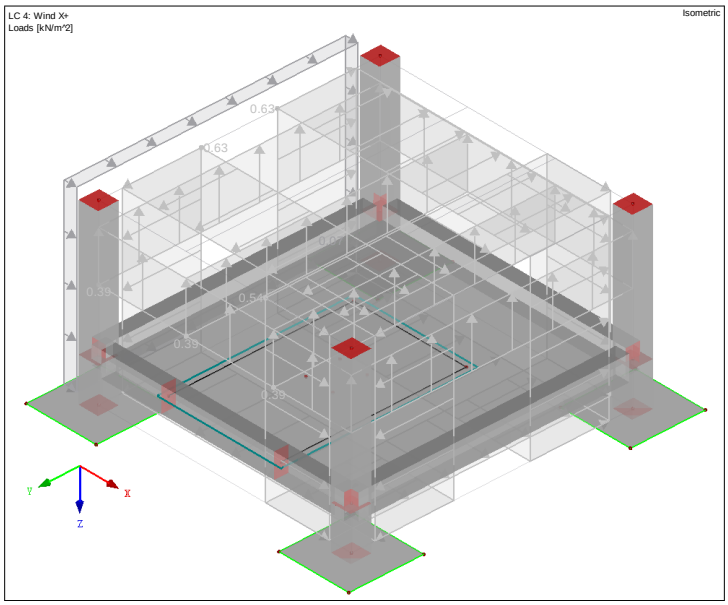


Figure 43 Wind in x-, +Cpi

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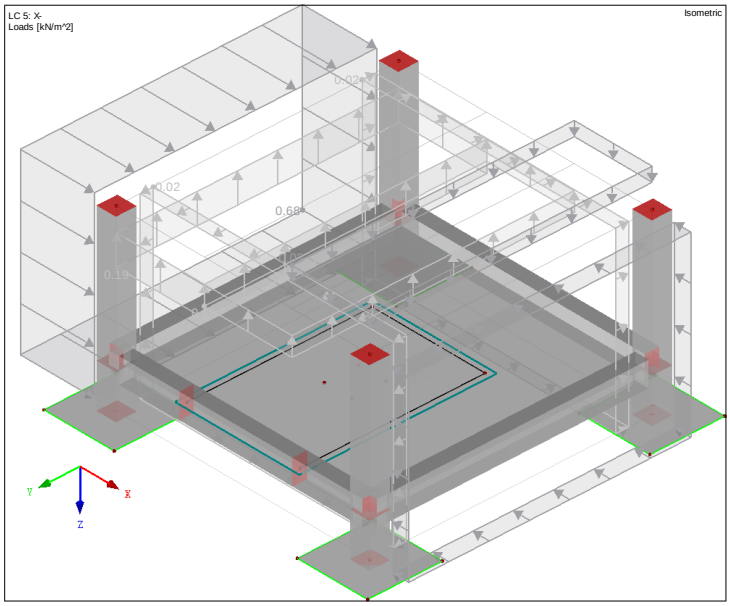


Figure 44 Wind in +, -Cpi

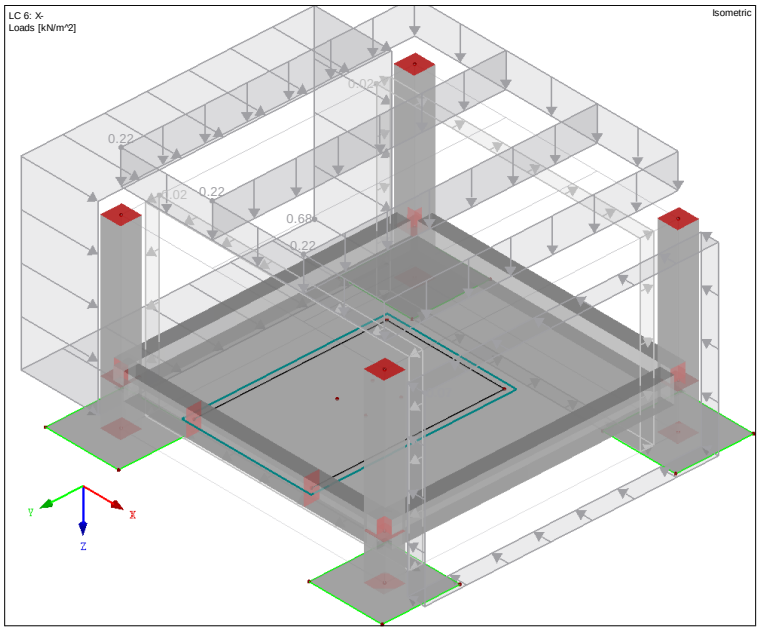


Figure 45 Wind in x-, -Cpi

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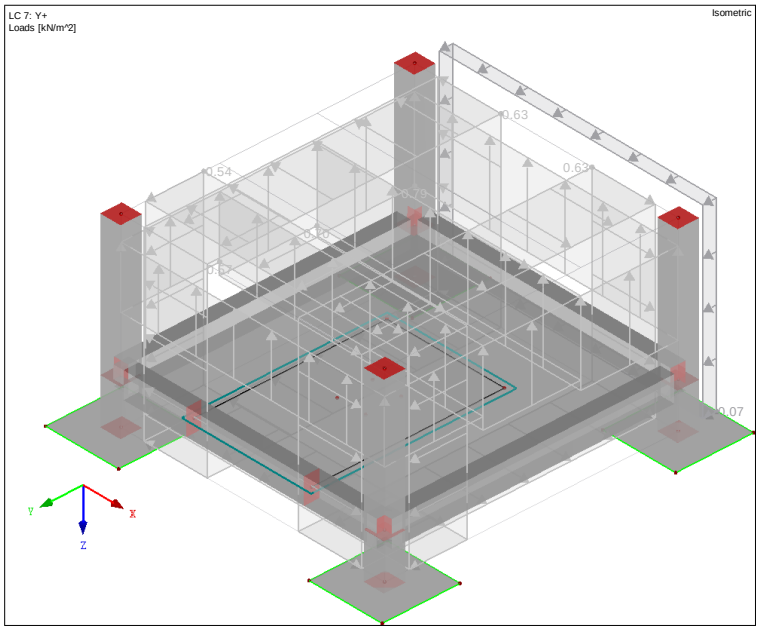


Figure 46 Wind in y+, +Cpi

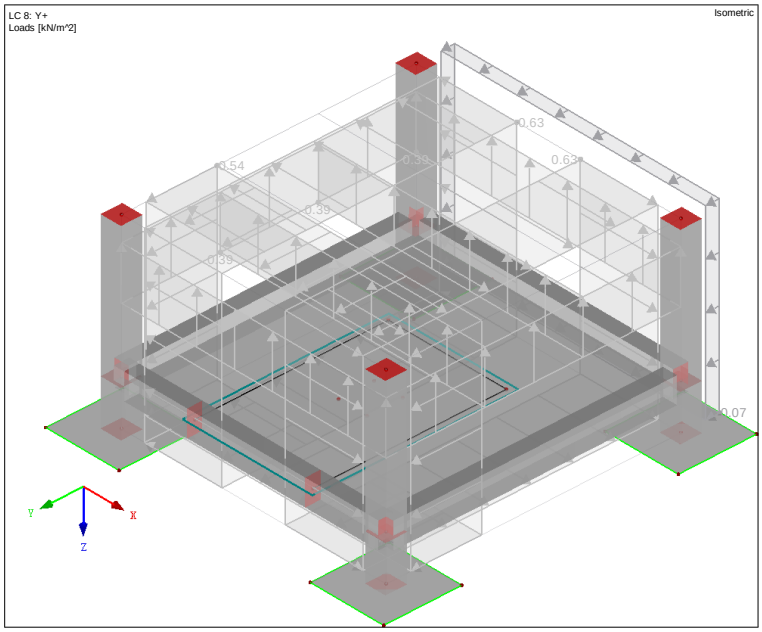


Figure 47 Wind in y-, +Cpi

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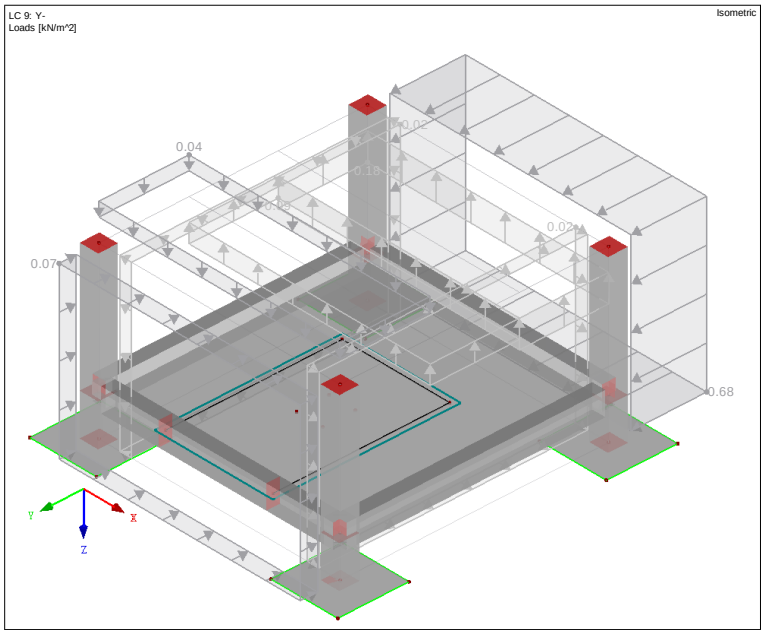


Figure 48 Wind in y+, -Cpi

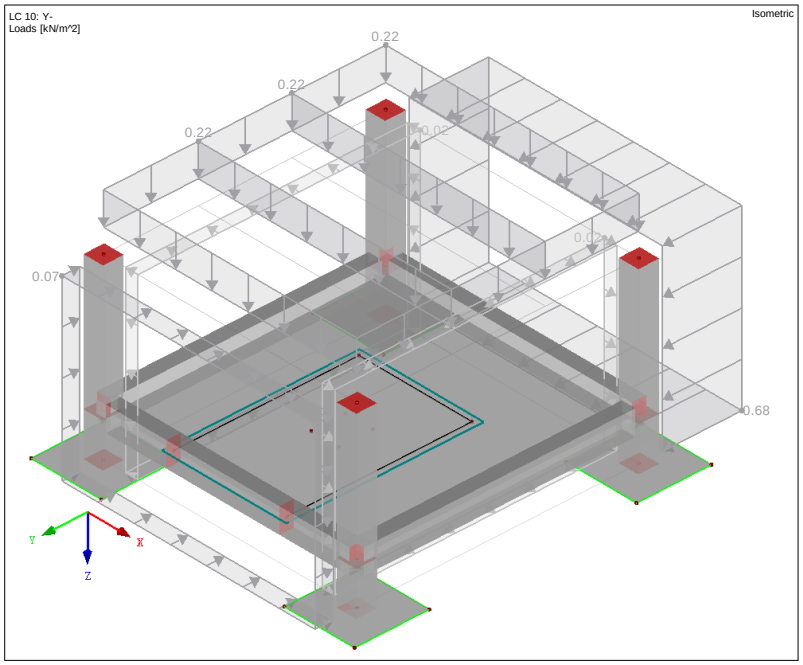


Figure 49 Wind in y-, -Cpi

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8.3. Analysis results

8.3.1. Global deformation

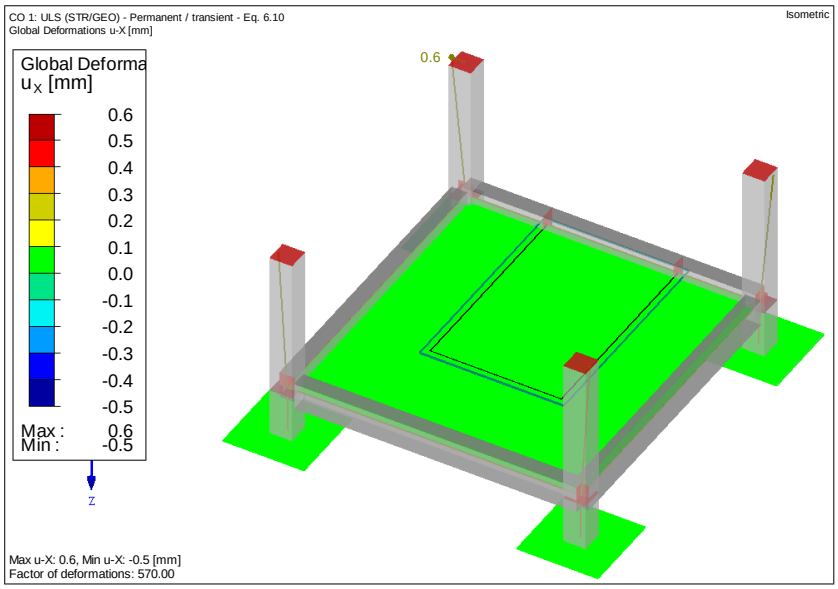
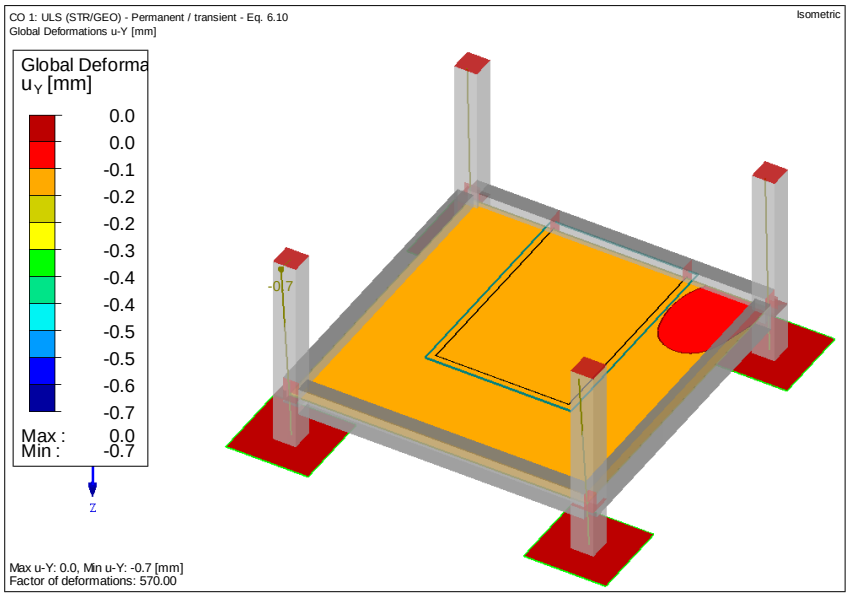


Figure 50 Global deformation u_x



C01

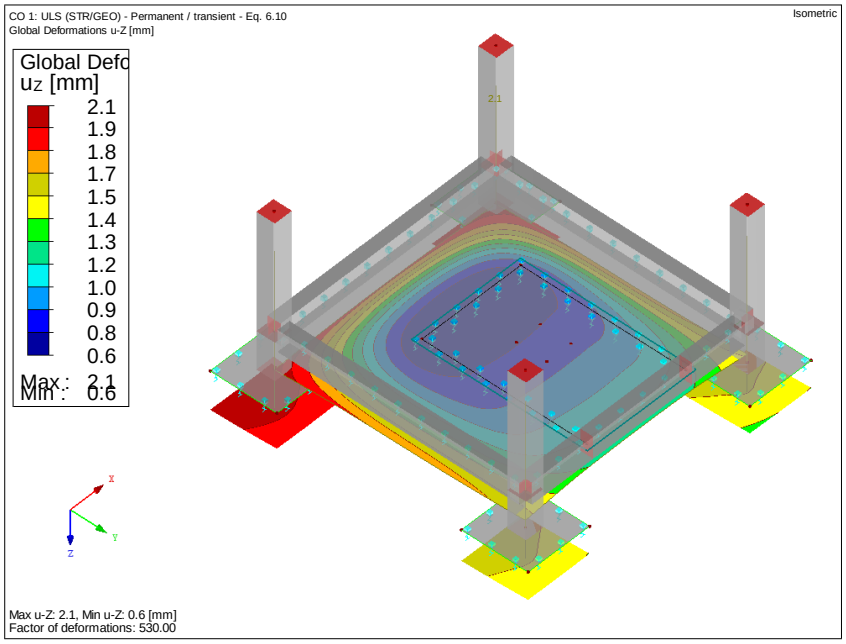
Figure 51 Global deformation u_x

u< u_{max}= 6775mm/360=18.8mm, ok

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C01

Figure 52 Global deformation u_z

u< u_{max}= 25mm (according to MD-101-SMEM-EG-ST-CRT-0001), ok

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8.3.2. Column design

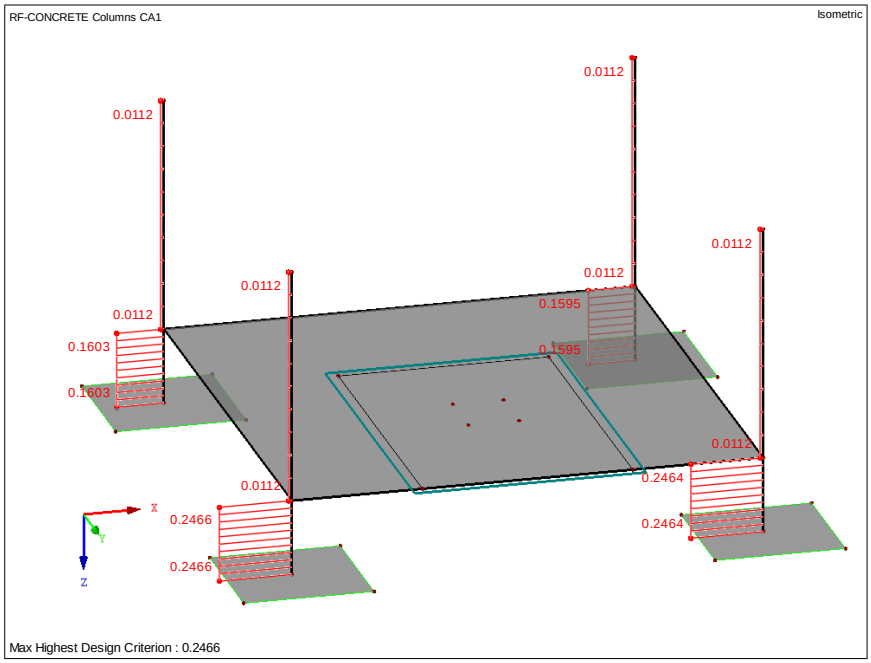


Figure 53 Utility ratio for columns

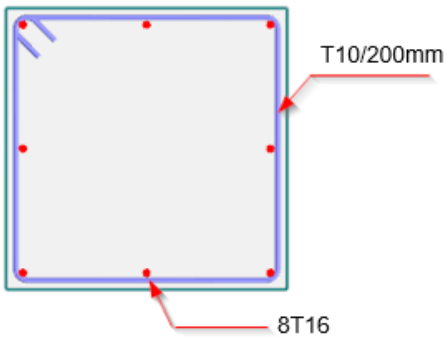


Figure 54 Sketch of reinforcement in column

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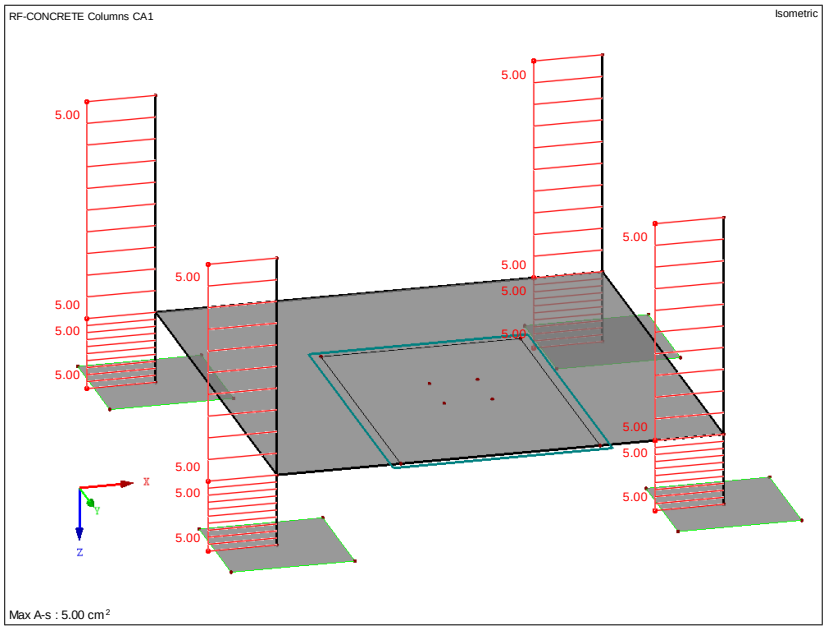


Figure 55 Required longitudinal reinforcement

Provide 8T16, As = 16.08cm^2

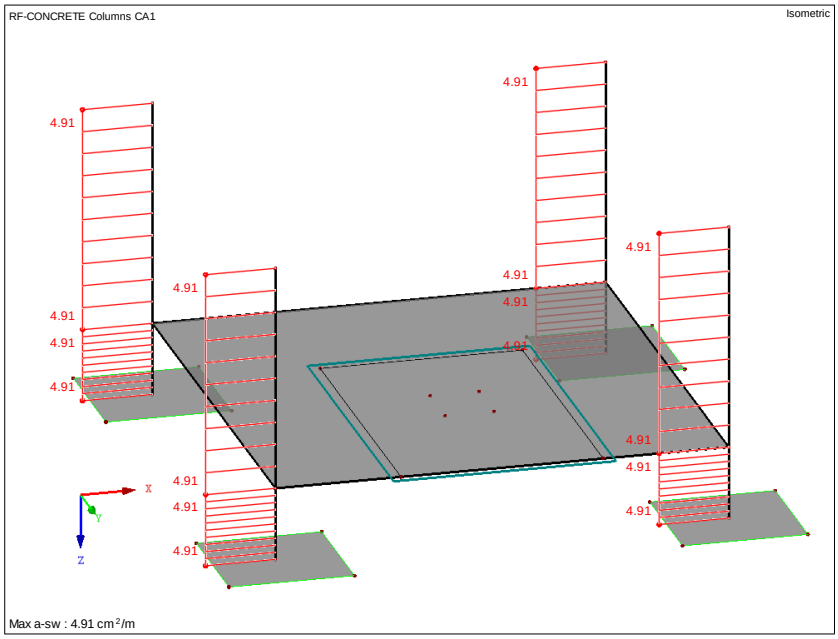


Figure 56 Required shear reinforcement

Provide T10/200mm, As = 7.85cm^2/m

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8.3.3. Plinth beam design

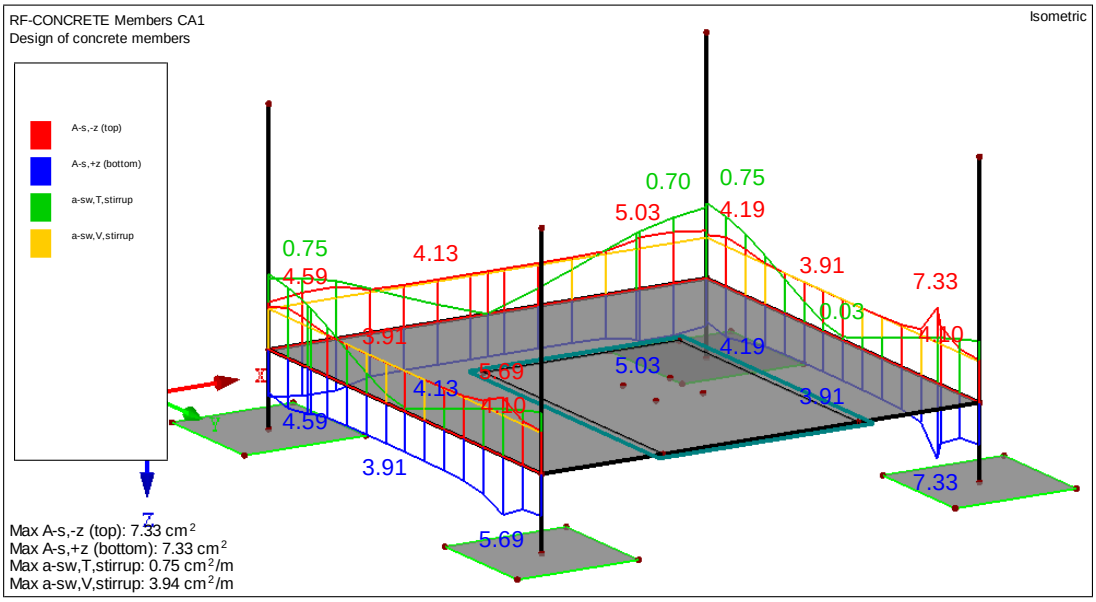


Figure 57 Required reinforcement

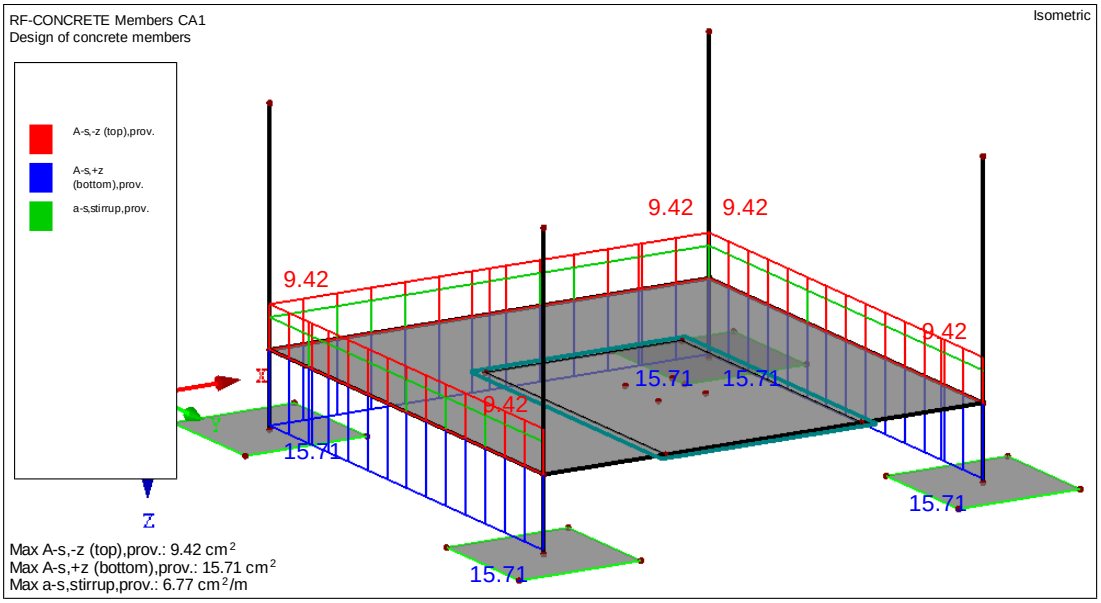


Figure 58 Provide reinforcement

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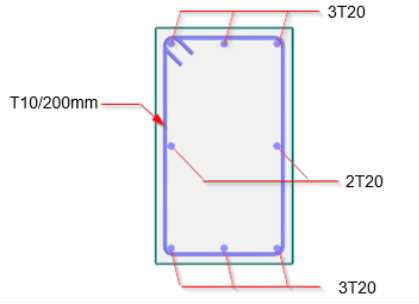


Figure 59 Sketch of reinforcement in beam

8.3.4. Ground slab design

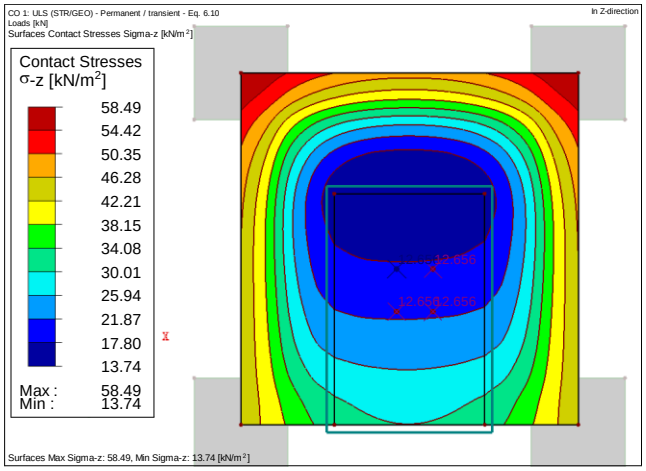


Figure 60 Soil contact stress at ULS

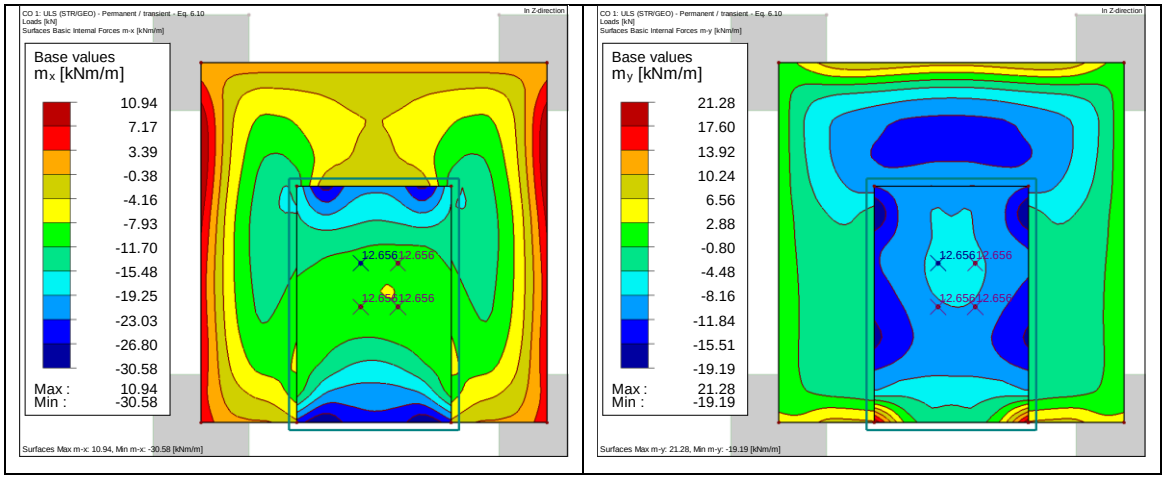


Figure 61 Bending moment

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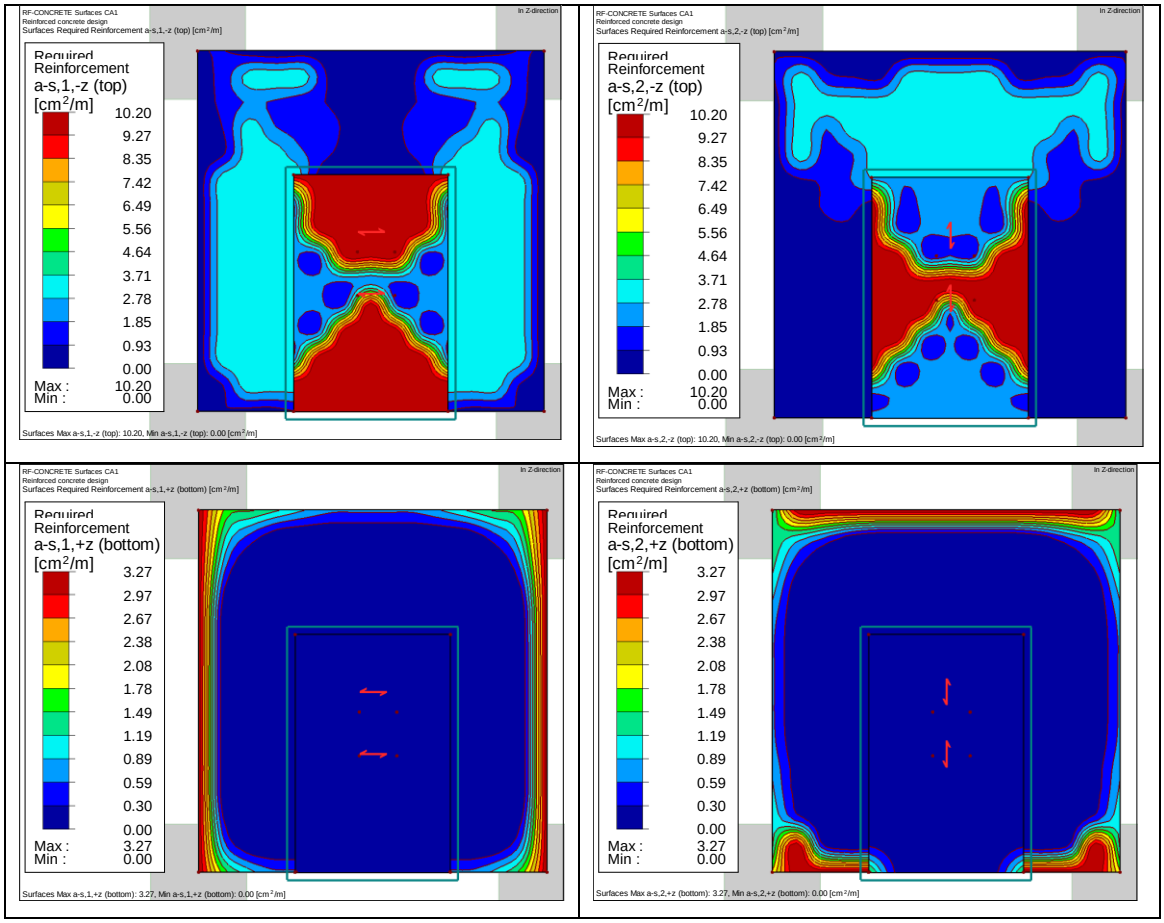


Figure 62 Required reinforcement

for top reinforcement:

Provide T12/200mm

for bottom reinforcement:

Provide T16/200mm for t=550mm slab, As = 10.05cm^2/m

Provide T12/200mm for t=200mm slab, As = 6.65cm^2/m

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8.3.5. Pad footing design

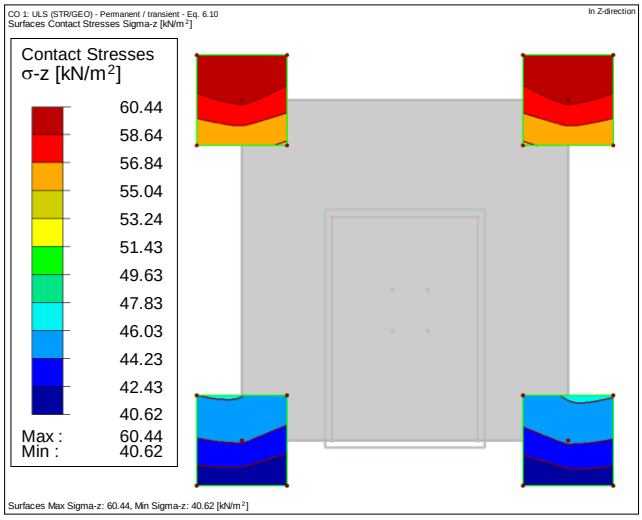


Figure 63 Soil contact stress at ULS

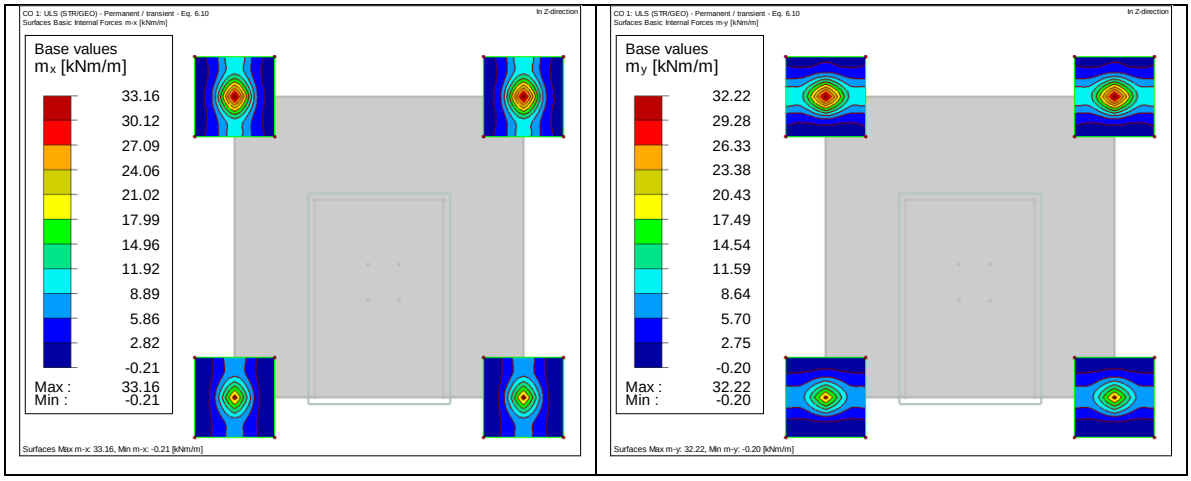


Figure 64 Bending moment

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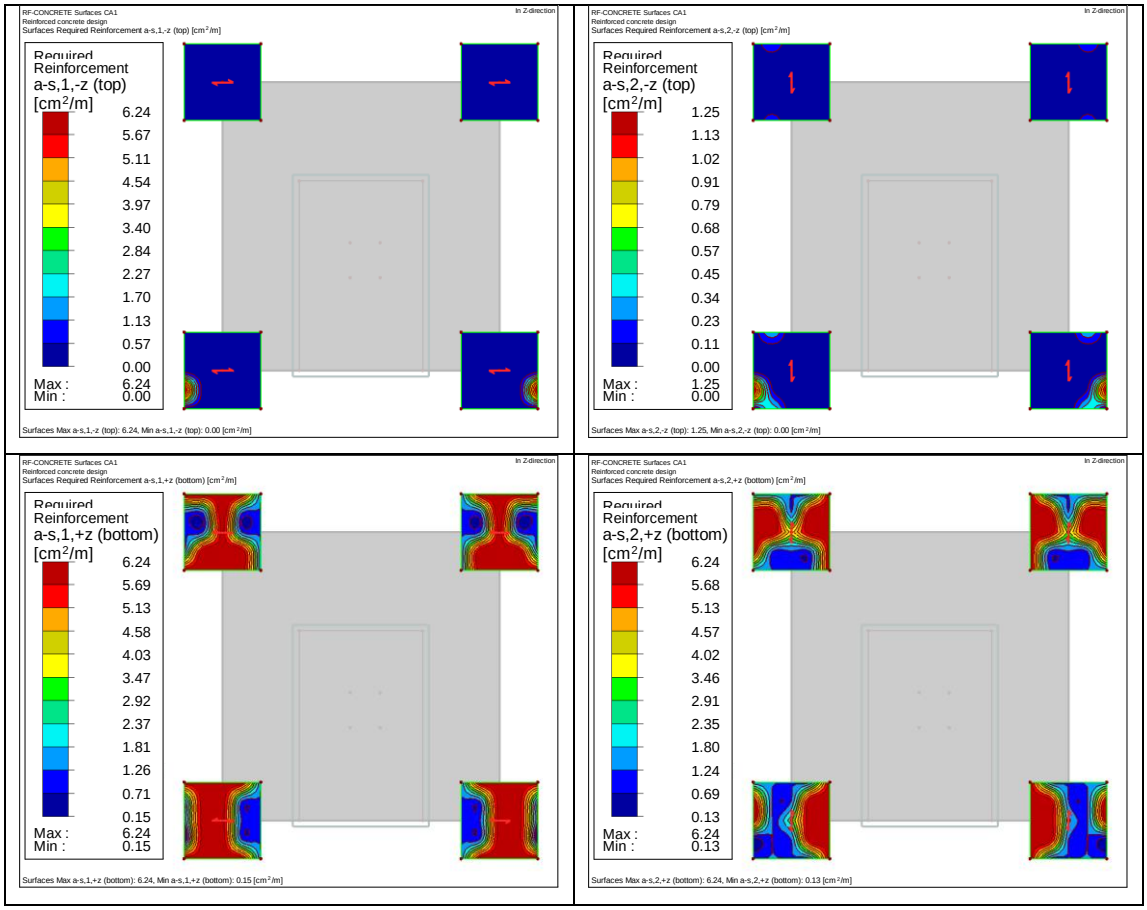


Figure 65 Required reinforcement for pad footing

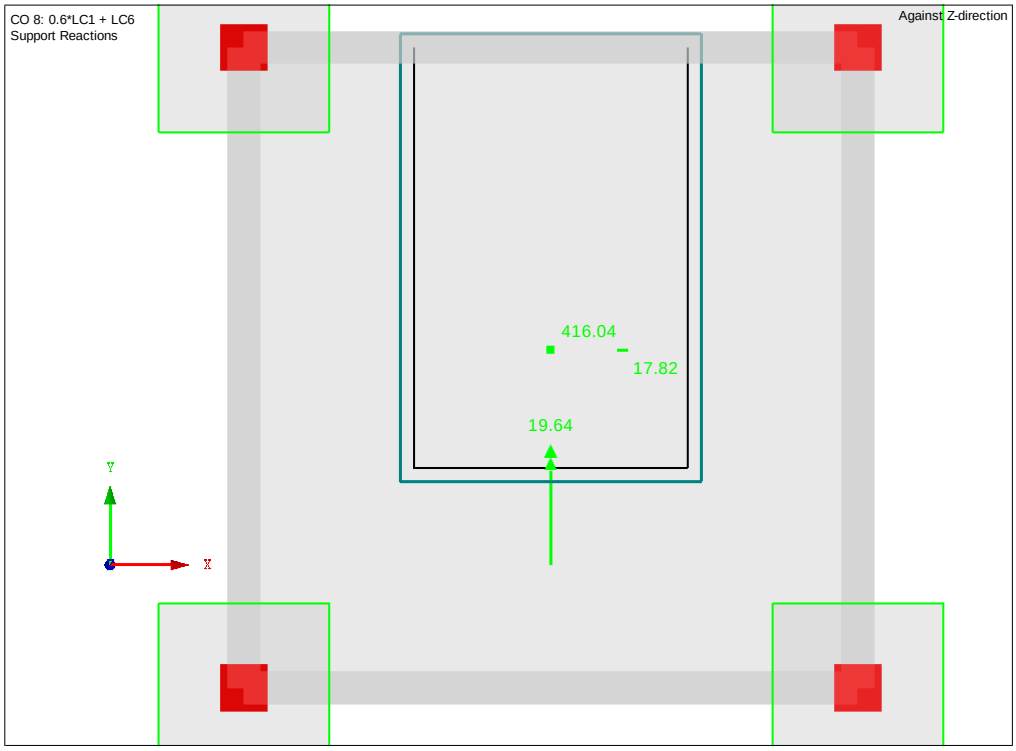
Provide T12/150mm, As = 7.54mm^2/m

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8.3.6. Stability check



for load combination 0.60D+ 1.00W_x,

eccentricity e= Mx/Ned= 19.64kNm/416.04kN = 0.05m < B/2 = 6.5m/2 = 3.25m,

F.O.S = (B/2) / e = 68.8 > 1.50, overturn OK!

resultant horizontal force on foundation

Hd = [$(Fx)^2$ + $(Fy)^2$]$^{1/2}$ = 17.82kN

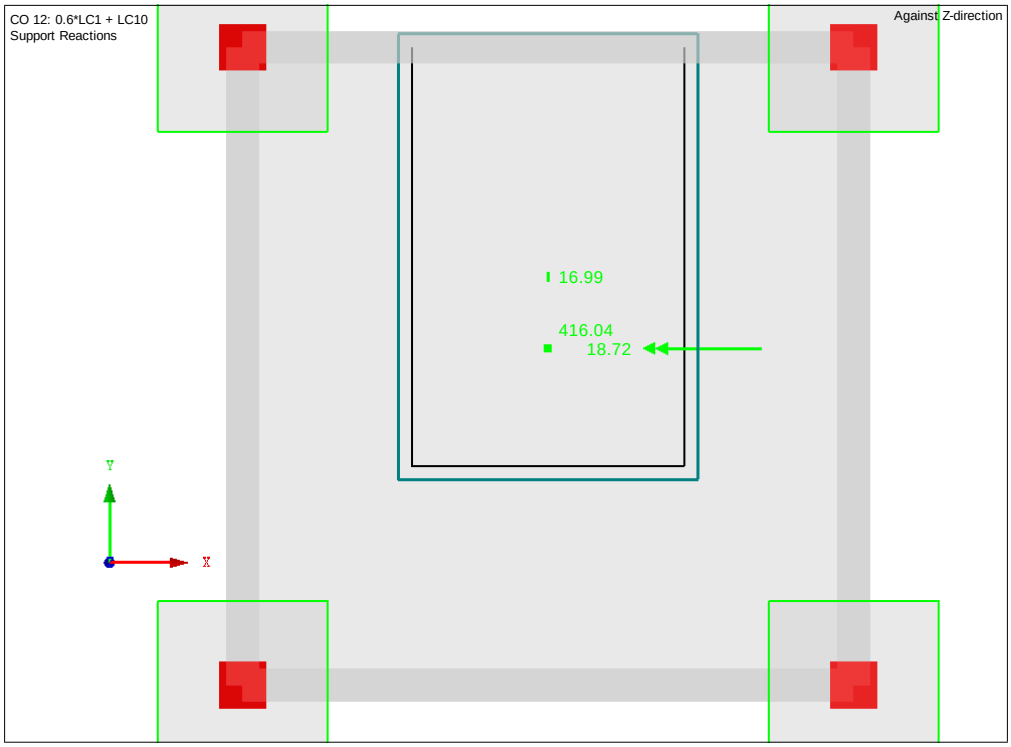
resistance to sliding Rd = μ (Ned) = 0.40* 416.04kN = 166.42kN

F.O.S = Rd/ Hd= 9.34 > 1.50, sliding ok!

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for critical load combination 0.60D+ 1.00W_y,

eccentricity e= Mx/Ned= 18.72kNm/416.04kN = 0.05m < B/2 = 3.38m,

F.O.S = (B/2) / e = 75 > 1.50, overturn OK!

resultant horizontal force on foundation

Hd = $[(Fx)^2 + (Fy)^2]^{1/2}$ = 16.99kN

resistance to sliding Rd = μ (Ned) = 0.40* 416.04kN = 166.42kN

F.O.S = Rd/ Hd= 9.8 > 1.50, sliding ok!

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9. REFERENCE

[1] Saudi Building Code (SBC) 301, Structural - Loading and Forces.

[2] Saudi Building Code (SBC) 304, Structural – Concrete Structures.

[3] Project Design Criteria, MD-216-1000-EG-ST-CRT-1001-A03.

[4] Overall geotechnical report, MD-216-3000-EG-CV-RPT-1008-C01.

[5] General arrangement drawing, raw water treatment plan, 1-8/E-J, MD-216-49W1-EG-CV-GAD-1001

[6] General arrangement drawing, raw water treatment plan, 7-12/E-J, MD-216-49W1-EG-CV-GAD-1002

[7] General arrangement drawing, raw water treatment plan, 1-10/A-E, MD-216-49W1-EG-CV-GAD-1003

[8] Equipment layout of prefab building, MD 216 49W1 EG EL GAD V079-1004

[9] General arrangement drawing, 1200KVA, 13.8/0.417 kV Transformer, MD-216-49W1-EG-EL-GAD-V079-1001-A01

[10] Structural Design Criteria - Process Buildings and Structures, MD-101-SMEM-EG-ST-CRT-0001

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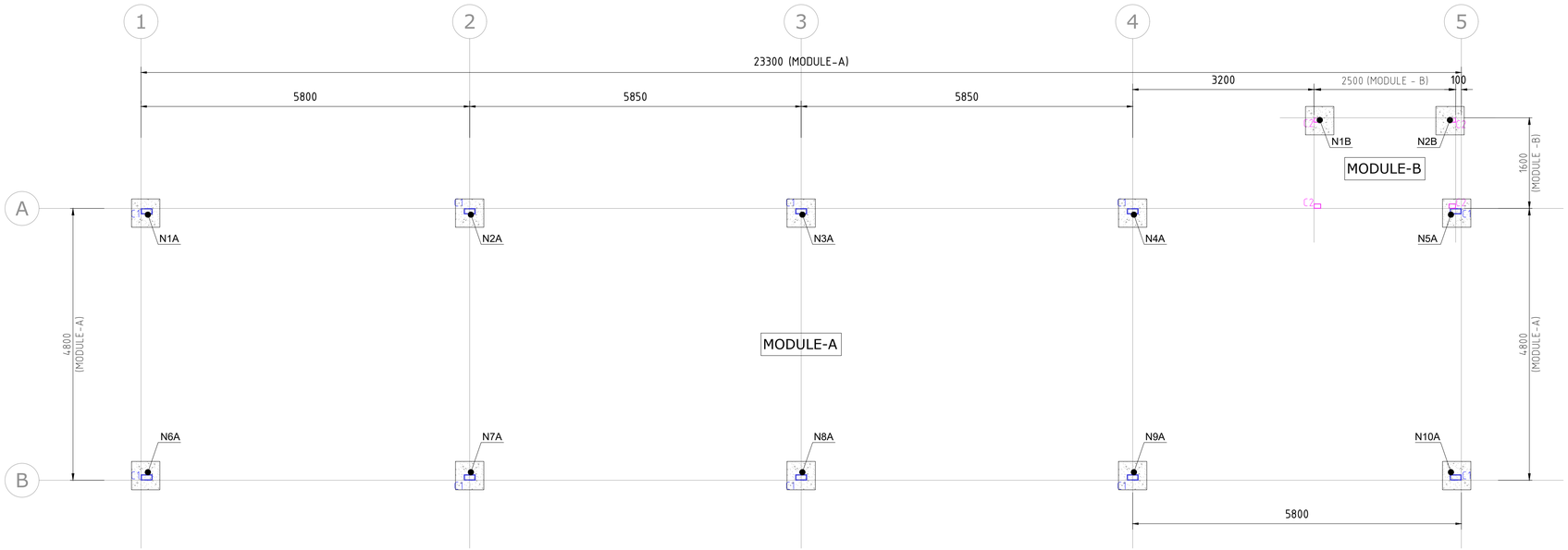
ATTACHMENT 1 INPUT DRAWING

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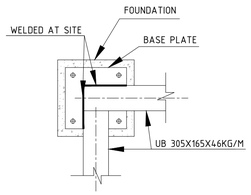
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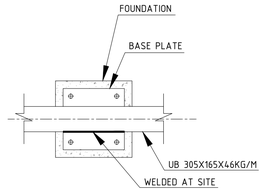


FOUNDATION LAYOUT

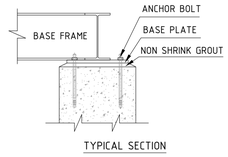
COLUMN DETAILS	
C1	RHS 250X100X12mm
C2	RHS 120X80X4mm



CORNER BASE PLATE



INTERMEDIATE BASE PLATE



TYPICAL SECTION

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	04/11/2021	ISSUED FOR APPROVAL	--	MehAky	--	JusRii	
		A01	29/09/2021	ISSUED FOR APPROVAL	--	MehAky	--	JusRii	

SIEMENS Ingenuity for life

DRAWING NUMBER					REVISION
SASL-RS-DWG-ST-21K025-000007					01
DATE	04/11/2021	DATE	04/11/2021	DATE	04/11/2021
DRAWN	S.S.	CHECKED	F.S.	APPROVED	S.J.T.

LARSEN & TOUBRO DRAWING NUMBER	REVISION
	1
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DRAWN	--	--
CHECKED	MehAhy	27/09/2021
ENGINEER	--	--
APPROVED	JusRii	29/09/2021
EMPLOYER		
SCALE	NTS	A1

MANSOURAH - MASSARAH GOLD PROJECT
LAYOUT
RAW WATER TREATMENT PLANT
EQUIPMENT LAYOUT OF PREFAB BUILDING

DRAWING NUMBER: MD-216-49W1-EG-EL-GAD-V079-1004 | REV: A02

DRAWING NUMBER: MD-216-49W1-EG-EL-GAD-V079-1004 REV A02

	DEAD LOAD				LIVE LOAD					EQUIPMENT LOAD			
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N2A	-4.184	59.995	1.382		N2A	-3.001	108.523	2.618		N2A	-1.223	68.517	1.405
N3A	0.005	51.058	1.328		N3A	0.119	104.504	2.486		N3A	0.02	47.231	0.919
N4A	0.456	59.66	1.449		N4A	0.785	126.414	3.013		N4A	-0.193	30.239	0.679
N5A	-2.255	38.998	0.462		N5A	-4.323	88.655	0.486		N5A	-0.386	16.658	0.123
N6A	1.973	25.303	-0.53		N6A	3.007	50.809	-1.228		N6A	1.305	16.262	-0.292
N7A	-0.458	56.181	-1.404		N7A	-0.857	109.834	-2.581		N7A	0.088	65.769	-1.321
N8A	4.736	54.16	-1.336		N8A	3.496	95.855	-2.449		N8A	0.029	56.26	-0.972
N9A	-4.394	59.415	-1.444		N9A	-2.547	121.763	-2.977		N9A	-0.861	30.985	-0.617
N10A	-1.955	25.191	-0.526		N10A	-3.282	46.299	-0.722		N10A	-0.435	10.13	-0.141
N1B	0.18	6.746	0.15		N1B	0.622	14.199	0.55		N1B	0.058	0.32	0.059
N2B	0.092	1.46	-0.102		N2B	0.422	-6.118	-0.414		N2B	0.054	-4.507	-0.057

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	04/11/2021	ISSUED FOR APPROVAL	--	MehAky	--	JusRii	
		A01	29/09/2021	ISSUED FOR APPROVAL	--	MehAky	--	JusRii	

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DRAWING NUMBER: SASL-RS-DWG-ST-21K025-000008 REVISION: 01

DATE	04/11/2021	DATE	04/11/2021	DATE	04/11/2021
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OUTOTEC DRAWING NUMBER: 900707 _GAU01 _AG11 _MTL01 _49010 DOCUMENT ID: OU100728448 REVISION 1

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ENGINEER	--	--
APPROVED	JusRii	29/09/2021
EMPLOYER		
SCALE	NTS	A1

MANSOURAH - MASSARAH GOLD PROJECT
LAYOUT
RAW WATER TREATMENT PLANT
EQUIPMENT LAYOUT OF PREFAB BUILDING

DRAWING NUMBER: MD-216-49W1-EG-EL-GAD-V079-1004 REV: A02

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	SEISMIC+X					SEISMIC-X					SEISMIC+Z					SEISMIC-Z		
Node	Fx kN	Fy kN	Fz kN		Node	Fx kN	Fy kN	Fz kN		Node	Fx kN	Fy kN	Fz kN		Node	Fx kN	Fy kN	Fz kN
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N2A	-8.039	13.054	0.022		N2A	8.039	-13.054	-0.022		N2A	0.725	-4.017	-2.258		N2A	-0.725	4.017	2.258
N3A	-0.151	0.106	-0.005		N3A	0.151	-0.106	0.005		N3A	-0.046	-4.355	-2.176		N3A	0.046	4.355	2.176
N4A	-0.293	-0.563	0.003		N4A	0.293	0.563	-0.003		N4A	0.081	-3.545	-1.997		N4A	-0.081	3.545	1.997
N5A	-0.607	0.587	0.137		N5A	0.607	-0.587	-0.137		N5A	-0.039	-0.089	-1.605		N5A	0.039	0.089	1.605
N6A	-0.042	-0.127	0.097		N6A	0.042	0.127	-0.097		N6A	0.177	2.749	-1.804		N6A	-0.177	-2.749	1.804
N7A	-0.205	0.03	0.041		N7A	0.205	-0.03	-0.041		N7A	-0.163	3.997	-2.417		N7A	0.163	-3.997	2.417
N8A	-8.311	-11.588	0.043		N8A	8.311	11.588	-0.043		N8A	0.672	4.287	-2.741		N8A	-0.672	-4.287	2.741
N9A	-9.011	11.542	-0.024		N9A	9.011	-11.542	0.024		N9A	-0.614	4.012	-2.72		N9A	0.614	-4.012	2.72
N10A	-0.023	0.143	-0.137		N10A	0.023	-0.143	0.137		N10A	-0.028	2.403	-1.577		N10A	0.028	-2.403	1.577
N1B	-0.403	-1.262	-0.079		N1B	0.403	1.262	0.079		N1B	0.062	-0.895	-0.338		N1B	-0.062	0.895	0.338
N2B	-0.43	1.261	0.052		N2B	0.43	-1.261	-0.052		N2B	0.048	-1.827	-0.285		N2B	-0.048	1.827	0.285

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	04/11/2021	ISSUED FOR APPROVAL	--	MehAky	--	JusRii	
		A01	29/09/2021	ISSUED FOR APPROVAL	--	MehAky	--	JusRii	

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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DRAWN	S.S.	04/11/2021	
CHECKED	F.S.	04/11/2021	
APPROVED	S.J.T.	04/11/2021	

DRAWING NUMBER: SASL-RS-DWG-ST-21K025-000009 REVISION: 01

LARSEN & TOUBRO DRAWING NUMBER: REVISION: 1

OUTOTEC DRAWING NUMBER: 900707 _GAU01 _AG11 _MTL01 _49010 DOCUMENT ID: OU100728448 REVISION: 1

DRAWN	--	--
CHECKED	MehAhy	27/09/2021
ENGINEER	--	--
APPROVED	JusRii	29/09/2021
EMPLOYER		
SCALE	NTS	A1

MANSOURAH - MASSARAH GOLD PROJECT
LAYOUT
RAW WATER TREATMENT PLANT
EQUIPMENT LAYOUT OF PREFAB BUILDING

DRAWING NUMBER: MD-216-49W1-EG-EL-GAD-V079-1004 REV: A02

	WINDWARD+X				WINDWARD-X				WINDWARD+Z				WINDWARD-Z		
Node	Fx kN	Fy kN	Fz kN		Node	Fx kN	Fy kN	Fz kN	Node	Fx kN	Fy kN	Fz kN	Node	Fx kN	Fy kN	Fz kN
N1A	-12.422	-6.222	-0.053		N1A	4.664	6.067	0.048	N1A	-1.658	-7.097	-17.841	N1A	3.752	11.548	5.687
N2A	-4.524	6.132	0.008		N2A	3.582	-5.967	-0.007	N2A	2.543	-14.772	-21.745	N2A	-2.907	16.993	9.87
N3A	-0.114	0.288	0.002		N3A	0.098	-0.281	-0.002	N3A	0.277	-16.451	-8.219	N3A	-0.061	17.812	9.39
N4A	-0.874	-1.537	0.107		N4A	0.506	1.59	-0.039	N4A	0.705	-4.705	-4.949	N4A	-0.11	16.749	8.52
N5A	-2.432	1.337	0.057		N5A	11.875	-1.407	-0.099	N5A	-0.593	13.411	-3.913	N5A	-0.128	6.817	6.201
N6A	-8.365	-0.095	0.031		N6A	0.018	0.047	-0.033	N6A	0.311	8.376	-5.536	N6A	-0.606	-11.534	17.646
N7A	-0.128	0.05	0.016		N7A	0.077	-0.007	-0.016	N7A	0.022	13.698	-18.105	N7A	0.554	-16.978	30.323
N8A	-3.614	-5.421	0.029		N8A	4.508	5.501	-0.02	N8A	0.646	14.336	-38.197	N8A	-2.77	-17.978	39.786
N9A	-4.653	5.398	0.055		N9A	3.754	-5.433	-0.021	N9A	-2.945	10.993	-25.318	N9A	2.571	-16.51	39.417
N10A	-0.118	0.069	-0.193		N10A	8.385	-0.108	0.099	N10A	0.16	3.153	-2.117	N10A	-0.048	-10.25	16.722
N1B	-3.781	-3.964	-0.448		N1B	1.52	3.956	0.358	N1B	0.201	-9.137	-7.163	N1B	-0.182	-0.018	0.022
N2B	-1.649	3.966	0.389		N2B	3.684	-3.958	-0.268	N2B	0.33	-11.805	-6.395	N2B	-0.067	3.349	0.128

DRAWING NUMBER: MD-216-49W1-EG-EL-GAD-V079-1004 REV: A02

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	04/11/2021	ISSUED FOR APPROVAL	--	MehAky	--	JusRii	
		A01	29/09/2021	ISSUED FOR APPROVAL	--	MehAky	--	JusRii	

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DRAWING NUMBER						REVISION
SASL-RS-DWG-ST-21K025-000010						01
DATE	04/11/2021	DATE	04/11/2021	DATE	04/11/2021	
DRAWN	S.S.	CHECKED	F.S.	APPROVED	S.J.T.	

LARSEN & TOUBRO DRAWING NUMBER: REVISION 1

OUTOTEC DRAWING NUMBER: 900707 _GAU01 _AG11 _MTL01 _49010 | DOCUMENT ID: OU100728448 | REVISION 1

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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	MehAhy	27/09/2021	LAYOUT
ENGINEER	--	--	RAW WATER TREATMENT PLANT
APPROVED	JusRii	29/09/2021	EQUIPMENT LAYOUT OF PREFAB BUILDING
EMPLOYER			
SCALE	NTS	A1	DRAWING NUMBER: MD-216-49W1-EG-EL-GAD-V079-1004 REV: A02

	LEEWARD+X				LEEWARD-X				LEEWARD+Z				LEEWARD-Z			
Node	Fx kN	Fy kN	Fz kN		Node	Fx kN	Fy kN	Fz kN	Node	Fx kN	Fy kN	Fz kN	Node	Fx kN	Fy kN	Fz kN
N1A	-4.664	-6.067	-0.048		N1A	12.422	6.222	0.053	N1A	-3.752	-11.548	-5.687	N1A	1.658	7.097	17.841
N2A	-3.582	5.967	0.007		N2A	4.524	-6.132	-0.008	N2A	2.907	-16.993	-9.87	N2A	-2.543	14.772	21.745
N3A	-0.098	0.281	0.002		N3A	0.114	-0.288	-0.002	N3A	0.061	-17.812	-9.39	N3A	-0.277	16.451	8.219
N4A	-0.506	-1.59	0.039		N4A	0.874	1.537	-0.107	N4A	0.11	-16.749	-8.52	N4A	-0.705	4.705	4.949
N5A	-11.875	1.407	0.099		N5A	2.432	-1.337	-0.057	N5A	0.128	-6.817	-6.201	N5A	0.593	-13.411	3.913
N6A	-0.018	-0.047	0.033		N6A	8.365	0.095	-0.031	N6A	0.606	11.534	-17.646	N6A	-0.311	-8.376	5.536
N7A	-0.077	0.007	0.016		N7A	0.128	-0.05	-0.016	N7A	-0.554	16.978	-30.323	N7A	-0.022	-13.698	18.105
N8A	-4.508	-5.501	0.02		N8A	3.614	5.421	-0.029	N8A	2.77	17.978	-39.786	N8A	-0.646	-14.336	38.197
N9A	-3.754	5.433	0.021		N9A	4.653	-5.398	-0.055	N9A	-2.571	16.51	-39.417	N9A	2.945	-10.993	25.318
N10A	-8.385	0.108	-0.099		N10A	0.118	-0.069	0.193	N10A	0.048	10.25	-16.722	N10A	-0.16	-3.153	2.117
N1B	-1.52	-3.956	-0.358		N1B	3.781	3.964	0.448	N1B	0.182	0.018	-0.022	N1B	-0.201	9.137	7.163
N2B	-3.684	3.958	0.268		N2B	1.649	-3.966	-0.389	N2B	0.067	-3.349	-0.128	N2B	-0.33	11.805	6.395

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	04/11/2021	ISSUED FOR APPROVAL	--	MehAky	--	JusRii	
		A01	29/09/2021	ISSUED FOR APPROVAL	--	MehAky	--	JusRii	

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DRAWING NUMBER	SASL-RS-DWG-ST-21K025-000011		REVISION 01
DATE	04/11/2021	DATE 04/11/2021	DATE 04/11/2021
DRAWN	S.S.	CHECKED F.S.	APPROVED S.J.T.

LARSEN & TOUBRO DRAWING NUMBER		REVISION 1
OUTOTEC DRAWING NUMBER 900707 __GAU01 __AG11 __MTL01 __49010	DOCUMENT ID OU100728448	REVISION 1

DRAWN	--	--
CHECKED	MehAhy	27/09/2021
ENGINEER	--	--
APPROVED	JusRii	29/09/2021
EMPLOYER		
SCALE	NTS	A1

MANSOURAH - MASSARAH GOLD PROJECT
LAYOUT
RAW WATER TREATMENT PLANT
EQUIPMENT LAYOUT OF PREFAB BUILDING

DRAWING NUMBER MD-216-49W1-EG-EL-GAD-V079-1004 REV A02

ATTACHMENT 2 COMMENT RESOLUTION SHEET

When comments are received from Employer, they need to be placed in a comment resolution table, and the filled table is always attached to the end of the document.

COMMENT TABLE	Document Name:	Document Number:	Revision No.	Comment Date:
General Comment:				

Sr. No	Chapter	Regarding	Comment	Resolution
1	5.1	raft thickness	raft thickness	shown
2	7.2	max drift	Drift limit is too big, please update it to height/300 = 10.3mm	corrected
3	7.2	max settlement	Shall be 25mm per 6.3.1 of MD-101-SMEM-EG-ST-CRT-0001	corrected
4	8.1	embedment& live load	Embedment = ? Please explain why live load is ignored	dimension shown. underground depth please see drawing. live load added
5	8.3.1	max drift& settlement	Shall be 6775/360 = 18.8mm per Table 7-4 of MD-101-SMEM-EG-ST-CRT-0001	corrected

1 Remember also to always mark the made changes with a triangle to the left side of the document.

You can also reply to the comments on an excel template, then it will be coded as a separate document and submitted to Employer together with the revised document itself. You can find the instruction and the excel template from “Document Codes and Numbering Procedure” [1].

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4.0 RESULTS - SUMMARY

Description	Value	Unit	Comment
Load Case LC1 - DL			
Sum of loads in X	0.00	kN	
Sum of support reactions in X	0.00	kN	
Sum of loads in Y	0.00	kN	
Sum of support reactions in Y	0.00	kN	
Sum of loads in Z	4167.36	kN	
Sum of support reactions in Z	4167.36	kN	Deviation 0.00%
Resultant of reactions about X	174.838	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
Resultant of reactions about Y	419.838	kNm	At center of gravity of model
Resultant of reactions about Z	-0.024	kNm	At center of gravity of model
Max. displacement in X	-0.9	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
Max. displacement in Y	-0.4	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
Max. displacement in Z	1.0	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
Max. vector displacement	1.2	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
Max. rotation about X	-0.1	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
Max. rotation about Y	0.3	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
Method of analysis	Linear		Geometrically linear analysis
Reduction of stiffness			Cross-sections, Members, Surfaces
Number of load increments	1		
Number of iterations	1		
Maximum value of element of stiffness matrix on diagonal	1.399E+11		
Minimum value of element of stiffness matrix on diagonal	1.0E+04		
Stiffness matrix determinant	1.639E+107465		
Infinity Norm	2.798E+11		
Load Case LC2 - LL			
Sum of loads in X	-0.09	kN	
Sum of support reactions in X	-0.09	kN	Deviation 0.00%
Sum of loads in Y	0.27	kN	
Sum of support reactions in Y	0.27	kN	Deviation -0.00%
Sum of loads in Z	909.05	kN	
Sum of support reactions in Z	909.05	kN	Deviation 0.00%
Resultant of reactions about X	34.064	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
Resultant of reactions about Y	-153.990	kNm	At center of gravity of model
Resultant of reactions about Z	-1.588	kNm	At center of gravity of model
Max. displacement in X	-0.8	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
Max. displacement in Y	-0.6	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
Max. displacement in Z	0.4	mm	FE Mesh Node No. 1771 (X: 5.646, Y: -1.000, Z: 3.500 m)
Max. vector displacement	0.9	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
Max. rotation about X	-0.2	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
Max. rotation about Y	0.3	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
Method of analysis	Linear		Geometrically linear analysis
Reduction of stiffness			Cross-sections, Members, Surfaces
Number of load increments	1		
Number of iterations	1		
Maximum value of element of stiffness matrix on diagonal	1.399E+11		
Minimum value of element of stiffness matrix on diagonal	1.0E+04		
Stiffness matrix determinant	1.639E+107465		
Infinity Norm	2.798E+11		
Load Case LC3 - EQUIPMENT			
Sum of loads in X	0.00	kN	
Sum of support reactions in X	0.00	kN	
Sum of loads in Y	-0.74	kN	
Sum of support reactions in Y	-0.74	kN	Deviation 0.00%
Sum of loads in Z	357.04	kN	
Sum of support reactions in Z	357.04	kN	Deviation 0.00%
Resultant of reactions about X	83.203	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
Resultant of reactions about Y	771.888	kNm	At center of gravity of model
Resultant of reactions about Z	4.739	kNm	At center of gravity of model
Max. displacement in X	-0.2	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
Max. displacement in Y	-0.3	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
Max. displacement in Z	0.2	mm	FE Mesh Node No. 1794 (X: -5.458, Y: -1.000, Z: 3.500 m)
Max. vector displacement	0.4	mm	Member No. 6, x: 0.000 m
Max. rotation about X	-0.1	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
Max. rotation about Y	0.1	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
Max. rotation about Z	0.0	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
Method of analysis	Linear		Geometrically linear analysis
Reduction of stiffness			Cross-sections, Members, Surfaces
Number of load increments	1		
Number of iterations	1		
Maximum value of element of stiffness matrix on diagonal	1.399E+11		
Minimum value of element of stiffness matrix on diagonal	1.0E+04		
Stiffness matrix determinant	1.639E+107465		
Infinity Norm	2.798E+11		
Load Case LC4 - SEISMIC X+			
Sum of loads in X	37.35	kN	
Sum of support reactions in X	37.35	kN	Deviation 0.00%
Sum of loads in Y	0.00	kN	
Sum of support reactions in Y	0.00	kN	
Sum of loads in Z	-1.17	kN	
Sum of support reactions in Z	-1.17	kN	Deviation 0.00%
Resultant of reactions about X	2.571	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Resultant of reactions about Y	-260.563	kNm	At center of gravity of model
	Resultant of reactions about Z	-1.285	kNm	At center of gravity of model
	Max. displacement in X	1.2	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	0.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	-0.2	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	1.2	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	0.0	mrad	FE Mesh Node No. 5 (X: -12.525, Y: 0.943, Z: 3.500 m)
	Max. rotation about Y	-0.4	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Case	LC5 - SEISMIC X-			
	Sum of loads in X	-37.35	kN	
	Sum of support reactions in X	-37.35	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	0.002	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
	Resultant of reactions about Y	273.600	kNm	At center of gravity of model
	Resultant of reactions about Z	1.285	kNm	At center of gravity of model
	Max. displacement in X	-1.2	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	0.2	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	1.2	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.0	mrad	FE Mesh Node No. 5 (X: -12.525, Y: 0.943, Z: 3.500 m)
	Max. rotation about Y	0.4	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Case	LC6 - SEISMIC Z+			
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	21.25	kN	
	Sum of support reactions in Y	21.25	kN	Deviation 0.00%
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	155.922	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
	Resultant of reactions about Y	-0.264	kNm	At center of gravity of model
	Resultant of reactions about Z	-4.367	kNm	At center of gravity of model
	Max. displacement in X	0.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	0.3	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	-0.0	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	0.3	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	0.1	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 293 (X: -11.443, Y: 0.008, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Case	LC7 - SEISMIC Z-			
	Sum of loads in X	-0.04	kN	
	Sum of support reactions in X	-0.04	kN	Deviation -0.00%
	Sum of loads in Y	-21.25	kN	
	Sum of support reactions in Y	-21.25	kN	Deviation 0.00%
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	-155.923	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
	Resultant of reactions about Y	0.408	kNm	At center of gravity of model
	Resultant of reactions about Z	4.484	kNm	At center of gravity of model
	Max. displacement in X	-0.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.3	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	0.0	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	0.3	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.0	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 293 (X: -11.443, Y: 0.008, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Case LC8 - WINDWARD X+				
	Sum of loads in X	42.67	kN	
	Sum of support reactions in X	42.67	kN	Deviation 0.00%
	Sum of loads in Y	0.21	kN	
	Sum of support reactions in Y	0.21	kN	Deviation 0.00%
	Sum of loads in Z	0.19	kN	
	Sum of support reactions in Z	0.19	kN	Deviation 0.00%
	Resultant of reactions about X	1.178	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
	Resultant of reactions about Y	-220.225	kNm	At center of gravity of model
	Resultant of reactions about Z	22.391	kNm	At center of gravity of model
	Max. displacement in X	1.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	0.1	mm	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. displacement in Z	-0.2	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	1.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	0.0	mrad	FE Mesh Node No. 5 (X: -12.525, Y: 0.943, Z: 3.500 m)
	Max. rotation about Y	-0.5	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1867 (X: -11.526, Y: 4.732, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Case LC9 - WINDWARD X-				
	Sum of loads in X	-42.85	kN	
	Sum of support reactions in X	-42.85	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	0.003	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
	Resultant of reactions about Y	223.104	kNm	At center of gravity of model
	Resultant of reactions about Z	-21.894	kNm	At center of gravity of model
	Max. displacement in X	-1.4	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.1	mm	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. displacement in Z	-0.2	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.4	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.0	mrad	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. rotation about Y	0.5	mrad	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1905 (X: 11.776, Y: 4.731, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Case LC10 - WINDWARD Z+				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	159.50	kN	
	Sum of support reactions in Y	159.50	kN	Deviation 0.00%
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	785.291	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
	Resultant of reactions about Y	-2.104	kNm	At center of gravity of model
	Resultant of reactions about Z	-215.907	kNm	At center of gravity of model
	Max. displacement in X	0.3	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	4.4	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	0.3	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	4.4	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	1.4	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 293 (X: -11.443, Y: 0.008, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on d	1.0E+04		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	diagonal			
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Case LC11 - WINDWARD Z-				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	-183.71	kN	
	Sum of support reactions in Y	-183.71	kN	Deviation 0.00%
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	-943.335	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
	Resultant of reactions about Y	0.323	kNm	At center of gravity of model
	Resultant of reactions about Z	56.885	kNm	At center of gravity of model
	Max. displacement in X	-0.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-4.6	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	-0.4	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	4.6	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-1.5	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.2	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Case LC12 - LEEWARD X+				
	Sum of loads in X	42.67	kN	
	Sum of support reactions in X	42.67	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	-0.003	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
	Resultant of reactions about Y	-222.530	kNm	At center of gravity of model
	Resultant of reactions about Z	21.211	kNm	At center of gravity of model
	Max. displacement in X	1.4	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	0.1	mm	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. displacement in Z	0.2	mm	FE Mesh Node No. 81 (X: 12.775, Y: -1.700, Z: 3.500 m)
	Max. vector displacement	1.4	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	0.0	mrad	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. rotation about Y	-0.5	mrad	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1905 (X: 11.776, Y: 4.731, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Case LC13 - LEEWARD X-				
	Sum of loads in X	-42.67	kN	
	Sum of support reactions in X	-42.67	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	0.001	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
	Resultant of reactions about Y	222.542	kNm	At center of gravity of model
	Resultant of reactions about Z	-21.256	kNm	At center of gravity of model
	Max. displacement in X	-1.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.1	mm	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. displacement in Z	0.2	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	1.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.0	mrad	FE Mesh Node No. 5 (X: -12.525, Y: 0.943, Z: 3.500 m)
	Max. rotation about Y	0.5	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1867 (X: -11.526, Y: 4.732, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Case LC14 - LEEWARD Z+				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Sum of loads in Y	183.71	kN	
	Sum of support reactions in Y	183.71	kN	Deviation 0.00%
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	943.335	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
	Resultant of reactions about Y	-0.323	kNm	At center of gravity of model
	Resultant of reactions about Z	-56.885	kNm	At center of gravity of model
	Max. displacement in X	0.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	4.6	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	0.4	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	4.6	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	1.5	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	-0.2	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Case LC15 - LEEWARD Z-				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	-159.50	kN	
	Sum of support reactions in Y	-159.50	kN	Deviation 0.00%
	Sum of loads in Z	-9.41	kN	
	Sum of support reactions in Z	-9.41	kN	Deviation -0.00%
	Resultant of reactions about X	-764.673	kNm	At center of gravity of model (X:0.670, Y:2.191, Z:3.192 m)
	Resultant of reactions about Y	52.024	kNm	At center of gravity of model
	Resultant of reactions about Z	215.907	kNm	At center of gravity of model
	Max. displacement in X	-0.3	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	-4.4	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	-0.3	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	4.4	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-1.4	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.1	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 293 (X: -11.443, Y: 0.008, Z: 3.500 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.639E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO1 - 1.4*LC1 + 1.4*LC3				
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	-1.04	kN	
	Sum of support reactions in Y	-1.04	kN	Deviation 0.00%
	Sum of loads in Z	6334.16	kN	
	Sum of support reactions in Z	6334.16	kN	Deviation 0.00%
	Resultant of reactions about X	361.2	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1668.3	kNm	At center of gravity of model
	Resultant of reactions about Z	6.6	kNm	At center of gravity of model
	Max. displacement in X	-1.6	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.9	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.5	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.3	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	0.5	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.470E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO2 - 1.4*LC1 + 1.7*LC2 + 1.4*LC3				
	Sum of loads in X	-0.16	kN	
	Sum of support reactions in X	-0.16	kN	Deviation 0.00%
	Sum of loads in Y	-0.58	kN	
	Sum of support reactions in Y	-0.58	kN	Deviation 0.00%
	Sum of loads in Z	7879.55	kN	
	Sum of support reactions in Z	7879.55	kN	Deviation 0.00%

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Resultant of reactions about X	419.0	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1406.3	kNm	At center of gravity of model
	Resultant of reactions about Z	3.9	kNm	At center of gravity of model
	Max. displacement in X	-3.0	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-1.7	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	2.1	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	3.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.5	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	1.0	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.301E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO3 - 1.2*LC1 + LC2 + 1.2*LC3 + 1.6*LC8				
	Sum of loads in X	68.19	kN	
	Sum of support reactions in X	68.19	kN	Deviation 0.00%
	Sum of loads in Y	-0.28	kN	
	Sum of support reactions in Y	-0.28	kN	Deviation 0.00%
	Sum of loads in Z	6338.64	kN	
	Sum of support reactions in Z	6338.64	kN	Deviation 0.00%
	Resultant of reactions about X	345.5	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	922.3	kNm	At center of gravity of model
	Resultant of reactions about Z	39.9	kNm	At center of gravity of model
	Max. displacement in X	1.8	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-1.3	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.6	mm	FE Mesh Node No. 1794 (X: -5.458, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.7	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.4	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	-0.6	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.389E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO4 - 1.2*LC1 + LC2 + 1.2*LC3 + 1.6*LC9				
	Sum of loads in X	-68.65	kN	
	Sum of support reactions in X	-68.65	kN	Deviation 0.00%
	Sum of loads in Y	-0.62	kN	
	Sum of support reactions in Y	-0.62	kN	Deviation -0.00%
	Sum of loads in Z	6338.33	kN	
	Sum of support reactions in Z	6338.33	kN	Deviation 0.00%
	Resultant of reactions about X	343.6	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1634.1	kNm	At center of gravity of model
	Resultant of reactions about Z	-31.0	kNm	At center of gravity of model
	Max. displacement in X	-3.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-1.2	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.8	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	3.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.4	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	1.0	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1883 (X: 0.125, Y: 4.738, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.389E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO5 - 1.2*LC1 + LC2 + 1.2*LC3 + 1.6*LC10				
	Sum of loads in X	-0.09	kN	
	Sum of support reactions in X	-0.09	kN	Deviation -0.00%
	Sum of loads in Y	254.58	kN	
	Sum of support reactions in Y	254.58	kN	Deviation 0.00%

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Sum of loads in Z	6338.33	kN	
	Sum of support reactions in Z	6338.33	kN	Deviation 0.00%
	Resultant of reactions about X	1606.6	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1272.5	kNm	At center of gravity of model
	Resultant of reactions about Z	-341.4	kNm	At center of gravity of model
	Max. displacement in X	-1.8	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	8.2	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	2.0	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	8.5	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	2.7	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.6	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1881 (X: 0.063, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.390E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO6 - 1.2*LC1 + LC2 + 1.2*LC3 + 1.6*LC11				
	Sum of loads in X	-0.09	kN	
	Sum of support reactions in X	-0.09	kN	Deviation 0.00%
	Sum of loads in Y	-294.56	kN	
	Sum of support reactions in Y	-294.56	kN	Deviation 0.00%
	Sum of loads in Z	6338.33	kN	
	Sum of support reactions in Z	6338.33	kN	Deviation 0.00%
	Resultant of reactions about X	-1172.8	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1276.4	kNm	At center of gravity of model
	Resultant of reactions about Z	95.1	kNm	At center of gravity of model
	Max. displacement in X	-3.0	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-6.2	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.9	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	6.4	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-2.1	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	1.0	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.388E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO7 - 1.2*LC1 + LC2 + 1.2*LC3 + 1.6*LC12				
	Sum of loads in X	68.18	kN	
	Sum of support reactions in X	68.18	kN	Deviation -0.00%
	Sum of loads in Y	-0.62	kN	
	Sum of support reactions in Y	-0.62	kN	Deviation -0.00%
	Sum of loads in Z	6338.33	kN	
	Sum of support reactions in Z	6338.33	kN	Deviation 0.00%
	Resultant of reactions about X	343.6	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	918.6	kNm	At center of gravity of model
	Resultant of reactions about Z	38.0	kNm	At center of gravity of model
	Max. displacement in X	3.5	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-1.3	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.7	mm	FE Mesh Node No. 79 (X: 12.775, Y: -0.700, Z: 3.500 m)
	Max. vector displacement	3.8	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.4	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	-1.1	mrad	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.391E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO8 - 1.2*LC1 + LC2 + 1.2*LC3 + 1.6*LC14				
	Sum of loads in X	-0.10	kN	
	Sum of support reactions in X	-0.10	kN	Deviation 0.00%

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Sum of loads in Y	293.32	kN	
	Sum of support reactions in Y	293.32	kN	Deviation 0.00%
	Sum of loads in Z	6338.33	kN	
	Sum of support reactions in Z	6338.33	kN	Deviation 0.00%
	Resultant of reactions about X	1861.1	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1275.4	kNm	At center of gravity of model
	Resultant of reactions about Z	-86.9	kNm	At center of gravity of model
	Max. displacement in X	-1.5	mm	FE Mesh Node No. 33 (X: -11.525, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	8.6	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	2.1	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	8.9	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	2.8	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.5	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.392E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO9 - 1.2*LC1 + LC2 + 1.2*LC3 + 1.6*LC15				
	Sum of loads in X	-0.09	kN	
	Sum of support reactions in X	-0.09	kN	Deviation 0.00%
	Sum of loads in Y	-255.82	kN	
	Sum of support reactions in Y	-255.82	kN	Deviation 0.00%
	Sum of loads in Z	6323.28	kN	
	Sum of support reactions in Z	6323.28	kN	Deviation 0.00%
	Resultant of reactions about X	-886.1	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1359.1	kNm	At center of gravity of model
	Resultant of reactions about Z	349.5	kNm	At center of gravity of model
	Max. displacement in X	-2.6	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-5.9	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	2.1	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	6.1	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-2.0	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.8	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 293 (X: -11.443, Y: 0.008, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.391E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO10 - 1.2*LC1 + LC2 + 1.2*LC3 + LC4				
	Sum of loads in X	37.25	kN	
	Sum of support reactions in X	37.25	kN	Deviation 0.00%
	Sum of loads in Y	-0.62	kN	
	Sum of support reactions in Y	-0.62	kN	Deviation -0.00%
	Sum of loads in Z	6337.16	kN	
	Sum of support reactions in Z	6337.16	kN	Deviation 0.00%
	Resultant of reactions about X	346.2	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1014.4	kNm	At center of gravity of model
	Resultant of reactions about Z	2.8	kNm	At center of gravity of model
	Max. displacement in X	1.9	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-1.3	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.6	mm	FE Mesh Node No. 1794 (X: -5.458, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.8	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.4	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	-0.6	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.390E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO11 - 1.2*LC1 + LC2 + 1.2*LC3 + LC5				

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Sum of loads in X	-37.44	kN	
	Sum of support reactions in X	-37.44	kN	Deviation 0.00%
	Sum of loads in Y	-0.62	kN	
	Sum of support reactions in Y	-0.62	kN	Deviation 0.00%
	Sum of loads in Z	6338.33	kN	
	Sum of support reactions in Z	6338.33	kN	Deviation 0.00%
	Resultant of reactions about X	343.6	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1550.4	kNm	At center of gravity of model
	Resultant of reactions about Z	5.4	kNm	At center of gravity of model
	Max. displacement in X	-3.4	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-1.2	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.8	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	3.7	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.4	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	1.1	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1883 (X: 0.125, Y: 4.738, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.390E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO12 - 1.2*LC1 + LC2 + 1.2*LC3 + LC6				
	Sum of loads in X	-0.09	kN	
	Sum of support reactions in X	-0.09	kN	Deviation -0.00%
	Sum of loads in Y	20.62	kN	
	Sum of support reactions in Y	20.62	kN	Deviation 0.00%
	Sum of loads in Z	6338.33	kN	
	Sum of support reactions in Z	6338.33	kN	Deviation 0.00%
	Resultant of reactions about X	500.1	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1275.6	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.3	kNm	At center of gravity of model
	Max. displacement in X	-2.0	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	1.5	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.6	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	0.5	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.7	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.390E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO13 - 1.2*LC1 + LC2 + 1.2*LC3 + LC7				
	Sum of loads in X	-0.14	kN	
	Sum of support reactions in X	-0.14	kN	Deviation 0.00%
	Sum of loads in Y	-21.87	kN	
	Sum of support reactions in Y	-21.87	kN	Deviation 0.00%
	Sum of loads in Z	6338.33	kN	
	Sum of support reactions in Z	6338.33	kN	Deviation 0.00%
	Resultant of reactions about X	187.1	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1276.3	kNm	At center of gravity of model
	Resultant of reactions about Z	8.6	kNm	At center of gravity of model
	Max. displacement in X	-2.3	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-1.5	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.7	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.8	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.5	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	0.7	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.390E+107465		
	Infinity Norm	2.798E+11		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
Load Combination CO14 - 0.9*LC1 + 1.6*LC8				
	Sum of loads in X	68.28	kN	
	Sum of support reactions in X	68.28	kN	Deviation 0.00%
	Sum of loads in Y	0.34	kN	
	Sum of support reactions in Y	0.34	kN	Deviation 0.00%
	Sum of loads in Z	3750.93	kN	
	Sum of support reactions in Z	3750.93	kN	Deviation 0.00%
	Resultant of reactions about X	159.2	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	25.1	kNm	At center of gravity of model
	Resultant of reactions about Z	35.8	kNm	At center of gravity of model
	Max. displacement in X	1.6	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.4	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.1	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.7	mm	Member No. 8, x: 0.000 m
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 76 (X: 12.775, Y: 0.800, Z: 3.500 m)
	Max. rotation about Y	-0.5	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.566E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO15 - 0.9*LC1 + 1.6*LC9				
	Sum of loads in X	-68.56	kN	
	Sum of support reactions in X	-68.56	kN	Deviation 0.00%
	Sum of loads in Y	-0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	3750.63	kN	
	Sum of support reactions in Z	3750.63	kN	Deviation 0.00%
	Resultant of reactions about X	157.3	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	735.2	kNm	At center of gravity of model
	Resultant of reactions about Z	-35.1	kNm	At center of gravity of model
	Max. displacement in X	-1.9	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.5	mm	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. displacement in Z	1.0	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.0	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. rotation about Y	0.6	mrad	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.566E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO16 - 0.9*LC1 + 1.6*LC10				
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	255.20	kN	
	Sum of support reactions in Y	255.20	kN	Deviation 0.00%
	Sum of loads in Z	3750.63	kN	
	Sum of support reactions in Z	3750.63	kN	Deviation 0.00%
	Resultant of reactions about X	1415.6	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	374.4	kNm	At center of gravity of model
	Resultant of reactions about Z	-345.5	kNm	At center of gravity of model
	Max. displacement in X	0.9	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	7.3	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.3	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	7.4	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	2.4	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	-0.3	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1881 (X: 0.063, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.568E+10746		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
		65		
	Infinity Norm	2.798E+11		
Load Combination CO17 - 0.9*LC1 + 1.6*LC11				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	-293.94	kN	
	Sum of support reactions in Y	-293.94	kN	Deviation 0.00%
	Sum of loads in Z	3750.63	kN	
	Sum of support reactions in Z	3750.63	kN	Deviation 0.00%
	Resultant of reactions about X	-1353.5	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	378.3	kNm	At center of gravity of model
	Resultant of reactions about Z	91.0	kNm	At center of gravity of model
	Max. displacement in X	-1.6	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-7.1	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	7.1	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-2.3	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.5	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.565E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO18 - 0.9*LC1 + 1.6*LC12				
	Sum of loads in X	68.27	kN	
	Sum of support reactions in X	68.27	kN	Deviation 0.00%
	Sum of loads in Y	-0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	3750.63	kN	
	Sum of support reactions in Z	3750.63	kN	Deviation 0.00%
	Resultant of reactions about X	157.3	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	21.4	kNm	At center of gravity of model
	Resultant of reactions about Z	33.9	kNm	At center of gravity of model
	Max. displacement in X	2.6	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.4	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	2.8	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 76 (X: 12.775, Y: 0.800, Z: 3.500 m)
	Max. rotation about Y	-0.9	mrad	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1905 (X: 11.776, Y: 4.731, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.568E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO19 - 0.9*LC1 + 1.6*LC13				
	Sum of loads in X	-68.28	kN	
	Sum of support reactions in X	-68.28	kN	Deviation 0.00%
	Sum of loads in Y	-0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	3750.62	kN	
	Sum of support reactions in Z	3750.62	kN	Deviation 0.00%
	Resultant of reactions about X	157.3	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	734.3	kNm	At center of gravity of model
	Resultant of reactions about Z	-34.0	kNm	At center of gravity of model
	Max. displacement in X	-3.2	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.5	mm	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	3.3	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. rotation about Y	1.0	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1867 (X: -11.526, Y: 4.732, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on d	1.0E+04		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	diagonal			
	Stiffness matrix determinant	1.568E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO20 - 0.9*LC1 + 1.6*LC14				
	Sum of loads in X	-0.01	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	293.94	kN	
	Sum of support reactions in Y	293.94	kN	Deviation 0.00%
	Sum of loads in Z	3750.63	kN	
	Sum of support reactions in Z	3750.63	kN	Deviation 0.00%
	Resultant of reactions about X	1669.1	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	377.3	kNm	At center of gravity of model
	Resultant of reactions about Z	-91.0	kNm	At center of gravity of model
	Max. displacement in X	-0.9	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	7.7	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.3	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	7.8	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	2.5	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.3	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.569E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO21 - 0.9*LC1 + 1.6*LC15				
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	-255.20	kN	
	Sum of support reactions in Y	-255.20	kN	Deviation 0.00%
	Sum of loads in Z	3735.57	kN	
	Sum of support reactions in Z	3735.57	kN	Deviation 0.00%
	Resultant of reactions about X	-1067.7	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	461.1	kNm	At center of gravity of model
	Resultant of reactions about Z	345.4	kNm	At center of gravity of model
	Max. displacement in X	-1.2	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-6.7	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.3	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	6.7	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-2.2	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.4	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 293 (X: -11.443, Y: 0.008, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.568E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO22 - 0.9*LC1 + LC4				
	Sum of loads in X	37.34	kN	
	Sum of support reactions in X	37.34	kN	Deviation 0.00%
	Sum of loads in Y	-0.00	kN	
	Sum of support reactions in Y	-0.00	kN	
	Sum of loads in Z	3749.45	kN	
	Sum of support reactions in Z	3749.45	kN	Deviation 0.00%
	Resultant of reactions about X	159.9	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	117.0	kNm	At center of gravity of model
	Resultant of reactions about Z	-1.3	kNm	At center of gravity of model
	Max. displacement in X	1.3	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	0.3	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	0.9	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.6	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	0.1	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	-0.5	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix o	1.399E+11		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	on diagonal			
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.567E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO23 - 0.9*LC1 + LC5				
	Sum of loads in X	-37.35	kN	
	Sum of support reactions in X	-37.35	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	3750.63	kN	
	Sum of support reactions in Z	3750.63	kN	Deviation 0.00%
	Resultant of reactions about X	157.3	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	651.7	kNm	At center of gravity of model
	Resultant of reactions about Z	1.3	kNm	At center of gravity of model
	Max. displacement in X	-2.0	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.4	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.1	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.2	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 5 (X: -12.525, Y: 0.943, Z: 3.500 m)
	Max. rotation about Y	0.7	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1883 (X: 0.125, Y: 4.738, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.567E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO24 - 0.9*LC1 + LC6				
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	21.25	kN	
	Sum of support reactions in Y	21.25	kN	Deviation 0.00%
	Sum of loads in Z	3750.63	kN	
	Sum of support reactions in Z	3750.63	kN	Deviation 0.00%
	Resultant of reactions about X	313.4	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	377.6	kNm	At center of gravity of model
	Resultant of reactions about Z	-4.4	kNm	At center of gravity of model
	Max. displacement in X	-0.7	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	0.6	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	0.9	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.1	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	0.2	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.2	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.567E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO25 - 0.9*LC1 + LC7				
	Sum of loads in X	-0.05	kN	
	Sum of support reactions in X	-0.05	kN	Deviation 0.00%
	Sum of loads in Y	-21.25	kN	
	Sum of support reactions in Y	-21.25	kN	Deviation 0.00%
	Sum of loads in Z	3750.63	kN	
	Sum of support reactions in Z	3750.63	kN	Deviation 0.00%
	Resultant of reactions about X	1.3	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	378.2	kNm	At center of gravity of model
	Resultant of reactions about Z	4.5	kNm	At center of gravity of model
	Max. displacement in X	-0.9	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.6	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	0.9	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.3	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.2	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	0.3	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 293 (X: -11.443, Y: 0.008, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.567E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO26 - LC1 + LC2 + LC3				
	Sum of loads in X	-0.09	kN	
	Sum of support reactions in X	-0.09	kN	Deviation 0.00%
	Sum of loads in Y	-0.47	kN	
	Sum of support reactions in Y	-0.47	kN	Deviation -0.00%
	Sum of loads in Z	5433.45	kN	
	Sum of support reactions in Z	5433.45	kN	Deviation 0.00%
	Resultant of reactions about X	292.0	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1037.6	kNm	At center of gravity of model
	Resultant of reactions about Z	3.1	kNm	At center of gravity of model
	Max. displacement in X	-1.9	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-1.1	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.4	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.3	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.4	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	0.6	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.412E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO27 - LC1 + 0.75*LC2 + LC3				
	Sum of loads in X	-0.07	kN	
	Sum of support reactions in X	-0.07	kN	Deviation 0.00%
	Sum of loads in Y	-0.54	kN	
	Sum of support reactions in Y	-0.54	kN	Deviation -0.00%
	Sum of loads in Z	5206.19	kN	
	Sum of support reactions in Z	5206.19	kN	Deviation 0.00%
	Resultant of reactions about X	283.5	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	1076.1	kNm	At center of gravity of model
	Resultant of reactions about Z	3.5	kNm	At center of gravity of model
	Max. displacement in X	-1.7	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-1.0	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.3	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.3	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	0.6	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.437E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO28 - LC1 + LC8				
	Sum of loads in X	42.67	kN	
	Sum of support reactions in X	42.67	kN	Deviation 0.00%
	Sum of loads in Y	0.21	kN	
	Sum of support reactions in Y	0.21	kN	Deviation 0.00%
	Sum of loads in Z	4167.55	kN	
	Sum of support reactions in Z	4167.55	kN	Deviation 0.00%
	Resultant of reactions about X	176.0	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	199.3	kNm	At center of gravity of model
	Resultant of reactions about Z	22.4	kNm	At center of gravity of model
	Max. displacement in X	0.9	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	-0.4	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.1	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.3	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	0.1	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	-0.3	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.559E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO29 - LC1 + LC9				
	Sum of loads in X	-42.85	kN	
	Sum of support reactions in X	-42.85	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	4167.36	kN	
	Sum of support reactions in Z	4167.36	kN	Deviation 0.00%
	Resultant of reactions about X	174.8	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	643.2	kNm	At center of gravity of model
	Resultant of reactions about Z	-21.9	kNm	At center of gravity of model
	Max. displacement in X	-1.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.5	mm	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. displacement in Z	1.1	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	1.7	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. rotation about Y	0.5	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1883 (X: 0.125, Y: 4.738, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.559E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO30 - LC1 + LC10				
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	159.50	kN	
	Sum of support reactions in Y	159.50	kN	Deviation 0.00%
	Sum of loads in Z	4167.36	kN	
	Sum of support reactions in Z	4167.36	kN	Deviation 0.00%
	Resultant of reactions about X	961.3	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	417.7	kNm	At center of gravity of model
	Resultant of reactions about Z	-215.9	kNm	At center of gravity of model
	Max. displacement in X	0.7	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	4.7	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.1	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	4.8	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	1.6	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	-0.3	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1881 (X: 0.063, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.560E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO31 - LC1 + LC11				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	-183.71	kN	
	Sum of support reactions in Y	-183.71	kN	Deviation 0.00%
	Sum of loads in Z	4167.36	kN	
	Sum of support reactions in Z	4167.36	kN	Deviation 0.00%
	Resultant of reactions about X	-769.7	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	420.1	kNm	At center of gravity of model
	Resultant of reactions about Z	56.9	kNm	At center of gravity of model
	Max. displacement in X	-1.4	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-4.3	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	4.3	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-1.4	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.4	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.558E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO32 - LC1 + LC12				
	Sum of loads in X	42.67	kN	
	Sum of support reactions in X	42.67	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	4167.36	kN	
	Sum of support reactions in Z	4167.36	kN	Deviation 0.00%
	Resultant of reactions about X	174.8	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	197.0	kNm	At center of gravity of model
	Resultant of reactions about Z	21.2	kNm	At center of gravity of model
	Max. displacement in X	1.8	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.4	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	2.1	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 76 (X: 12.775, Y: 0.800, Z: 3.500 m)
	Max. rotation about Y	-0.6	mrad	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.560E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO33 - LC1 + LC13				
	Sum of loads in X	-42.68	kN	
	Sum of support reactions in X	-42.68	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	4167.36	kN	
	Sum of support reactions in Z	4167.36	kN	Deviation 0.00%
	Resultant of reactions about X	174.8	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	642.6	kNm	At center of gravity of model
	Resultant of reactions about Z	-21.3	kNm	At center of gravity of model
	Max. displacement in X	-2.4	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.5	mm	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. displacement in Z	1.1	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.6	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. rotation about Y	0.8	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1867 (X: -11.526, Y: 4.732, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.560E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO34 - LC1 + LC14				
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	183.71	kN	
	Sum of support reactions in Y	183.71	kN	Deviation 0.00%
	Sum of loads in Z	4167.36	kN	
	Sum of support reactions in Z	4167.36	kN	Deviation 0.00%
	Resultant of reactions about X	1119.7	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	419.5	kNm	At center of gravity of model
	Resultant of reactions about Z	-56.9	kNm	At center of gravity of model
	Max. displacement in X	-0.8	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	4.9	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 106 (X: -12.525, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	5.1	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	1.6	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.3	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.561E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO35 - LC1 + LC15				
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	-159.50	kN	
	Sum of support reactions in Y	-159.50	kN	Deviation 0.00%
	Sum of loads in Z	4157.95	kN	
	Sum of support reactions in Z	4157.95	kN	Deviation 0.00%
	Resultant of reactions about X	-591.0	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	471.8	kNm	At center of gravity of model
	Resultant of reactions about Z	215.9	kNm	At center of gravity of model
	Max. displacement in X	-1.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-4.0	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	4.1	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-1.3	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.4	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 293 (X: -11.443, Y: 0.008, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.560E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO36 - LC1 + 0.7*LC4				
	Sum of loads in X	26.14	kN	
	Sum of support reactions in X	26.14	kN	Deviation 0.00%
	Sum of loads in Y	-0.00	kN	
	Sum of support reactions in Y	-0.00	kN	
	Sum of loads in Z	4166.54	kN	
	Sum of support reactions in Z	4166.54	kN	Deviation 0.00%
	Resultant of reactions about X	176.6	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	237.2	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.9	kNm	At center of gravity of model
	Max. displacement in X	1.1	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	-0.4	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.0	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.4	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	-0.4	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.559E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO37 - LC1 + 0.7*LC5				
	Sum of loads in X	-26.14	kN	
	Sum of support reactions in X	-26.14	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	4167.36	kN	
	Sum of support reactions in Z	4167.36	kN	Deviation 0.00%
	Resultant of reactions about X	174.8	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	611.5	kNm	At center of gravity of model
	Resultant of reactions about Z	0.9	kNm	At center of gravity of model
	Max. displacement in X	-1.7	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.4	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.1	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.0	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Max. rotation about Y	0.6	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1883 (X: 0.125, Y: 4.738, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.559E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO38 - LC1 + 0.7*LC6				
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	14.87	kN	
	Sum of support reactions in Y	14.87	kN	Deviation 0.00%
	Sum of loads in Z	4167.36	kN	
	Sum of support reactions in Z	4167.36	kN	Deviation 0.00%
	Resultant of reactions about X	284.1	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	419.6	kNm	At center of gravity of model
	Resultant of reactions about Z	-3.1	kNm	At center of gravity of model
	Max. displacement in X	-0.8	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	0.6	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.0	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.2	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	0.2	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.3	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.559E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO39 - LC1 + 0.7*LC7				
	Sum of loads in X	-0.03	kN	
	Sum of support reactions in X	-0.03	kN	Deviation 0.00%
	Sum of loads in Y	-14.87	kN	
	Sum of support reactions in Y	-14.87	kN	Deviation 0.00%
	Sum of loads in Z	4167.36	kN	
	Sum of support reactions in Z	4167.36	kN	Deviation 0.00%
	Resultant of reactions about X	65.6	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	420.1	kNm	At center of gravity of model
	Resultant of reactions about Z	3.1	kNm	At center of gravity of model
	Max. displacement in X	-1.0	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.5	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.0	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.3	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.2	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	0.3	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.559E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO40 - LC1 + 0.75*LC2 + 0.75*LC8				
	Sum of loads in X	31.94	kN	
	Sum of support reactions in X	31.94	kN	Deviation 0.00%
	Sum of loads in Y	0.36	kN	
	Sum of support reactions in Y	0.36	kN	Deviation 0.00%
	Sum of loads in Z	4849.29	kN	
	Sum of support reactions in Z	4849.29	kN	Deviation 0.00%
	Resultant of reactions about X	201.2	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	138.7	kNm	At center of gravity of model
	Resultant of reactions about Z	15.6	kNm	At center of gravity of model
	Max. displacement in X	1.1	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.8	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Max. vector displacement	1.7	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-0.2	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	-0.4	mrad	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.478E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO41 - LC1 + 0.75*LC2 + 0.75*LC9				
	Sum of loads in X	-32.21	kN	
	Sum of support reactions in X	-32.21	kN	Deviation 0.00%
	Sum of loads in Y	0.20	kN	
	Sum of support reactions in Y	0.20	kN	Deviation 0.00%
	Sum of loads in Z	4849.15	kN	
	Sum of support reactions in Z	4849.15	kN	Deviation 0.00%
	Resultant of reactions about X	200.3	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	472.0	kNm	At center of gravity of model
	Resultant of reactions about Z	-17.6	kNm	At center of gravity of model
	Max. displacement in X	-1.9	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.8	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.3	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.3	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.3	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	0.6	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1883 (X: 0.125, Y: 4.738, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.477E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO42 - LC1 + 0.75*LC2 + 0.75*LC10				
	Sum of loads in X	-0.07	kN	
	Sum of support reactions in X	-0.07	kN	Deviation -0.00%
	Sum of loads in Y	119.83	kN	
	Sum of support reactions in Y	119.83	kN	Deviation 0.00%
	Sum of loads in Z	4849.15	kN	
	Sum of support reactions in Z	4849.15	kN	Deviation 0.00%
	Resultant of reactions about X	791.1	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	302.7	kNm	At center of gravity of model
	Resultant of reactions about Z	-163.1	kNm	At center of gravity of model
	Max. displacement in X	-1.3	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	4.0	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.3	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	4.2	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	1.3	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.4	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1881 (X: 0.063, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.478E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO43 - LC1 + 0.75*LC2 + 0.75*LC11				
	Sum of loads in X	-0.07	kN	
	Sum of support reactions in X	-0.07	kN	Deviation -0.00%
	Sum of loads in Y	-137.58	kN	
	Sum of support reactions in Y	-137.58	kN	Deviation 0.00%
	Sum of loads in Z	4849.15	kN	
	Sum of support reactions in Z	4849.15	kN	Deviation 0.00%
	Resultant of reactions about X	-509.2	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	304.5	kNm	At center of gravity of model
	Resultant of reactions about Z	41.4	kNm	At center of gravity of model
	Max. displacement in X	-1.9	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Max. displacement in Y	-2.8	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.4	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	3.0	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-0.9	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.6	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.477E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO44 - LC1 + 0.75*LC2 + 0.75*LC12				
	Sum of loads in X	31.93	kN	
	Sum of support reactions in X	31.93	kN	Deviation 0.00%
	Sum of loads in Y	0.20	kN	
	Sum of support reactions in Y	0.20	kN	Deviation -0.00%
	Sum of loads in Z	4849.15	kN	
	Sum of support reactions in Z	4849.15	kN	Deviation 0.00%
	Resultant of reactions about X	200.3	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	137.0	kNm	At center of gravity of model
	Resultant of reactions about Z	14.7	kNm	At center of gravity of model
	Max. displacement in X	1.9	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.8	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.3	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	2.3	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.2	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	-0.6	mrad	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.479E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO45 - LC1 + 0.75*LC2 + 0.75*LC13				
	Sum of loads in X	-32.08	kN	
	Sum of support reactions in X	-32.08	kN	Deviation 0.00%
	Sum of loads in Y	0.20	kN	
	Sum of support reactions in Y	0.20	kN	Deviation -0.00%
	Sum of loads in Z	4849.15	kN	
	Sum of support reactions in Z	4849.15	kN	Deviation 0.00%
	Resultant of reactions about X	200.3	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	471.5	kNm	At center of gravity of model
	Resultant of reactions about Z	-17.2	kNm	At center of gravity of model
	Max. displacement in X	-2.6	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.8	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.4	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.9	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.3	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	0.9	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1867 (X: -11.526, Y: 4.732, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.478E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO46 - LC1 + 0.75*LC2 + 0.75*LC14				
	Sum of loads in X	-0.07	kN	
	Sum of support reactions in X	-0.07	kN	Deviation 0.00%
	Sum of loads in Y	137.99	kN	
	Sum of support reactions in Y	137.99	kN	Deviation 0.00%
	Sum of loads in Z	4849.15	kN	
	Sum of support reactions in Z	4849.15	kN	Deviation 0.00%
	Resultant of reactions about X	910.1	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	304.0	kNm	At center of gravity of model

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Resultant of reactions about Z	-43.9	kNm	At center of gravity of model
	Max. displacement in X	-1.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	4.2	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.4	mm	FE Mesh Node No. 106 (X: -12.525, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	4.4	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	1.4	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.4	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.479E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO47 - LC1 + 0.75*LC2 + 0.75*LC15				
	Sum of loads in X	-0.07	kN	
	Sum of support reactions in X	-0.07	kN	Deviation 0.00%
	Sum of loads in Y	-119.42	kN	
	Sum of support reactions in Y	-119.42	kN	Deviation 0.00%
	Sum of loads in Z	4842.09	kN	
	Sum of support reactions in Z	4842.09	kN	Deviation 0.00%
	Resultant of reactions about X	-374.9	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	343.3	kNm	At center of gravity of model
	Resultant of reactions about Z	160.7	kNm	At center of gravity of model
	Max. displacement in X	-1.7	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-2.6	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.4	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.9	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.9	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.5	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 293 (X: -11.443, Y: 0.008, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.478E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO48 - LC1 + 0.75*LC2 + 0.53*LC4				
	Sum of loads in X	19.72	kN	
	Sum of support reactions in X	19.72	kN	Deviation 0.00%
	Sum of loads in Y	0.20	kN	
	Sum of support reactions in Y	0.20	kN	Deviation 0.00%
	Sum of loads in Z	4848.53	kN	
	Sum of support reactions in Z	4848.53	kN	Deviation 0.00%
	Resultant of reactions about X	201.7	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	165.9	kNm	At center of gravity of model
	Resultant of reactions about Z	-1.9	kNm	At center of gravity of model
	Max. displacement in X	1.1	mm	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.8	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.7	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-0.3	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	-0.4	mrad	FE Mesh Node No. 12 (X: -5.725, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.478E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO49 - LC1 + 0.75*LC2 + 0.53*LC5				
	Sum of loads in X	-19.86	kN	
	Sum of support reactions in X	-19.86	kN	Deviation 0.00%
	Sum of loads in Y	0.20	kN	
	Sum of support reactions in Y	0.20	kN	Deviation 0.00%
	Sum of loads in Z	4849.15	kN	
	Sum of support reactions in Z	4849.15	kN	Deviation 0.00%

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Resultant of reactions about X	200.3	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	449.6	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.5	kNm	At center of gravity of model
	Max. displacement in X	-2.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.8	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.3	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.4	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.3	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	0.7	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1883 (X: 0.125, Y: 4.738, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.478E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO50 - LC1 + 0.75*LC2 + 0.53*LC6				
	Sum of loads in X	-0.07	kN	
	Sum of support reactions in X	-0.07	kN	Deviation -0.00%
	Sum of loads in Y	11.46	kN	
	Sum of support reactions in Y	11.46	kN	Deviation 0.00%
	Sum of loads in Z	4849.15	kN	
	Sum of support reactions in Z	4849.15	kN	Deviation 0.00%
	Resultant of reactions about X	283.2	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	304.1	kNm	At center of gravity of model
	Resultant of reactions about Z	-3.5	kNm	At center of gravity of model
	Max. displacement in X	-1.4	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	0.9	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	1.8	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	0.3	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.5	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.478E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO51 - LC1 + 0.75*LC2 + 0.53*LC7				
	Sum of loads in X	-0.09	kN	
	Sum of support reactions in X	-0.09	kN	Deviation -0.00%
	Sum of loads in Y	-11.06	kN	
	Sum of support reactions in Y	-11.06	kN	Deviation 0.00%
	Sum of loads in Z	4849.15	kN	
	Sum of support reactions in Z	4849.15	kN	Deviation 0.00%
	Resultant of reactions about X	117.5	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	304.5	kNm	At center of gravity of model
	Resultant of reactions about Z	1.2	kNm	At center of gravity of model
	Max. displacement in X	-1.6	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.9	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	1.9	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.3	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	0.5	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.478E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO52 - 0.6*LC1 + LC8				
	Sum of loads in X	42.67	kN	
	Sum of support reactions in X	42.67	kN	Deviation 0.00%
	Sum of loads in Y	0.21	kN	
	Sum of support reactions in Y	0.21	kN	Deviation 0.00%

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Sum of loads in Z	2500.61	kN	
	Sum of support reactions in Z	2500.61	kN	Deviation 0.00%
	Resultant of reactions about X	106.1	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	31.5	kNm	At center of gravity of model
	Resultant of reactions about Z	22.4	kNm	At center of gravity of model
	Max. displacement in X	1.0	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.2	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	0.7	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 76 (X: 12.775, Y: 0.800, Z: 3.500 m)
	Max. rotation about Y	-0.3	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1898 (X: 5.972, Y: 4.727, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.590E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO53 - 0.6*LC1 + LC9				
	Sum of loads in X	-42.85	kN	
	Sum of support reactions in X	-42.85	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	2500.42	kN	
	Sum of support reactions in Z	2500.42	kN	Deviation 0.00%
	Resultant of reactions about X	104.9	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	475.2	kNm	At center of gravity of model
	Resultant of reactions about Z	-21.9	kNm	At center of gravity of model
	Max. displacement in X	-1.2	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.3	mm	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. displacement in Z	0.7	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	1.3	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. rotation about Y	0.4	mrad	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.590E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO54 - 0.6*LC1 + LC10				
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	159.50	kN	
	Sum of support reactions in Y	159.50	kN	Deviation 0.00%
	Sum of loads in Z	2500.42	kN	
	Sum of support reactions in Z	2500.42	kN	Deviation 0.00%
	Resultant of reactions about X	890.9	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	249.8	kNm	At center of gravity of model
	Resultant of reactions about Z	-215.9	kNm	At center of gravity of model
	Max. displacement in X	0.6	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	4.6	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	0.8	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	4.6	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	1.5	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	-0.2	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1881 (X: 0.063, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.591E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO55 - 0.6*LC1 + LC11				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Sum of loads in Y	-183.71	kN	
	Sum of support reactions in Y	-183.71	kN	Deviation 0.00%
	Sum of loads in Z	2500.42	kN	
	Sum of support reactions in Z	2500.42	kN	Deviation 0.00%
	Resultant of reactions about X	-839.1	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	252.2	kNm	At center of gravity of model
	Resultant of reactions about Z	56.9	kNm	At center of gravity of model
	Max. displacement in X	-1.0	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-4.4	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	0.8	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	4.4	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-1.5	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.3	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.589E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO56 - 0.6*LC1 + LC12				
	Sum of loads in X	42.67	kN	
	Sum of support reactions in X	42.67	kN	Deviation 0.00%
	Sum of loads in Y	-0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	2500.42	kN	
	Sum of support reactions in Z	2500.42	kN	Deviation 0.00%
	Resultant of reactions about X	104.9	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	29.2	kNm	At center of gravity of model
	Resultant of reactions about Z	21.2	kNm	At center of gravity of model
	Max. displacement in X	1.6	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.2	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	0.8	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.8	mm	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 76 (X: 12.775, Y: 0.800, Z: 3.500 m)
	Max. rotation about Y	-0.5	mrad	FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 1905 (X: 11.776, Y: 4.731, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.591E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO57 - 0.6*LC1 + LC13				
	Sum of loads in X	-42.68	kN	
	Sum of support reactions in X	-42.68	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	2500.42	kN	
	Sum of support reactions in Z	2500.42	kN	Deviation 0.00%
	Resultant of reactions about X	104.9	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	474.6	kNm	At center of gravity of model
	Resultant of reactions about Z	-21.3	kNm	At center of gravity of model
	Max. displacement in X	-2.0	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.3	mm	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. displacement in Z	0.8	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	2.1	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 35 (X: 9.175, Y: -1.600, Z: 0.000 m)
	Max. rotation about Y	0.7	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1867 (X: -11.526, Y: 4.732, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.591E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO58 - 0.6*LC1 + LC14				

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	183.71	kN	
	Sum of support reactions in Y	183.71	kN	Deviation 0.00%
	Sum of loads in Z	2500.42	kN	
	Sum of support reactions in Z	2500.42	kN	Deviation 0.00%
	Resultant of reactions about X	1049.2	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	251.6	kNm	At center of gravity of model
	Resultant of reactions about Z	-56.9	kNm	At center of gravity of model
	Max. displacement in X	-0.6	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	4.8	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	0.9	mm	FE Mesh Node No. 1761 (X: -0.123, Y: 5.800, Z: 3.500 m)
	Max. vector displacement	4.9	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	1.6	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.2	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.592E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO59 - 0.6*LC1 + LC15				
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	-159.50	kN	
	Sum of support reactions in Y	-159.50	kN	Deviation 0.00%
	Sum of loads in Z	2491.01	kN	
	Sum of support reactions in Z	2491.01	kN	Deviation 0.00%
	Resultant of reactions about X	-660.4	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	303.9	kNm	At center of gravity of model
	Resultant of reactions about Z	215.9	kNm	At center of gravity of model
	Max. displacement in X	-0.8	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-4.2	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	0.9	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	4.2	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	-1.4	mrad	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.3	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 293 (X: -11.443, Y: 0.008, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.591E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO60 - 0.6*LC1 + 0.7*LC4				
	Sum of loads in X	26.14	kN	
	Sum of support reactions in X	26.14	kN	Deviation 0.00%
	Sum of loads in Y	-0.00	kN	
	Sum of support reactions in Y	-0.00	kN	
	Sum of loads in Z	2499.60	kN	
	Sum of support reactions in Z	2499.60	kN	Deviation 0.00%
	Resultant of reactions about X	106.7	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	69.4	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.9	kNm	At center of gravity of model
	Max. displacement in X	0.9	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Y	0.2	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	0.6	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	1.1	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	0.1	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	-0.3	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.591E+107465		
	Infinity Norm	2.798E+11		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
Load Combination CO61 - 0.6*LC1 + 0.7*LC5				
	Sum of loads in X	-26.14	kN	
	Sum of support reactions in X	-26.14	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	2500.42	kN	
	Sum of support reactions in Z	2500.42	kN	Deviation 0.00%
	Resultant of reactions about X	104.9	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	443.5	kNm	At center of gravity of model
	Resultant of reactions about Z	0.9	kNm	At center of gravity of model
	Max. displacement in X	-1.4	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.2	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	0.7	mm	FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	1.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 5 (X: -12.525, Y: 0.943, Z: 3.500 m)
	Max. rotation about Y	0.4	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1883 (X: 0.125, Y: 4.738, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.591E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO62 - 0.6*LC1 + 0.7*LC6				
	Sum of loads in X	-0.00	kN	
	Sum of support reactions in X	-0.00	kN	
	Sum of loads in Y	14.87	kN	
	Sum of support reactions in Y	14.87	kN	Deviation 0.00%
	Sum of loads in Z	2500.42	kN	
	Sum of support reactions in Z	2500.42	kN	Deviation 0.00%
	Resultant of reactions about X	214.1	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	251.7	kNm	At center of gravity of model
	Resultant of reactions about Z	-3.1	kNm	At center of gravity of model
	Max. displacement in X	-0.5	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	0.4	mm	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	0.6	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	0.7	mm	FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	0.1	mrad	FE Mesh Node No. 20 (X: 5.975, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	0.1	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 334 (X: 5.973, Y: 4.873, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.591E+107465		
	Infinity Norm	2.798E+11		
Load Combination CO63 - 0.6*LC1 + 0.7*LC7				
	Sum of loads in X	-0.03	kN	
	Sum of support reactions in X	-0.03	kN	Deviation -0.00%
	Sum of loads in Y	-14.87	kN	
	Sum of support reactions in Y	-14.87	kN	Deviation 0.00%
	Sum of loads in Z	2500.42	kN	
	Sum of support reactions in Z	2500.42	kN	Deviation 0.00%
	Resultant of reactions about X	-4.3	kNm	At center of gravity of model (X:0.7, Y:2.2, Z:3.2 m)
	Resultant of reactions about Y	252.2	kNm	At center of gravity of model
	Resultant of reactions about Z	3.1	kNm	At center of gravity of model
	Max. displacement in X	-0.6	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	-0.4	mm	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. displacement in Z	0.6	mm	FE Mesh Node No. 104 (X: 12.775, Y: -2.200, Z: 3.500 m)
	Max. vector displacement	0.8	mm	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 3 (X: 5.975, Y: 0.000, Z: 0.000 m)
	Max. rotation about Y	0.2	mrad	FE Mesh Node No. 16 (X: -11.525, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 293 (X: -11.443, Y: 0.008, Z: 3.500 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.399E+11		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.591E+10746		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
		65		
	Infinity Norm	2.798E+11		

Summary				
	Max. displacement in X	3.5	mm	CO7, FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. displacement in Y	8.6	mm	CO8, FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. displacement in Z	2.1	mm	CO9, FE Mesh Node No. 45 (X: -12.525, Y: -1.000, Z: 3.500 m)
	Max. vector displacement	8.9	mm	CO8, FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about X	2.8	mrad	CO8, FE Mesh Node No. 18 (X: 0.125, Y: 4.800, Z: 0.000 m)
	Max. rotation about Y	-1.1	mrad	CO7, FE Mesh Node No. 39 (X: 11.775, Y: 0.000, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	CO8, FE Mesh Node No. 317 (X: 0.187, Y: 4.800, Z: 3.500 m)
	Other Settings:			
	Number of 1D finite elements	168		
	Number of 2D finite elements	1722		
	Number of 3D finite elements	0		
	Number of FE mesh nodes	1923		
	Number of equations	11538		
	Internal forces referred to deformed system for...:			
	Max. number of iterations	100		
	Number of divisions for member results	10		
	Division of cable/foundation/tapered members	10		
	Number of member divisions for searching maximum values	10		
	Subdivisions of FE mesh for graphical results	0		
	Percentage of iterations according to Picard method in combination with Newton-Raphson method	5	%	
	Options:			
	Activate shear stiffness of members (Ay, Az)	☒		
	Activate member divisions for large deformation or post-critical analysis	☒		
	Activate entered stiffness modifications	☒		
	Ignore rotational degrees of freedom	☐		
	Check of critical forces of members	☒		
	Nonsymmetric direct solver if demanded by nonlinear model	☐		
	Method for the system of equations	Direct		
	Plate bending theory	Mindlin		
	Solver version	64-bit		
	Precision and Tolerance:			
	Change default setting	☐		

4.3 LINES - SUPPORT FORCES

Line No.	LC/CO	Node No.	Location x [m]	Support Forces [kN/m] p_X	p_Y	p_Z	Support Moments [kNm/m] m_X	m_Y	m_Z
1	LC1	103	0.000	0.04	0.01	0.00	0.00	0.00	0.00
			0.252	0.04	0.01	0.00	0.00	0.00	0.00
			0.505	0.04	0.01	0.00	0.00	0.00	0.00
			1.010	0.03	0.01	0.00	0.00	0.00	0.00
			1.514	0.03	0.01	0.00	0.00	0.00	0.00
			2.019	0.03	0.01	0.00	0.00	0.00	0.00
			2.524	0.02	0.02	0.00	0.00	0.00	0.00
			3.029	0.02	0.02	0.00	0.00	0.00	0.00
			3.533	0.02	0.03	0.00	0.00	0.00	0.00
			4.038	0.02	0.03	0.00	0.00	0.00	0.00
			4.543	0.02	0.04	0.00	0.00	0.00	0.00
			5.048	0.02	0.04	0.00	0.00	0.00	0.00
			5.552	0.02	0.04	0.00	0.00	0.00	0.00
			6.057	0.02	0.04	0.00	0.00	0.00	0.00
			6.562	0.02	0.04	0.00	0.00	0.00	0.00
			7.067	0.02	0.03	0.00	0.00	0.00	0.00
			7.571	0.02	0.03	0.00	0.00	0.00	0.00
			8.076	0.02	0.02	0.00	0.00	0.00	0.00
			8.581	0.02	0.02	0.00	0.00	0.00	0.00
			9.086	0.02	0.01	0.00	0.00	0.00	0.00
			9.590	0.02	0.01	0.00	0.00	0.00	0.00
			10.095	0.01	0.01	0.00	0.00	0.00	0.00
			10.600	0.01	0.01	0.00	0.00	0.00	0.00
			11.105	0.01	0.01	0.00	0.00	0.00	0.00
			11.610	0.01	0.00	0.00	0.00	0.00	0.00
			12.114	0.02	0.00	0.00	0.00	0.00	0.00
			12.619	0.02	-0.01	0.00	0.00	0.00	0.00
			13.124	0.01	-0.02	0.00	0.00	0.00	0.00
			13.629	0.01	-0.03	0.00	0.00	0.00	0.00
			14.133	0.00	-0.03	0.00	0.00	0.00	0.00
			14.638	-0.01	-0.03	0.00	0.00	0.00	0.00
			15.143	-0.02	-0.03	0.00	0.00	0.00	0.00
			15.648	-0.03	-0.03	0.00	0.00	0.00	0.00
			16.152	-0.04	-0.02	0.00	0.00	0.00	0.00
			16.657	-0.05	-0.02	0.00	0.00	0.00	0.00
			17.162	-0.06	-0.02	0.00	0.00	0.00	0.00
			17.667	-0.08	-0.02	0.00	0.00	0.00	0.00
			18.171	-0.09	-0.02	0.00	0.00	0.00	0.00
			18.676	-0.11	-0.01	0.00	0.00	0.00	0.00
			19.181	-0.12	-0.01	0.00	0.00	0.00	0.00
			19.686	-0.13	-0.01	0.00	0.00	0.00	0.00
			20.190	-0.14	-0.01	0.00	0.00	0.00	0.00